
O-RAN Work Group 3 (Near-RT RIC and E2 Interface)

E2 Application Protocol (E2AP)

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Foreword

This Technical Specification (TS) has been produced by WG3 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the Near-RT RIC layer signalling protocol for the E2 interface.

The E2 interface provides means for interconnecting a Near-RT RIC and an E2 Node. The E2 Application Protocol (E2AP) supports the functions of E2 interface by signalling procedures defined in the present document. E2AP is developed in accordance with the general principles stated in O-RAN.WG3.TS.E2GAP [2].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

- | | |
|------|--|
| [1] | Void. |
| [2] | O-RAN WG3.TS.E2GAP: "O-RAN E2 General Aspects and Principles (E2GAP)". |
| [3] | O-RAN WG3.TS.E2SM: "O-RAN E2 Service Model (E2SM)". |
| [4] | Void. |
| [5] | Void. |
| [6] | Void. |
| [7] | Void. |
| [8] | Void. |
| [9] | Void. |
| [10] | Void. |
| [11] | Void. |
| [12] | Void. |
| [13] | Void. |
| [14] | Void. |
| [15] | Recommendation ITU-T X.691 (02/2021): "Information technology – ASN.1 encoding rules: Specification of Packed Encoding Rules (PER)". |
| [16] | Recommendation ITU-T X.680 (02/2021): "Information technology – Abstract Syntax Notation One (ASN.1): Specification of basic notation". |
| [17] | Recommendation ITU-T X.681 (02/2021): "Information technology – Abstract Syntax Notation One (ASN.1): Information object specification". |
| [18] | Void. |
| [19] | 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)" |

- [20] 3GPP TS 38.423: "NG-RAN; Xn application protocol (XnAP)"
- [21] 3GPP TS 37.483: "E1 Application Protocol (E1AP)"
- [22] 3GPP TS 38.473: "NG-RAN; F1 application protocol (F1AP)"
- [23] 3GPP TS 37.473: "W1 interface; Application Protocol (W1AP)"
- [24] 3GPP TS 36.413: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP)"
- [25] 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 application protocol (X2AP) "
- [26] IETF RFC 5905: "Network Time Protocol Version 4: Protocol and Algorithms Specification"
- [27] O-RAN WG1.TS.OAD: "O-RAN Architecture Description"

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications"
- [i.2] 3GPP TR 25.921: "Guidelines and principles for protocol description and error handling".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [i.1] apply.

In addition, the terms defined in O-RAN WG3.TS.E2GAP [2], O-RAN WG1.TS.OAD [27], and the following also apply:

NOTE: A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [i.1], O-RAN WG3.TS.E2GAP [2], and O-RAN WG1.TS.OAD [27].

E2 Node Component ID: local identifier used to uniquely identify an E2 Node component

Elementary Procedure: E2AP protocol consists of Elementary Procedures (EPs)

NOTE: An E2AP Elementary Procedure is a unit of interaction between the Near-RT RIC and an E2 Node. An EP consists of an initiating message and possibly a response message. Two kinds of EPs are used:

Class 1: Elementary Procedures with response (success or failure),

Class 2: Elementary Procedures without response.

Global E2 Node ID: global identifier of an E2 Node. Defined as the global eNB or gNB identifier and an optional local identifier of an CU-UP or DU which is required when and if an individual DU or CU-UP supports a direct E2 interface

Global RIC ID: global identifier of a Near-RT RIC

RAN Function ID: local identifier of a specific RAN Function within an E2 Node that supports one or more RIC Services using a specific E2 Service Model

RAN Function OID: RAN Function Object Identifier used to identify specific RAN function definition (i.e. E2SM used by specific RAN Function)

RIC Action ID: local identifier used Near-RT RIC to identify a specific RIC Service Action within a specific RIC Subscription Request, used by E2 Node in subsequent RIC Indication messages

RIC Call Process ID: local identifier used by E2 Node to identify the associated procedure during an Insert RIC Service Action, used by Near-RT RIC in subsequent RIC Control procedure

RIC Request ID: local identifier used to identify a specific RIC Functional procedure among all ongoing parallel procedures of the same type initiated by the same protocol peer.

NOTE: Messages belonging to the same procedure use the same RIC Request ID. The RIC Request ID is determined by the initiating peer of a RIC Functional Procedure.

Transaction ID: local identifier used to uniquely identify a Global Procedure among all ongoing parallel procedures of the same type initiated by the same protocol peer

NOTE: Messages belonging to the same procedure use the same Transaction ID. The Transaction ID is determined by the initiating peer of a Global Procedure (Near-RT RIC or E2 Node).

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [i.1] apply.

In addition, the abbreviations defined in O-RAN WG1.TS.OAD [18] and the following also apply:

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [i.1] and O-RAN WG1.TS.OAD [27].

EP Elementary Procedure

4 General

4.1 Procedure specification principles

The principle for specifying the procedure logic is to specify the functional behaviour of the terminating node exactly and completely. Any rule that specifies the behaviour of the originating node shall be possible to be verified with information that is visible within the system.

The following specification principles have been applied for the procedure text in clause 8:

- The procedure text discriminates between:

- 1) Functionality which "shall" be executed.

The procedure text indicates that the receiving node "shall" perform a certain function Y under a certain condition. If the receiving node supports procedure X but cannot perform functionality Y requested in the REQUEST message of a Class 1 EP, the receiving node shall respond with the message used to report unsuccessful outcome for this procedure, containing an appropriate cause value.

- 2) Functionality which "shall, if supported" be executed.

The procedure text indicates that the receiving node "shall, if supported," perform a certain function Y under a certain condition. If the receiving node supports procedure X, but does not support functionality Y, the receiving node shall proceed with the execution of the EP, possibly informing the requesting node about the nonsupported functionality.

- Any required inclusion of an optional IE in a response message is explicitly indicated in the procedure text. If the procedure text does not explicitly indicate that an optional IE shall be included in a response message, the optional IE shall not be included. For requirements on including Criticality Diagnostics IE, see clause 10.

4.2 Forwards and backwards compatibility

The forwards and backwards compatibility of the protocol is assured by mechanism where all current and future messages, and IEs or groups of related IEs, include ID and criticality fields that are coded in a standard format that will not be changed in the future. These parts can always be decoded regardless of the standard version.

4.3 Specification notations

For the purposes of the present document, the following notations apply:

Procedure	When referring to an Elementary Procedure (EP) in the present document the Procedure Name is written with the first letters in each word in upper case characters followed by the word "procedure", e.g. Handover Preparation procedure.
Message	When referring to a message in the present document the MESSAGE NAME is written with all letters in upper case characters followed by the word "message", e.g. HANDOVER REQUEST message.
IE	When referring to an Information Element (IE) in the present document the <i>Information Element Name</i> is written with the first letters in each word in upper case characters and all letters in <i>Italic font</i> followed by the abbreviation "IE", e.g. <i>E-RAB ID</i> IE.
Value of an IE	When referring to the value of an Information Element (IE) in the present document the "Value" is written as it is specified in the present document enclosed by quotation marks, e.g. "Value".

5 E2AP Services

5.1 E2AP procedure modules

The E2 interface E2AP procedures are divided into two modules as follows:

1. RIC Functional Procedures;
2. Global Procedures.

The RIC functional procedures module contains procedures used to pass application specific messages between Near-RT RIC applications and a target RAN Function in an E2 node as specified in O-RAN WG3.TS.E2GAP [2].

The Global Procedures module contains procedures that are not directly related to a specific application.

5.2 Parallel transactions

Parallel transactions, that is, multiple ongoing E2AP procedures related to the same Application and E2 node, are supported.

6 Services expected from signalling transport

The signalling connection shall provide in sequence delivery of E2AP messages. E2AP shall be notified if the signalling connection breaks.

7 Functions of E2AP

The functions of E2AP are described in O-RAN WG3.TS.E2GAP [2].

8 E2AP procedures

8.1 Elementary procedures

In the Tables 8.1-1 and 8.1-2, all EPs are divided into Class 1 and Class 2 EPs.

Table 8.1-1: Class 1 Elementary Procedures

Initiated by	Elementary Procedure	Initiating Message	Successful Outcome Response message	Unsuccessful Outcome Response message
Near-RT RIC	RIC Subscription	RIC SUBSCRIPTION REQUEST	RIC SUBSCRIPTION RESPONSE	RIC SUBSCRIPTION FAILURE
Near-RT RIC	RIC Subscription Delete	RIC SUBSCRIPTION DELETE REQUEST	RIC SUBSCRIPTION DELETE RESPONSE	RIC SUBSCRIPTION DELETE FAILURE
Near-RT RIC	RIC Subscription Modification	RIC SUBSCRIPTION MODIFICATION REQUEST	RIC SUBSCRIPTION MODIFICATION RESPONSE	RIC SUBSCRIPTION MODIFICATION FAILURE
E2 Node	RIC Subscription Modification Required	RIC SUBSCRIPTION MODIFICATION REQUIRED	RIC SUBSCRIPTION MODIFICATION CONFIRM	RIC SUBSCRIPTION MODIFICATION REFUSE
Near-RT RIC	RIC Subscription State Control	RIC SUBSCRIPTION STATE CONTROL REQUEST	RIC SUBSCRIPTION STATE CONTROL RESPONSE	RIC SUBSCRIPTION STATE CONTROL FAILURE
Near-RT RIC	RIC Subscription Audit	RIC SUBSCRIPTION AUDIT REQUEST	RIC SUBSCRIPTION AUDIT RESPONSE	RIC SUBSCRIPTION AUDIT FAILURE
E2 Node	RIC Assistance	RIC ASSISTANCE REQUEST	RIC ASSISTANCE RESPONSE	RIC ASSISTANCE FAILURE
Near-RT RIC	RIC Control	RIC CONTROL REQUEST	RIC CONTROL ACKNOWLEDGE	RIC CONTROL FAILURE
Near-RT RIC	RIC Query	RIC QUERY REQUEST	RIC QUERY RESPONSE	RIC QUERY FAILURE
Near-RT RIC	RIC Service Load Status	RIC SERVICE LOAD STATUS REQUEST	RIC SERVICE LOAD STATUS RESPONSE	RIC SERVICE LOAD STATUS FAILURE
E2 Node	E2 Setup	E2 SETUP REQUEST	E2 SETUP RESPONSE	E2 SETUP FAILURE
E2 Node	RIC Service Update	RIC SERVICE UPDATE	RIC SERVICE UPDATE ACKNOWLEDGE	RIC SERVICE UPDATE FAILURE
E2 Node	E2 Node Configuration Update	E2 NODE CONFIGURATION UPDATE	E2 NODE CONFIGURATION UPDATE ACKNOWLEDGE	E2 NODE CONFIGURATION UPDATE FAILURE
Near-RT RIC	E2 Connection Update	E2 CONNECTION UPDATE	E2 CONNECTION UPDATE ACKNOWLEDGE	E2 CONNECTION UPDATE FAILURE
Near-RT RIC or E2 Node	Reset	RESET REQUEST	RESET RESPONSE	
Near-RT RIC or E2 Node	E2 Removal	E2 REMOVAL REQUEST	E2 REMOVAL RESPONSE	E2 REMOVAL FAILURE
Near-RT RIC	E2 Resource Status	E2 RESOURCE STATUS REQUEST	E2 RESOURCE STATUS RESPONSE	E2 RESOURCE STATUS FAILURE

Table 8.1-2: Class 2 Elementary Procedures

Initiated by	Elementary Procedure	Initiating Message
Near-RT RIC	RIC Assistance Indication	RIC ASSISTANCE INDICATION
E2 Node	RIC Assistance Halt	RIC ASSISTANCE HALT
E2 Node	RIC Indication	RIC INDICATION
Near-RT RIC	RIC Service Query	RIC SERVICE QUERY
E2 Node	RIC Subscription Delete Required	RIC SUBSCRIPTION DELETE REQUIRED
E2 Node	RIC Service Load Update	RIC SERVICE LOAD UPDATE
E2 Node or Near-RT RIC	Error Indication	ERROR INDICATION
E2 Node	E2 Resource Status Update	E2 RESOURCE STATUS UPDATE

8.2 RIC Functional procedures

8.2.1 RIC Subscription procedure

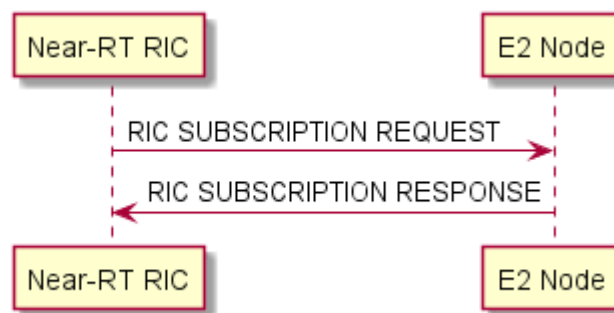
8.2.1.1 General

This procedure is used to establish RIC Subscriptions on E2 Node consisting of an event trigger and a sequence of RIC Service Actions.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.1.2 Successful operation


Figure 8.2.1.2-1: RIC Subscription procedure, successful operation

The Near-RT RIC initiates the procedure by sending a RIC SUBSCRIPTION REQUEST message, which shall contain a unique *RIC Request ID* IE, assigned by the Near-RT RIC, to the E2 Node.

When the Near-RT RIC sends the RIC SUBSCRIPTION REQUEST message, it shall start the timer $T_{\text{RICEVENTcreate}}$.

At reception of the RIC SUBSCRIPTION REQUEST message the E2 Node shall:

- Determine the target RAN Function using the information in the *RAN Function ID* IE and configure the requested event trigger using information in the *RIC Subscription Details* IE.
- If one or more Report, Insert and/or Policy RIC Service Actions are included in the *RIC Subscription Details* IE then the target RAN Function shall validate the event trigger and requested action sequence and, if accepted, store the required *RIC Request ID*, *RIC Event Trigger Definition* IE and sequence of RIC Service Actions.
- If optional *RIC Subscription Start Time* IE is present and has expired, then the E2 Node shall ignore the optional *RIC Subscription Start Time* IE.

If the requested trigger and at least one required RIC Service Action are accepted by the E2 Node, the E2 Node shall reserve for each admitted RIC Service Action the necessary resources and send the RIC SUBSCRIPTION RESPONSE message back to the Near-RT RIC.

The E2 Node shall include in the response message the RIC Service Actions for which resources have been prepared at the E2 Node in the *RIC Actions Admitted List IE*.

The E2 Node shall include the RIC Service Actions that have not been admitted in the *RIC Actions Not Admitted List IE* with an appropriate cause value.

Upon reception of the RIC SUBSCRIPTION RESPONSE message the Near-RT RIC shall stop timer $T_{RICEVENTcreate}$ and terminate the RIC Subscription procedure.

If more than one RIC Service Actions has been accepted by the E2 Node then, at each occurrence of the common Event Trigger, the sequence of RIC Service Actions shall be executed according to the following considerations:

- If optional *RIC Action Execution Order IE* is not present or is present and set to 0 ("Any order"), then the specific RIC Service Action in the sequence of RIC Service Actions may be executed in any order irrespective of the execution order of the other RIC Service Actions.
- If optional *RIC Action Execution Order IE* is present and set to a value greater than 0, then the specific RIC Service Action shall be executed in order according to the *RIC Action Execution Order IE*.
- If two or more RIC Service Actions have the same value for the optional *RIC Action Execution Order IE* then these RIC Service Actions shall be executed in parallel.

If the optional *RIC Subscription Start Time IE* is present, the E2 Node shall only enable the event trigger from the indicated start time.

If the optional *RIC Subscription End Time IE* is present, the E2 Node shall disable the event trigger when the indicated end time has expired.

Interactions with RIC Subscription Delete Required procedure:

If the optional *RIC Subscription End Time IE* is present and the indicated end time has expired, the E2 Node may send the RIC SUBSCRIPTION DELETE REQUIRED message to the Near-RT RIC with an appropriate cause value.

Interactions with RIC Subscription Delete procedure:

If the optional *RIC Subscription End Time IE* is present, the Near-RT RIC may initiate an RIC Subscription Delete procedure when the expected *RIC Subscription End Time* has expired.

8.2.1.3 Unsuccessful operation

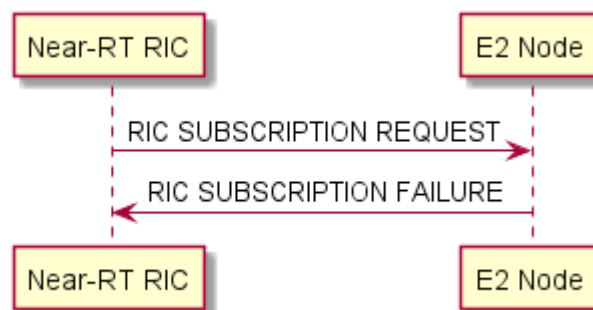


Figure 8.2.1.3-1: RIC Subscription procedure, unsuccessful operation

If a failure occurs during the RIC Subscription procedure the E2 Node shall send the RIC SUBSCRIPTION FAILURE message to the Near-RT RIC containing an appropriate cause value.

If the E2 Node admits none of the requested RIC Service Actions, or detects an issue with the requested sequence of RIC Service Actions, or in the optional *RIC Subsequent Action IE* definitions, the E2 Node shall send the RIC SUBSCRIPTION FAILURE message to the Near-RT RIC containing an appropriate cause value.

If the RIC Subscription procedure contains an invalid optional *RIC Subscription Start Time* IE and/or *RIC Subscription End Time* IE, the E2 Node shall send the RIC SUBSCRIPTION FAILURE message to the Near-RT RIC containing an appropriate cause value.

Upon reception of the RIC SUBSCRIPTION FAILURE message the Near-RT RIC shall stop the timer $T_{\text{RICEVENTcreate}}$ and terminate the RIC Subscription procedure.

Interactions with RIC Subscription Delete procedure:

If there is no response from the E2 Node to the RIC SUBSCRIPTION REQUEST message before timer $T_{\text{RICEVENTcreate}}$ expires in the Near-RT RIC, the Near-RT RIC shall initiate the RIC Subscription Delete procedure containing *RIC Request ID* IE that was previously assigned by the Near-RT RIC to cancel the RIC Subscription towards the E2 Node. The Near-RT RIC shall ignore any RIC SUBSCRIPTION RESPONSE or RIC SUBSCRIPTION FAILURE message containing *RIC Request ID* IE that was previously assigned by the Near-RT RIC received after the initiation of the RIC Subscription Delete procedure and release any resources related to the concerned E2 Node.

8.2.1.4 Abnormal conditions

If the E2 Node receives a RIC SUBSCRIPTION REQUEST message containing *RIC Subscription Details* IE that does not align with the specific E2 Service Model, see O-RAN WG3.TS.E2SM [3], the E2 Node shall send the RIC SUBSCRIPTION FAILURE message to the Near-RT RIC containing an appropriate cause value.

If the E2 Node receives a RIC SUBSCRIPTION REQUEST message which contains an unknown *RAN Function ID* IE, the E2 Node shall send the RIC SUBSCRIPTION FAILURE message to the Near-RT RIC containing an appropriate cause value.

If the E2 Node receives a RIC SUBSCRIPTION REQUEST message containing identical contents the E2 Node shall send the RIC SUBSCRIPTION FAILURE message to the Near-RT RIC containing an appropriate cause value.

8.2.2 RIC Subscription Delete procedure

8.2.2.1 General

This procedure is used to delete RIC Subscriptions on E2 Node.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.2.2 Successful operation

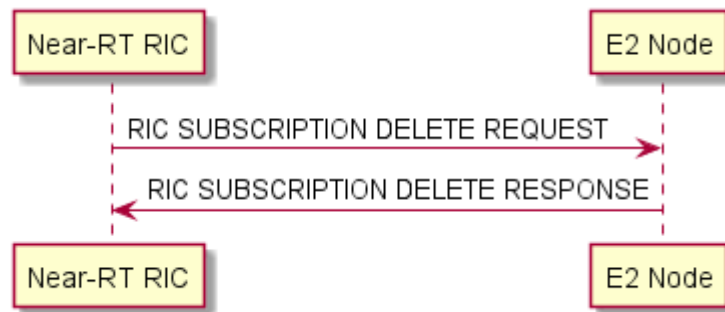


Figure 8.2.2.2-1: RIC Subscription Delete procedure, successful operation

The Near-RT RIC initiates the procedure by sending a RIC SUBSCRIPTION DELETE REQUEST message, containing *RIC Request ID* IE that was previously assigned by the Near-RT RIC during a successful RIC Subscription procedure, to the E2 Node.

When the Near-RT RIC sends the RIC SUBSCRIPTION DELETE REQUEST message, it shall start timer $T_{\text{RICEVENTdelete}}$.

At reception of the RIC SUBSCRIPTION DELETE REQUEST message the E2 Node shall delete the indicated RIC Subscription and release the corresponding necessary resources.

The E2 Node shall send the RIC SUBSCRIPTION DELETE RESPONSE message back to the Near-RT RIC.

Upon reception of the RIC SUBSCRIPTION DELETE RESPONSE message the Near-RT RIC shall stop timer $T_{\text{RICEVENTdelete}}$, release any necessary resources associated with that RIC Subscription and terminate the RIC Subscription Delete procedure.

8.2.2.3 Unsuccessful operation

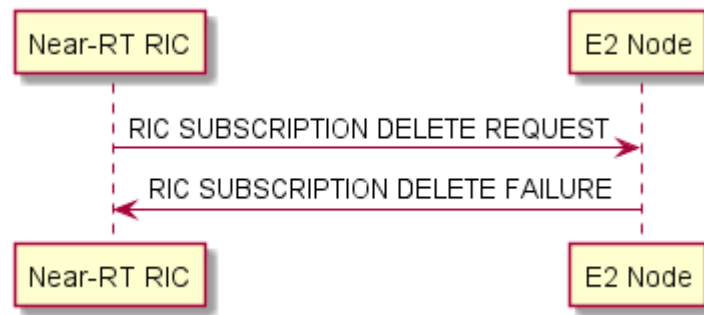


Figure 8.2.2.3-1: RIC Subscription Delete procedure, unsuccessful operation

If a failure occurs during the RIC Subscription Delete procedure, the E2 Node shall send the RIC SUBSCRIPTION DELETE FAILURE message to the Near-RT RIC containing an appropriate cause value.

If the *RIC Request ID* IE included in the RIC SUBSCRIPTION DELETE REQUEST message is unknown, the E2 Node shall send the RIC SUBSCRIPTION DELETE FAILURE message to the Near-RT RIC containing an appropriate cause value

Upon reception of the RIC SUBSCRIPTION DELETE FAILURE message the Near-RT RIC shall stop timer $T_{\text{RICEVENTdelete}}$ and terminate the RIC Subscription Delete procedure.

8.2.2.4 Abnormal conditions

If the E2 Node receives a RIC SUBSCRIPTION DELETE REQUEST message contains an unknown *RAN Function ID* IE, the E2 Node shall send the RIC SUBSCRIPTION DELETE FAILURE message to the Near-RT RIC containing an appropriate cause value.

8.2.2A RIC Subscription Delete Required procedure

8.2.2A.1 General

This procedure is used to enable the E2 Node to request deletion of the existing RIC Subscriptions in the E2 Node previously created for the Near-RT RIC.

This procedure shall be initiated by the E2 Node.

This procedure uses RIC Service signalling.

8.2.2A.2 Successful operation

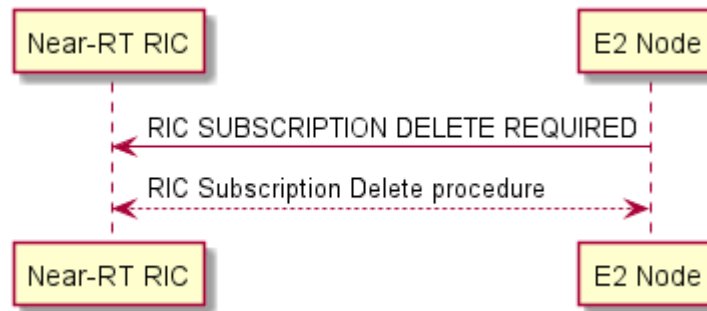


Figure 8.2.2A.2-1: RIC Subscription Delete Required procedure, successful operation

The E2 Node initiates the procedure by sending a RIC SUBSCRIPTION DELETE REQUIRED message, containing *RIC Request ID* IE that was previously assigned by the Near-RT RIC during a successful RIC Subscription procedure, to the Near-RT RIC.

The message shall contain an appropriate cause value for each RIC Subscription requesting to remove.

At reception of the RIC SUBSCRIPTION DELETE REQUIRED message, for each RIC Subscription associated with the included *RIC Request ID* IE and *RAN Function ID* IE, the Near-RT RIC may initiate the RIC Subscription Delete procedure toward the E2 Node.

8.2.2A.3 Abnormal conditions

If the Near-RT RIC receives a RIC SUBSCRIPTION DELETE REQUIRED message which contains an unknown *RIC Request ID* IE and *RAN Function ID* IE, the Near-RT RIC shall ignore the message.

8.2.2B RIC Subscription Audit procedure

8.2.2B.1 General

This procedure is used to audit the list of establish RIC Subscriptions on E2 Node.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.2B.2 Successful operation

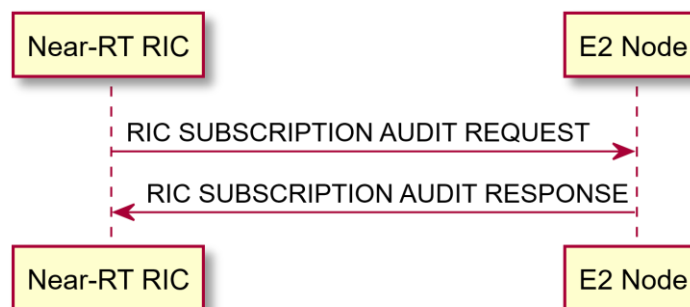


Figure 8.2.2B.2-1: RIC Subscription Audit procedure, successful operation

The Near-RT RIC initiates the procedure by sending a RIC SUBSCRIPTION AUDIT REQUEST message which shall contain a unique *RIC Request ID* IE, assigned by the Near-RT RIC to identify the procedure, and may contain the *RIC Subscription*

Audit List IE with each item containing a *RIC Request ID* IE, that was previously assigned by the Near-RT RIC during a successful RIC Subscription procedure, to the E2 Node.

When the Near-RT RIC sends the RIC SUBSCRIPTION AUDIT REQUEST message, it shall start the timer $T_{\text{RICEVENTcreate}}$.

At reception of the RIC SUBSCRIPTION AUDIT REQUEST message the E2 Node shall:

- If the *RIC Subscription Audit List* IE is present and contains at least one RIC Subscription, identified by the *RIC Request ID* IE, that is recognized by the E2 Node, then the E2 Node shall record the confirmed RIC Subscriptions in the *RIC Subscription Confirmed List* IE.
- If the *RIC Subscription Audit List* IE is present and contains at least one RIC Subscription, identified by the *RIC Request ID* IE, that is not known to the E2 Node, then the E2 Node shall record the unknown RIC Subscriptions in the *RIC Subscription Unknown List* IE.
- If the *RIC Subscription Audit List* IE is present and if the *RIC Subscription Audit Flag* IE is not present and the E2 Node holds at least one established RIC Subscription that is not included in the *RIC Subscription Audit List* IE, then the E2 Node shall record the missing RIC Subscriptions in the *RIC Subscription Missing List* IE.
- If the *RIC Subscription Audit List* IE is present and if the *RIC Subscription Audit Flag* IE is present and the *Listed Records Only* IE is present and set to TRUE, then the E2 Node shall only respond with respect to the RIC Subscriptions in the *RIC Subscription Audit List* IE.
- If the *RIC Subscription Audit List* IE is not present, then the E2 Node shall record all established RIC Subscriptions in the *RIC Subscription Missing List* IE.

After processing the RIC SUBSCRIPTION AUDIT REQUEST message, if the E2 Node has at least one item in *RIC Subscription Confirmed List* IE, *RIC Subscription Unknown List* IE or *RIC Subscription Missing List* IE, the E2 Node shall send the RIC SUBSCRIPTION AUDIT RESPONSE message back to the Near-RT RIC.

The E2 Node shall include in the response message the confirmed RIC Subscriptions in the *RIC Subscription Confirmed List* IE.

The E2 Node shall include in the response message the not known RIC Subscriptions in the *RIC Subscription Unknown List* IE.

The E2 Node shall include in the response message the missing RIC Subscriptions in the *RIC Subscription Missing List* IE.

Upon reception of the RIC SUBSCRIPTION AUDIT RESPONSE message the Near-RT RIC shall stop timer $T_{\text{RICEVENTcreate}}$ and terminate the RIC Subscription Audit procedure.

8.2.2B.3 Unsuccessful operation

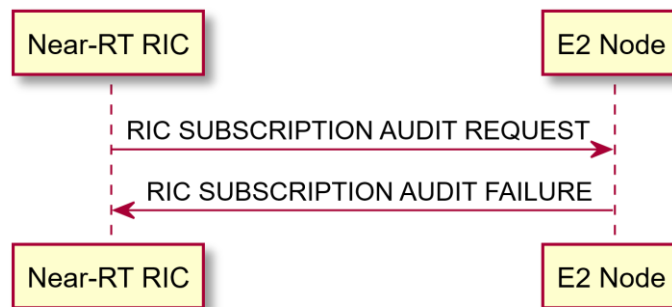


Figure 8.2.2B.3-1: RIC Subscription Audit procedure, unsuccessful operation

If a failure occurs during the RIC Subscription Audit procedure the E2 Node shall send the RIC SUBSCRIPTION AUDIT FAILURE message to the Near-RT RIC containing an appropriate cause value.

If the E2 Node after processing the RIC SUBSCRIPTION AUDIT REQUEST message, the E2 Node does not have at least one item in *RIC Subscription Confirmed List* IE, *RIC Subscription Unknown List* IE or *RIC Subscription Missing List* IE, the E2

Node shall send the RIC SUBSCRIPTION AUDIT FAILURE message to the Near-RT RIC containing an appropriate cause value.

Upon reception of the RIC SUBSCRIPTION AUDIT FAILURE message the Near-RT RIC shall stop the timer $T_{\text{RICEVENTcreate}}$ and terminate the RIC Subscription procedure.

8.2.2B.4 Abnormal conditions

If the E2 Node receives a RIC SUBSCRIPTION AUDIT REQUEST message which contains an unknown *RAN Function ID* IE, the E2 Node shall send the RIC SUBSCRIPTION AUDIT FAILURE message to the Near-RT RIC containing an appropriate cause value.

8.2.3 RIC Indication procedure

8.2.3.1 General

The purpose of the RIC Indication procedure is to transfer Report and/or Insert RIC Service Action associated with a RIC Subscription procedure.

This procedure shall be initiated by the E2 Node.

This procedure uses RIC Service signalling.

8.2.3.2 Successful operation

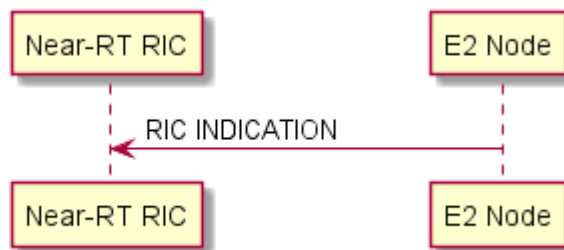


Figure 8.2.3.2-1: RIC Indication procedure, successful operation

An E2 Node initiates the procedure by sending RIC INDICATION message to the Near-RT RIC containing the *RIC Request ID* IE, that was previously assigned by the Near-RT RIC during a successful RIC Subscription procedure.

If the RIC Indication message is in response to an Insert RIC Service Action, then the E2 Node shall provide the *RIC Call Process ID* IE within the RIC INDICATION message, and the E2 Node shall start the associated *RIC Time to Wait* timer, and wait for the corresponding response from the Near-RT RIC.

Near-RT RIC may use the *RIC Call Process ID* IE in a subsequent RIC Control procedure.

If an *RIC Subsequent Action* IE was associated to the RIC Service Action then, after successful transmission of the RIC INDICATION message, the originating E2 Node shall progress accordingly:

- If the *RIC Subsequent Action Type* IE was set to Continue or Halt, and the associated *RIC Time to Wait* timer has not expired, and a RIC CONTROL REQUEST message is received with the same *RIC Call Process ID* IE, then the associated procedure shall process the RIC CONTROL REQUEST and continue to execute any remaining actions in the sequence of RIC Actions defined in the RIC Subscription procedure.
- If the *RIC Subsequent Action Type* IE was set to Continue and the associated *RIC Time to Wait* timer has expired, then the E2 Node shall continue to execute any remaining RIC Service Actions in the sequence of RIC Service Actions defined in the RIC Subscription procedure.
- If the *RIC Subsequent Action Type* IE was set to Halt and the associated *RIC Time to Wait* timer has expired, then the E2 Node shall abort further processing of the associated procedure in the E2 node. In this case, any remaining or

ongoing parallel RIC Service Actions in the sequence of RIC Actions defined in the RIC Subscription procedure shall also be aborted.

8.2.3.3 Unsuccessful operation

Not applicable.

8.2.3.4 Abnormal conditions

Not applicable.

8.2.4 RIC Control procedure

8.2.4.1 General

The purpose of the RIC Control procedure is to initiate or resume a specific functionality in the E2 Node.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.4.2 Successful operation

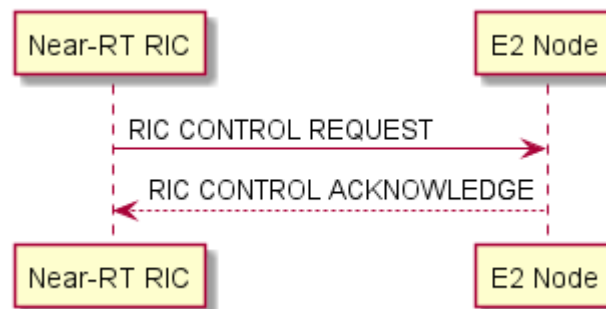


Figure 8.2.4.2-1: RIC Control procedure, successful operation

The Near-RT RIC initiates the procedure by sending a RIC CONTROL REQUEST message containing a unique *RIC Request ID* IE, assigned by the Near-RT RIC.

When the Near-RT RIC sends the RIC CONTROL REQUEST message and the optional *RIC Control Ack Request* IE has been set to "Ack", or is not present, the Near-RT RIC, it shall start the timer $T_{RICControl}$.

At reception of the RIC CONTROL REQUEST message the E2 Node shall:

- Determine the target RAN Function using the information in the *RAN Function ID* IE and initiate the requested RIC Control procedure action using information in the *RIC Control Message* IE.
- If the *RIC Call Process ID* IE is included in the RIC CONTROL REQUEST message, the E2 Node shall use this IE to identify a specific call process that was indicated in the RIC INDICATION message.
- If the RIC CONTROL REQUEST message contains the optional *RIC Control Ack Request* IE set to "Ack", or if the optional *RIC Control Ack Request* IE is not present, and the E2 Node has successfully processed the requested RIC Control procedure action, then the E2 Node shall respond with the RIC CONTROL ACKNOWLEDGE message.
- If the RIC CONTROL REQUEST message contains the optional *RIC Control Ack Request* IE set to "NoAck" and the E2 Node has successfully processed the requested RIC Control procedure action, then the E2 Node shall not send the RIC CONTROL ACKNOWLEDGE message.

Upon reception of the RIC CONTROL ACKNOWLEDGE message, the Near-RT RIC shall stop timer $T_{RICcontrol}$ and terminate the RIC Control procedure.

The Near-RT RIC may use the information contained in the optional *RIC Control Outcome* IE to determine subsequent actions.

8.2.4.3 Unsuccessful operation

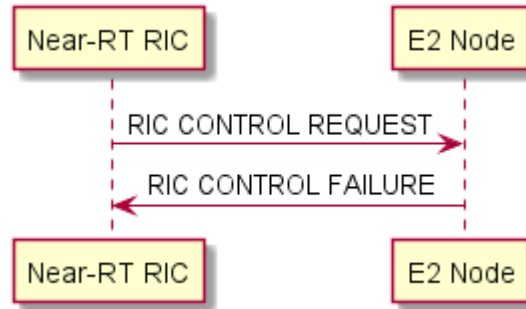


Figure 8.2.4.3-1: RIC Control procedure, unsuccessful operation

If the RIC CONTROL REQUEST message contains an invalid *RIC Call Process ID* IE, then the E2 Node shall respond with the RIC CONTROL FAILURE message with an appropriate cause value.

If the RIC CONTROL REQUEST message contains the optional *RIC Call Process ID* IE for which the associated *RIC Time to Wait* timer had expired, then the E2 Node shall respond with the RIC CONTROL FAILURE message with an appropriate cause value.

If the E2 Node fails to execute the requested RIC Control procedure E2SM specific action, then the E2 Node shall respond with the RIC CONTROL FAILURE message with an appropriate cause value.

If the E2 Node detects an encoding or functional error in the E2SM specific IEs contained in the RIC CONTROL REQUEST message, then the E2 Node shall respond with the RIC CONTROL FAILURE message with an appropriate cause value.

If the E2 Node receives a RIC CONTROL REQUEST message which contains an unknown *RAN Function ID* IE the E2 Node shall respond with the RIC CONTROL FAILURE message containing an appropriate cause value.

If the E2 Node does not support the specific RIC Control procedure action, then the E2 Node shall respond with the RIC CONTROL FAILURE message containing an appropriate cause value.

Upon reception of the RIC CONTROL FAILURE message the Near-RT RIC shall stop timer $T_{RICcontrol}$, if running, and terminate the RIC Control procedure.

The Near-RT RIC may use the information contained in the *Cause* IE and optional *RIC Control Outcome* IE to determine subsequent actions.

8.2.4.4 Abnormal conditions

Upon reception of the ERROR INDICATION message including the *RIC Request ID* IE associated to the RIC CONTROL REQUEST message, the Near-RT RIC shall stop timer $T_{RICcontrol}$, if running, and terminate the RIC Control procedure.

If timer $T_{RICcontrol}$ was set when sending the RIC CONTROL REQUEST message and there was no response from the E2 node before the timer expiry, the Near-RT RIC shall send an ERROR INDICATION with the appropriate value for the *Cause* IE.

8.2.5 RIC Subscription Modification procedure

8.2.5.1 General

The purpose of the RIC Subscription Modification procedure is to modify an existing RIC subscription on an E2 node, in terms of its event trigger definition and/or the sequence of actions.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.5.2 Successful operation

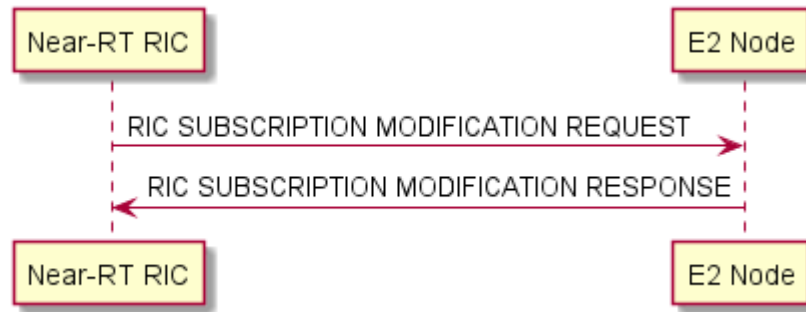


Figure 8.2.5.2-1: RIC Subscription Modification procedure, successful operation

The Near-RT RIC initiates this procedure by sending the RIC SUBSCRIPTION MODIFICATION REQUEST message to the E2 node, containing the *RIC Request ID* IE to uniquely identify the existing RIC Subscription in the E2 node.

When the Near-RT RIC sends the RIC SUBSCRIPTION MODIFICATION REQUEST message, it shall start timer $T_{\text{RICEVENTmodify}}$.

Upon reception of the RIC SUBSCRIPTION MODIFICATION REQUEST message, the E2 node shall determine the existing RIC Subscription and the target RAN Function from the *RIC Request ID* IE and the *RAN Function ID* IE, respectively.

If the *RIC Event Trigger Definition to be Modified* IE is included, then the E2 node shall validate and modify the event trigger defined for the existing RIC subscription based on the contents of the IE.

If the *RIC Actions to be Removed List* IE is included, then for every *RIC Action ID* IE included in the list, the E2 node shall delete the requested action and release the corresponding necessary resources.

If the *RIC Actions to be Modified List* IE is included, then for every *RIC Action ID* IE included in the list for which there exists a corresponding *RIC Action Definition* IE and/or *RIC Subsequent Action* IE, the E2 node shall modify the existing behaviour for the action with the requested modification in the respective IEs and modify the corresponding necessary resources.

If the *RIC Actions to be Modified List* IE is included, then for every *RIC Action ID* IE included in the list for which there exists a *RIC Action Execution Order* IE, the E2 node shall replace the current execution order for the action in the sequence of actions with the new execution order for the action in the sequence, as given in the *RIC Action Execution Order* IE.

If the *RIC Actions to be Added List* IE is included, then the E2 node shall validate and add the requested actions to the existing sequence of RIC Actions in order of the *RIC Action Execution Order* IE and reserve the necessary resources for the new actions.

The E2 node shall send the RIC SUBSCRIPTION MODIFICATION RESPONSE message back to the Near-RT RIC when one of the following cases is successfully executed:

- If the *RIC Event Trigger Definition to be Modified* IE is present in the RIC SUBSCRIPTION MODIFICATION REQUEST message and if the requested modification for the event trigger definition was successfully performed by the E2 node, or
- If the *RIC Event Trigger Definition to be Modified* IE is not present (i.e., no modification to the event trigger definition was requested) in the RIC SUBSCRIPTION MODIFICATION REQUEST message and if at least one of the requested actions to be added or modified or removed as requested by the Near-RT RIC in *RIC Actions To Be Added List* IE or *RIC Actions to be Modified List* IE or *RIC Actions to be Removed List* IE, respectively, in the RIC SUBSCRIPTION MODIFICATION REQUEST message, was successfully performed by the E2 node.

The E2 node shall report the result of all the requested modifications to the sequence of actions, if any, back to the Near-RT RIC in the RIC SUBSCRIPTION MODIFICATION RESPONSE message as follows:

- A list of actions requested to be removed, indexed by the *RIC Action ID* IE, which are successfully removed by the E2 node, shall be included in the *RIC Actions Removed List* IE.

- A list of actions requested to be removed, indexed by the *RIC Action ID* IE, which failed to get removed from the sequence by the E2 node, shall be included in the *RIC Actions Failed to be Removed List* IE, with appropriate cause values.
- A list of actions requested to be modified, indexed by the *RIC Action ID* IE, which are successfully modified by the E2 node, shall be included in the *RIC Actions Modified List* IE.
- A list of actions requested to be modified, indexed by the *RIC Action ID* IE, which failed to get modified by the E2 node, shall be included in the *RIC Actions Failed to be Modified List* IE with appropriate cause values.
- A list of actions requested to be added, indexed by the *RIC Action ID* IE, which are successfully added by the E2 node, shall be included in the *RIC Actions Added List* IE.
- A list of actions requested to be added, indexed by the *RIC Action ID* IE, which failed to get added to the sequence by the E2 node, shall be included in the *RIC Actions Failed to be Added List* IE with appropriate cause values.

If, for a given *RIC Action ID* IE in the *RIC Actions to be Modified List* IE, more than one modification to the RIC Service Action is requested in the form of *RIC Action Definition* IE and/or *RIC Action Execution Order* IE and/or *RIC Subsequent Action* IE, then the E2 node shall report that the requested action modification is successfully performed and shall include the action in the *RIC Actions Modified List* IE, if and only if, all the requested modifications to the action are successfully performed by the E2 node.

If one of the requested modifications to the RIC Service Action is not successfully performed by the E2 node, then the E2 node shall include the RIC Service Action in the *RIC Actions Failed to be Modified List* IE, along with an appropriate cause, to indicate failure for the requested modification to the RIC Service Action.

If, for a given *RIC Action ID* IE in the *RIC Actions to be Added* IE, either the action type in the *RIC Action Type* IE or the action definition in the *RIC Action Definition* IE or the action execution order in the *RIC Action Execution Order* IE or the subsequent action, if included, in the *RIC Subsequent Action* IE is not successfully processed by the E2 node, then the E2 node shall include the action in the *RIC Actions Failed to be Added List* IE with an appropriate cause, indicating failure to add the requested action to the existing sequence of actions.

If, after processing the RIC Subscription Modification procedure, more than one RIC Service Action remains accepted by the E2 Node then, at each occurrence of the common Event Trigger, the sequence of RIC Service Actions shall be executed according to the considerations defined in clause 8.2.1.2.

Upon reception of the RIC SUBSCRIPTION MODIFICATION RESPONSE message, the Near-RT RIC shall stop timer $T_{\text{RICEVENTmodify}}$ and terminate the RIC Subscription Modification procedure.

8.2.5.3 Unsuccessful operation

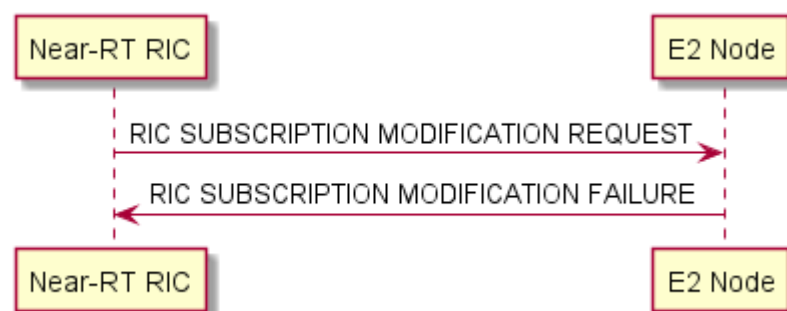


Figure 8.2.5.3-1: RIC Subscription Modification procedure, unsuccessful operation

If a failure occurs during the RIC Subscription Modification procedure, the E2 node shall send a RIC SUBSCRIPTION MODIFICATION FAILURE message to the Near-RT RIC with an appropriate cause value.

If the *RIC Event Trigger Definition to be Modified* IE is present in the RIC SUBSCRIPTION MODIFICATION REQUEST message and if the requested modification for the event trigger definition is not accepted by the E2 node, the E2 node shall send a RIC SUBSCRIPTION MODIFICATION FAILURE message to the Near-RT RIC with an appropriate cause value.

If the *RIC Event Trigger Definition to be Modified* IE is not present (i.e., no modification to the event trigger definition was requested) and if none of the requested modifications to the sequence of actions were successfully performed, that is if all of the following apply, the E2 node shall send a RIC SUBSCRIPTION MODIFICATION FAILURE message to the Near-RT RIC with an appropriate cause value:

- *RIC Actions to be Added List* IE was present in the RIC SUBSCRIPTION MODIFICATION REQUEST message and if none of the requested additions were successfully performed
- *RIC Actions to be Modified List* IE was present in the RIC SUBSCRIPTION MODIFICATION REQUEST message and if none of the requested modifications were successfully performed
- *RIC Actions to be Removed List* IE was present in the RIC SUBSCRIPTION MODIFICATION REQUEST message and if none of the requested removals were successfully performed

If the E2 node detects an issue with the resulting sequence of actions after processing the *RIC Actions to be Added List* IE and/or *RIC Actions to be Modified List* IE and/or *RIC Actions to be Removed List* IE, the E2 node shall send a RIC SUBSCRIPTION MODIFICATION FAILURE message to the Near-RT RIC with an appropriate cause value.

If all RIC Service Actions in the existing RIC Subscription are proposed for removal, the E2 node shall send a RIC SUBSCRIPTION MODIFICATION FAILURE message to the Near-RT RIC with an appropriate cause value.

Upon reception of the RIC SUBSCRIPTION FAILURE message, the Near-RT RIC shall stop the timer $T_{\text{RICEVENTmodify}}$ and terminate the RIC Subscription Modification procedure.

8.2.5.4 Abnormal conditions

If the E2 node receives a RIC SUBSCRIPTION MODIFICATION REQUEST message including an unknown *RAN Function ID* IE, the E2 node shall send the RIC SUBSCRIPTION MODIFICATION FAILURE message to the Near-RT RIC with an appropriate cause value.

If the E2 node receives a RIC SUBSCRIPTION MODIFICATION REQUEST message including an unknown *RIC Request ID* IE, the E2 node shall send the RIC SUBSCRIPTION MODIFICATION FAILURE message to the Near-RT RIC with an appropriate cause value.

If the E2 node receives a RIC SUBSCRIPTION MODIFICATION REQUEST message containing the same *RIC Action ID* IE value across the *RIC Actions to be Added List* IE, *RIC Actions to be Modified List* IE and/or *RIC Actions to be Removed List* IE, then the E2 node shall send the RIC SUBSCRIPTION MODIFICATION FAILURE message to the Near-RT RIC with an appropriate cause value.

8.2.6 RIC Subscription Modification Required procedure

8.2.6.1 General

This procedure is used by the E2 Node to request the Near-RT RIC for modifying an existing RIC Subscription in the E2 Node.

This procedure shall be initiated by the E2 Node.

This procedure uses RIC Service signalling.

8.2.6.2 Successful operation

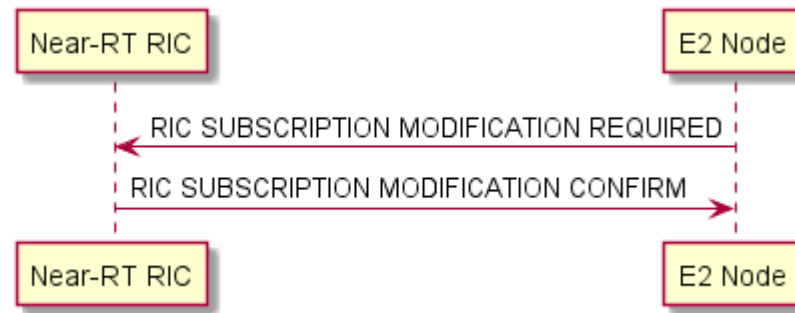


Figure 8.2.6.2-1: RIC Subscription Modification Required procedure, successful operation

The E2 Node initiates the procedure by sending the RIC SUBSCRIPTION MODIFICATION REQUIRED message, containing *RIC Request ID* IE that was previously assigned by the Near-RT RIC during a successful RIC Subscription procedure, to the Near-RT RIC.

Upon reception of the RIC SUBSCRIPTION MODIFICATION REQUIRED message, the Near-RT RIC shall determine the RIC Subscription from the *RIC Request ID* IE and the target RAN Function from the *RAN Function ID* IE.

If at least one of the requested actions in *RIC Actions Required to be Modified List* IE or *RIC Actions Required to be Removed List* IE is successfully confirmed by the Near-RT RIC, then the Near-RT RIC shall perform the required procedures to update the RIC Subscription and shall send the RIC SUBSCRIPTION MODIFICATION CONFIRM message to the E2 node.

The Near-RT RIC shall report the result to the E2 node in the RIC SUBSCRIPTION MODIFICATION CONFIRM as follows:

- A list of actions requested to be modified, indexed by the *RIC Action ID* IE, which are successfully confirmed for modification by the Near-RT RIC, shall be included in the *RIC Actions Confirmed for Modification List* IE.
- A list of actions requested to be modified, indexed by the *RIC Action ID* IE, which are refused to be modified by the Near-RT RIC, shall be included in the *RIC Actions Refused to be Modified List* IE with appropriate cause values.
- A list of actions requested to be removed, indexed by the *RIC Action ID* IE, which are successfully confirmed for removal by the Near-RT RIC, shall be included in the *RIC Actions Confirmed for Removal List* IE.
- A list of actions requested to be removed, indexed by the *RIC Action ID* IE, which are refused to be removed by the Near-RT RIC, shall be included in the *RIC Actions Refused for Removal List* IE with appropriate cause values.

If, after processing the RIC Subscription Modification Required procedure, more than one RIC Service Action remains in effect at the E2 node, then at each occurrence of the common Event Trigger, the sequence of RIC Service Actions shall be executed according to the considerations defined in clause 8.2.1.2.

Upon reception of the RIC SUBSCRIPTION MODIFICATION CONFIRM message, the E2 Node shall release the necessary resources for the actions that are confirmed for removal in the *RIC Actions Confirmed for Removal List* IE, if present, and shall modify the necessary resources for the actions that are confirmed for modification in the *RIC Actions Confirmed for Modification List* IE, if present.

8.2.6.3 Unsuccessful operation

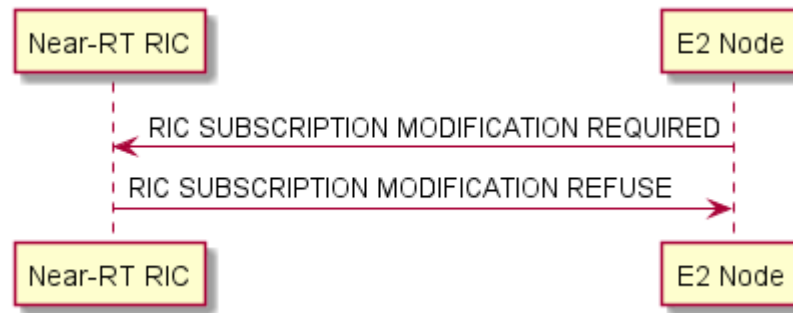


Figure 8.2.6.3-1: RIC Subscription Modification Required procedure, unsuccessful operation

If a failure occurs during the RIC Subscription Modification Required procedure the Near-RT RIC shall send the RIC SUBSCRIPTION MODIFICATION REFUSE message to the E2 node. If none of the requested modifications to the actions in the RIC SUBSCRIPTION MODIFICATION REQUIRED message (i.e., in the *RIC Actions Required to be Modified List* IE and the *RIC Actions Required to be Removed List* IE, if present) is successfully confirmed, the Near-RT RIC shall send the RIC SUBSCRIPTION MODIFICATION REFUSE message to the E2 node with an appropriate cause.

If the Near-RT RIC detects an issue with the requested sequence of actions after processing the requested *RIC Actions Required to be Modified List* IE and/or the *RIC Actions Required to be Removed List* IE respectively, if present, the Near-RT RIC shall send the RIC SUBSCRIPTION MODIFICATION REFUSE message to the E2 node with an appropriate cause.

If all RIC Service Action in the existing RIC Subscription are proposed for removal, the Near-RT RIC shall send a RIC SUBSCRIPTION MODIFICATION REFUSE message to the E2 Node with an appropriate cause value.

8.2.6.4 Abnormal conditions

If the Near-RT RIC receives a RIC SUBSCRIPTION MODIFICATION REQUIRED message which contains an unknown *RAN Function ID* IE, the Near-RT RIC shall send the RIC SUBSCRIPTION MODIFICATION REFUSE message to the E2 Node with an appropriate cause value.

If the Near-RT RIC receives a RIC SUBSCRIPTION MODIFICATION REQUIRED message containing an unknown *RIC Request ID* IE, the Near-RT RIC shall send the RIC SUBSCRIPTION MODIFICATION REFUSE message to the E2 Node with an appropriate cause value.

8.2.7 RIC Query procedure

8.2.7.1 General

This procedure is initiated by Near-RT RIC to request RAN and/or UE related information from E2 Node.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.7.2 Successful operation

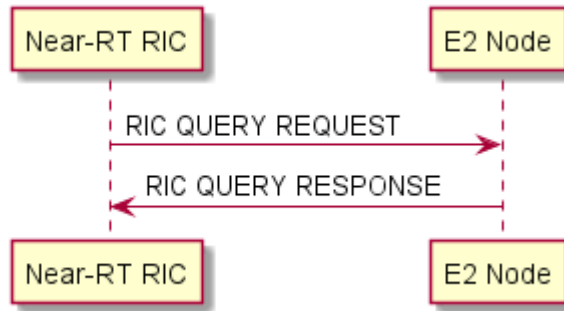


Figure 8.2.7.2-1: RIC Query procedure, successful operation

The Near-RT RIC initiates the procedure by sending the RIC QUERY REQUEST message, which shall contain a unique *RIC Request ID* IE, assigned by the Near-RT RIC, to the E2 Node. When the Near-RT RIC sends the RIC QUERY REQUEST message, it shall start timer $T_{RICQuery}$.

At reception of the RIC QUERY REQUEST message the E2 Node shall:

- Determine the target RAN Function using the information in the *RAN Function ID* IE.
- Validate the *RIC Query Header* IE and *RIC Query Definition* IE and if the requested information is available at E2 Node, then E2 Node shall respond back with RIC QUERY RESPONSE message containing the requested information.

Upon reception of the RIC QUERY RESPONSE message the Near-RT RIC shall stop timer $T_{RICQuery}$ and terminate the RIC Query procedure.

8.2.7.3 Unsuccessful operation

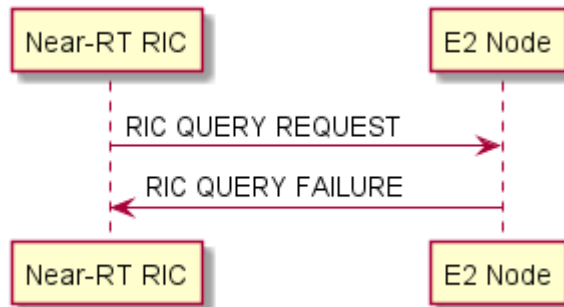


Figure 8.2.7.3-1: RIC Query procedure, unsuccessful operation

If the *RAN Function ID* IE in the RIC QUERY REQUEST message is not supported by E2 Node, then the E2 Node shall respond with the RIC QUERY FAILURE message to Near-RT RIC with an appropriate cause value.

If none of the requested information in the *RIC Query Definition* IE is available at E2 Node, then E2 Node shall respond with the RIC QUERY FAILURE message to Near-RT RIC with an appropriate cause value.

Upon reception of the RIC QUERY FAILURE message the Near-RT RIC shall stop timer $T_{RICQuery}$ and terminate the RIC Query Procedure.

8.2.7.4 Abnormal conditions

Upon reception of the ERROR INDICATION message including the *RIC Request ID* IE corresponding to the previous RIC QUERY REQUEST message, the Near-RT RIC shall stop timer $T_{RICQuery}$, if running, and terminate the RIC Query procedure.

8.2.8 RIC Service Load Status procedure

8.2.8.1 General

This procedure is used to control the reporting of RIC Service Load information for one or more RIC Services.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.8.2 Successful operation

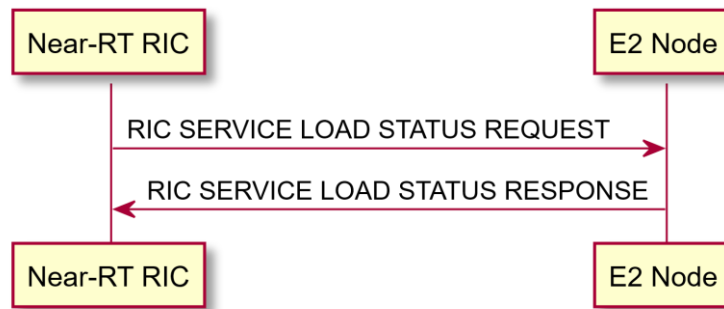


Figure 8.2.8.2-1: RIC Load Status procedure, successful operation

The Near-RT RIC initiates the procedure by sending a RIC LOAD STATUS REQUEST message to the E2 Node.

Upon reception of the RIC SERVICE LOAD STATUS REQUEST message the E2 Node shall process the message according to the following considerations:

The E2 Node:

- Shall initiate the requested load information reporting according to the parameters given in the request in case the *Registration Request* IE set to "start"; or
- Shall stop all measurements and terminate the reporting in case the *Registration Request* IE is set to "stop"; or
- Shall add load measurements indicated in the request to the measurements initiated before for the given measurement IDs, in case the *Registration Request* IE is set to "add". If measurements are already initiated for a requested load measurement, this information shall be ignored.

For each RAN Function in the *RAN Function Load Request List* IE:

- If the *RAN Function Load Request* IE is present, the E2 Node shall collect overall load information for the RAN Function identified by the corresponding *RAN Function ID* IE.
- If the *RIC Service Load Request* IE is present in the RIC SERVICE LOAD STATUS REQUEST message, the E2 Node shall collect load information for requested RIC Services.
- If the *RIC Subscription Load Request List* IE is present and the *RIC Subscription Load Request* IE is also included, the E2 Node shall collect load information for the RIC Subscription identified by the *RIC Request ID* IE. If the *RIC Action Load Request List* IE is also present for a given RIC Subscription, the E2 Node shall collect load information only for each RIC Action identified by the corresponding *RIC Action ID* IE in the list. If the *RIC Action Load Request List* IE is not present for a given RIC Subscription, the E2 Node shall collect load information for all the RIC Actions associated with the RIC Subscription

After processing the RIC SERVICE LOAD STATUS REQUEST message, if the E2 Node has accepted at least one item in the *RAN Function Load Request List* IE and/or *RIC Service Load Request List* IE and/or at least one item in the *RIC Subscription Load Request List* IE, the E2 Node shall send the RIC SERVICE LOAD STATUS RESPONSE message to the Near-RT RIC, and initiate load reporting using the RIC Service Load Update procedure.

8.2.8.3 Unsuccessful operation

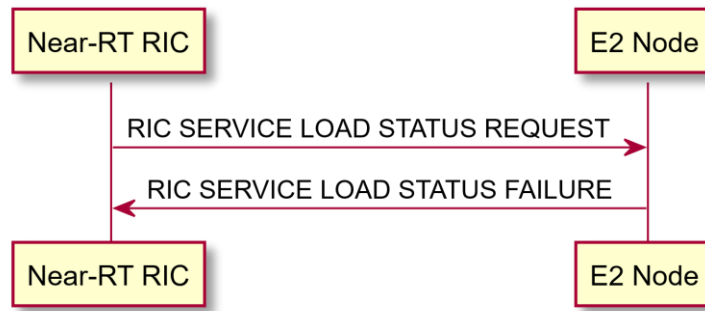


Figure 8.2.8.3-1: RIC Service Load Status procedure, unsuccessful operation

If the E2 Node cannot accept the RIC SERVICE LOAD STATUS REQUEST message, it shall respond with the RIC SERVICE LOAD STATUS FAILURE message with an appropriate cause value.

8.3.8.4 Abnormal conditions

Void.

8.2.9 RIC Service Load Update procedure

8.3.9.1 General

The purpose of the RIC Service Load Update procedure is to inform the Near-RT RIC of the load status in the E2 Node for one or more RIC Services.

This procedure shall be initiated by the E2 Node.

This procedure uses RIC Service signalling.

8.3.9.2 Successful operation

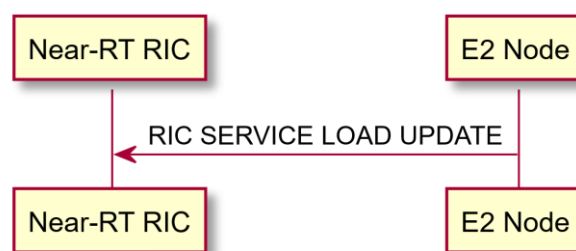


Figure 8.2.9.2-1: RIC Service Load Update procedure, successful operation

The E2 Node initiates the procedure by sending a RIC LOAD UPDATE message.

- If the *RAN Function Load Information* IE is included in the *RAN Function Load List* IE, the Near-RT RIC shall consider that the load information in the *RAN Function Load Information* IE is for the overall RAN Function load information associated with the corresponding *RAN Function ID* IE.
- If the *RIC Service Load Information* IE is included in the *RAN Function Load List* IE, the Near-RT RIC shall consider that the load information in the *RIC Service Load Information* IE is for the overall RIC Service load information associated with the corresponding *RIC Service type*.
- If the *RIC Subscription Load List* IE is included, and the *RIC Subscription Load Information* IE is also included in the *RIC Subscription Load List* IE, the Near-RT RIC shall consider that the load information in the *RIC Subscription*

Load Information IE is for the overall RIC Subscription load information associated with the corresponding *RIC Request ID* IE that was previously assigned by the Near-RT RIC during a successful RIC Subscription procedure. If the *RIC Action Load List* IE is also included in the *RIC Subscription Load List* IE, the Near-RT RIC shall consider that the load information in the *RIC Action Load Information* IE is for the overall RIC Action load information associated with the corresponding *RIC Action ID* IE.

8.3.9.3 Unsuccessful operation

Void.

8.3.9.4 Abnormal conditions

Void.

8.2.10 RIC Subscription State Control procedure

8.2.10.1 General

This procedure is used to suspend and/or resume existing RIC Subscriptions.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.10.2 Successful operation

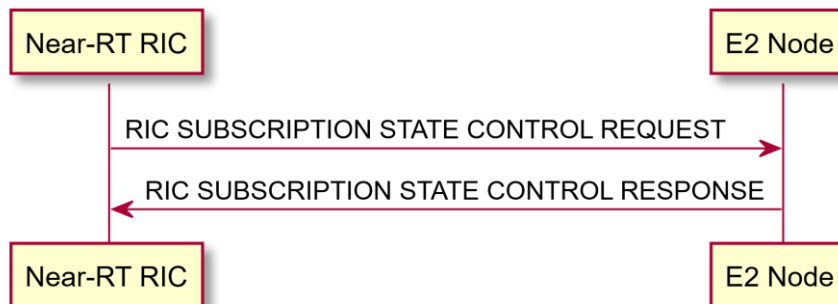


Figure 8.2.10.2-1: RIC Subscription State Control procedure, successful operation

The Near-RT RIC initiates the procedure by sending a *RIC SUBSCRIPTION STATE CONTROL REQUEST* message to the E2 Node.

Upon reception of the *RIC SUBSCRIPTION STATE CONTROL REQUEST* message the E2 Node shall process the message according to the following considerations for each *RAN Function State Control List* IE:

- If the *RIC Subscription To Be Suspended List* IE is present, the E2 Node shall suspend either the list of RIC Actions, if the corresponding *RIC Action List* IE is present, or the entire RIC Subscription.
- If the *RIC Subscription To Be Resumed List* IE is present, the E2 Node shall resume either the list of RIC Actions, if the corresponding *RIC Action List* IE is present, or the entire RIC Subscription.

After processing the *RIC SUBSCRIPTION STATE CONTROL REQUEST* message, if the E2 Node has at least one item to be included in the *RIC Subscription Suspended List* IE and/or the *RIC Subscription Resumed List* IE for any *RAN Function* requested in the *RAN Function State Control List* IE, the E2 Node shall send the *RIC SUBSCRIPTION STATE CONTROL RESPONSE* message to the Near-RT RIC.

If, after processing the *RIC SUBSCRIPTION STATE CONTROL REQUEST* message, one or more RIC Subscriptions have been accepted to be suspended by the E2 Node without the corresponding *RIC Action List* IE present then, the corresponding Event Trigger, defined in clause 8.2.1.2, shall be suspended.

If, after processing the RIC SUBSCRIPTION STATE CONTROL REQUEST message, one or more RIC Subscriptions have been accepted to be resumed by the E2 Node then, the corresponding Event Trigger, defined in clause 8.2.1.2, shall be resumed and at each occurrence of the corresponding Event Trigger, the sequence of RIC Service Actions shall be executed according to the considerations defined in clause 8.2.1.2.

8.2.10.3 Unsuccessful operation

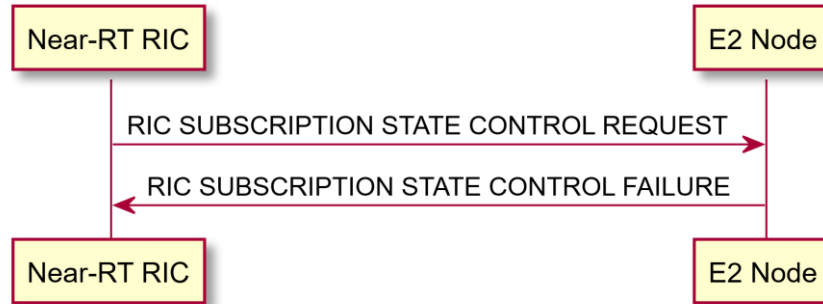


Figure 8.2.10.3-1: RIC Subscription State Control procedure, unsuccessful operation

If the E2 Node cannot accept the RIC SUBSCRIPTION STATE CONTROL REQUEST message it shall respond with the RIC SUBSCRIPTION STATE CONTROL FAILURE message with an appropriate cause value.

If the RIC Subscription State Control procedure failure occurs, the Near-RT RIC and E2 Node shall continue to operate with their existing configuration data.

8.2.10.4 Abnormal conditions

Void.

8.2.11 RIC Assistance procedure

8.2.11.1 General

This procedure is used to utilize an assistance service offered by the Near-RT RIC.

This procedure shall be initiated by the E2 Node.

This procedure uses RIC Service signalling.

8.2.11.2 Successful operation

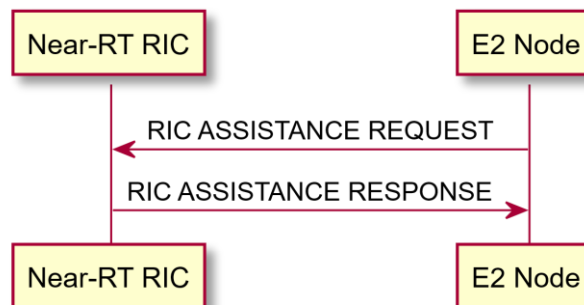


Figure 8.2.11.2-1: RIC Assistance procedure, successful operation

The E2 Node initiates the procedure by sending a RIC ASSISTANCE REQUEST message which shall contain a unique *RIC Request ID* IE, assigned by the E2 Node, to the Near-RT RIC. When the E2 Node sends the RIC ASSISTANCE REQUEST message, it shall start timer $T_{RICassist}$.

At reception of the RIC ASSISTANCE REQUEST message the Near-RT RIC shall:

- Consider the *RIC Assistance Header* IE and *RIC Assistance Message* IE to determine the requested service and if available at Near-RT RIC, then Near-RT RIC shall respond back with RIC ASSISTANCE RESPONSE message for the requested service with the result contained in the *RIC Assistance Header* IE and *RIC Assistance Outcome* IE.

Upon reception of the RIC ASSISTANCE RESPONSE message the E2 Node shall stop timer $T_{RICassist}$ and terminate the RIC Assistance procedure.

Interactions with RIC Assistance Indication procedure:

If the optional *RIC Assistance Update* IE is present, the Near-RT RIC shall use the *RIC Assistance Update Number* IE, if present, to determine the maximum number of RIC ASSISTANCE INDICATION messages to the E2 Node to provide updates for the requested assistance service offered by the Near-RT RIC.

If the *RIC Assistance Update Number* IE is not present, then the Near-RT RIC shall continue to send RIC ASSISTANCE INDICATION messages until reception of the RIC ASSISTANCE HALT message.

8.2.11.3 Unsuccessful operation

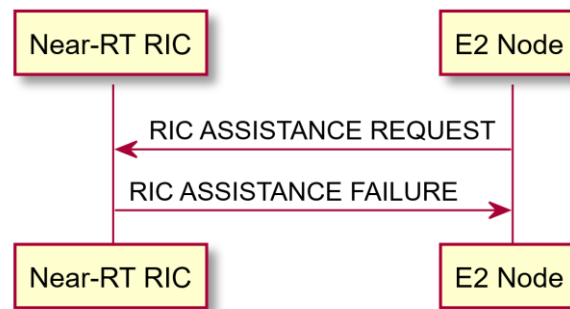


Figure 8.2.11.3-1: RIC Assistance procedure, unsuccessful operation

If the Near-RT RIC cannot accept the RIC ASSISTANCE REQUEST message it shall respond with the RIC ASSISTANCE FAILURE message with an appropriate cause value.

Upon reception of the RIC ASSISTANCE FAILURE message the E2 Node shall stop timer $T_{RICassist}$ and terminate the RIC Assistance Procedure.

8.2.11.4 Abnormal conditions

Void.

8.2.12 RIC Assistance Indication procedure

8.2.12.1 General

This procedure is used to provide an update for an assistance service offered by the Near-RT RIC.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses RIC Service signalling.

8.2.12.2 Successful operation

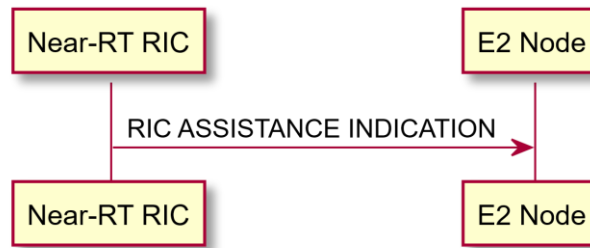


Figure 8.2.12.2-1: RIC Assistance Indication procedure, successful operation

The Near-RT RIC initiates the procedure by sending a RIC ASSISTANCE INDICATION message which shall contain a unique *RIC Request ID* IE, assigned by the E2 Node during a previous RIC Assistance procedure, to the E2 Node, and the updated assistance service result in *RIC Assistance Header* IE and *RIC Assistance Outcome* IE. Each update of the requested assistance service shall contain a unique *RIC Assistance SN* IE.

Interactions with RIC Assistance Halt procedure:

If the E2 Node sends a RIC ASSISTANCE HALT message, the Near-RT RIC shall halt sending RIC ASSISTANCE INDICATION messages corresponding to the *RIC Request ID* IE contained in the message.

8.2.12.3 Unsuccessful operation

Not applicable.

8.2.12.4 Abnormal conditions

Void.

8.2.13 RIC Assistance Halt procedure

8.2.13.1 General

This procedure is used to halt updates for an assistance service offered by the Near-RT RIC.

This procedure shall be initiated by the E2 Node.

This procedure uses RIC Service signalling.

8.2.13.2 Successful operation

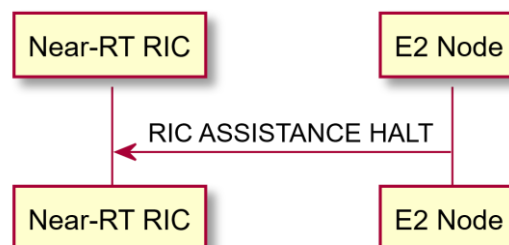


Figure 8.2.13.2-1: RIC Assistance Halt procedure, successful operation

The E2 Node initiates the procedure by sending a RIC ASSISTANCE HALT message which shall contain a unique *RIC Request ID* IE, assigned by the E2 Node during a previous RIC Assistance procedure, to the Near-RT RIC.

Upon reception the Near-RT RIC shall halt the requested updates to the RIC Assistance service.

8.2.13.3 Unsuccessful operation

Not applicable.

8.3.13.4 Abnormal conditions

If the Near-RT RIC receives a RIC Assistance Halt request from the E2 Node that does not correspond to an ongoing RIC Assistance service, then the Near-RT RIC shall ignore the message.

8.3 Global procedures

8.3.1 E2 Setup procedure

8.3.1.1 General

The purpose of the E2 Setup procedure is to exchange application level data needed for the E2 Node and Near-RT RIC to correctly interoperate on the E2 interface. This procedure shall be the first E2AP procedure triggered after the TNL association has become operational.

This procedure erases any existing application level configuration data in the two nodes and replace it by the one received.

This procedure shall be initiated by the E2 Node.

This procedure uses E2 Support Function signalling.

8.3.1.2 Successful operation

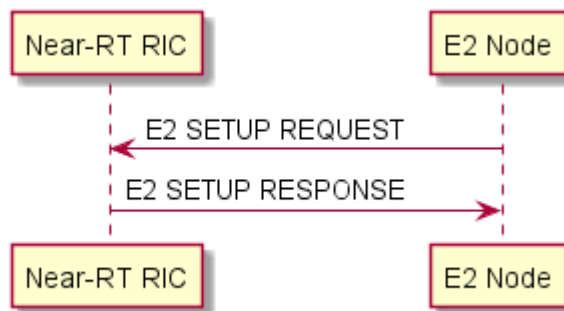


Figure 8.3.1.2-1: E2 Setup procedure, successful operation

The E2 Node initiates the procedure by sending the E2 SETUP REQUEST message including the appropriate data to a Near-RT RIC.

If the Near-RT RIC has successfully processed the *RAN Functions Added List* IE then Near-RT RIC shall contain, in the E2 SETUP RESPONSE message, the *RAN Functions Accepted List* IE and/or *RAN Functions Accepted with Modification List* IE and/or the *RAN Functions Rejected List* IE. If a RAN Function whose *RAN Function Definition* IE received in the *RAN Functions Added List* IE contains one or more RIC assistance services that the Near-RT RIC cannot support, but the RAN Function itself remains acceptable, the Near-RT RIC may, instead of rejecting the entire RAN Function, include it in the *RAN Functions Accepted with Modification List* IE of the E2 SETUP RESPONSE message, where the revised definition by the *Revised RAN Function Definition* IE excludes the unsupported RIC assistance services while retaining the rest of the services. The other RAN Functions in the *RAN Functions Added List* IE of the E2 SETUP REQUEST message that are accepted as they are from the Near-RT RIC are included in the *RAN Function Accepted List* IE of the E2 SETUP RESPONSE message.

If the Near-RT RIC has successfully processed the *E2 Node Component Configuration Addition List* IE then Near-RT RIC shall contain, in the E2 SETUP RESPONSE message, the *E2 Node Component Configuration Addition Acknowledge List* IE.

8.3.1.3 Unsuccessful operation

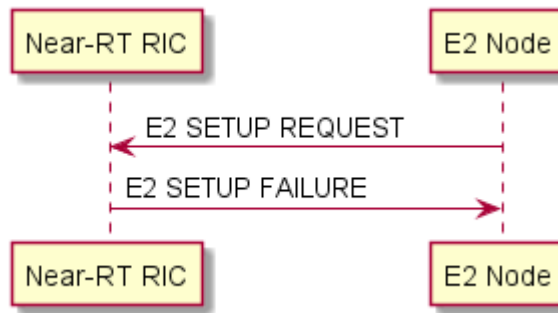


Figure 8.3.1.3-1: E2 Setup procedure, unsuccessful operation

If the Near-RT RIC cannot accept the setup it shall respond with an E2 SETUP FAILURE message with an appropriate cause value.

The Near-RT RIC may provide an alternative *Transport Layer Information* IE in the E2 SETUP FAILURE message for the E2 Node to use when reinitiating the E2 Setup procedure towards the Near-RT RIC.

If the E2 SETUP FAILURE message includes the *Time To Wait* IE, the E2 node shall wait at least for the indicated time before reinitiating the E2 Setup procedure towards the Near-RT RIC.

8.3.1.4 Abnormal conditions

If the first message received for a specific TNL association is not an E2 SETUP REQUEST, E2 SETUP RESPONSE, E2 SETUP FAILURE or E2 NODE CONFIGURATION UPDATE message then this shall be treated as a logical error.

8.3.2 Reset procedure

8.3.2.1 General

The purpose of the Reset procedure is to initialize or re-initialize the E2 Node in the event of Near-RT RIC failure or vice-versa.

This procedure does not affect the application level data exchanged during the E2 Setup procedure, E2 Node Configuration Update procedure and RIC Service Update procedure.

This procedure shall be initiated by the E2 Node or the Near-RT RIC.

This procedure uses E2 Support Function signalling.

8.3.2.2 Successful operation

This procedure may be initiated by either Near-RT RIC or E2 Node.

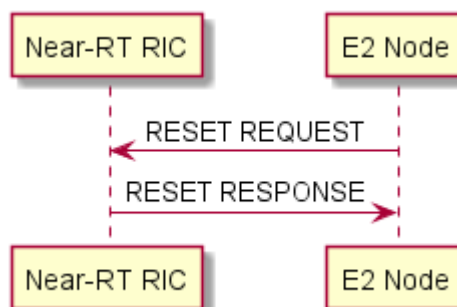


Figure 8.3.2.2-1: Reset, successful operation (E2 Node Initiated)

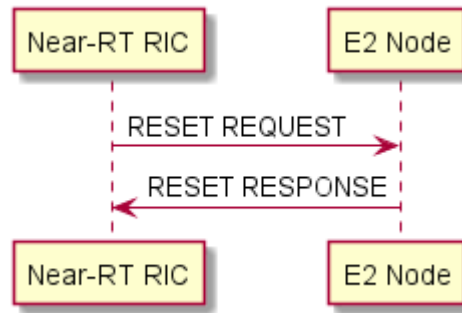


Figure 8.3.2.2-2: Reset, successful operation (Near-RT RIC Initiated)

When the Reset procedure is initiated, the Near-RT RIC and E2 Node shall:

- Delete any pre-established RIC Subscriptions.
- Gracefully terminate any ongoing Near-RT RIC call processes using Insert, Control or Policy RIC Service Actions while ensuring that impact to ongoing calls for connected UE is minimized.

After the Reset has been completed, the Near-RT RIC may re-issue any required RIC Subscriptions.

Interactions with other procedures:

If the RESET REQUEST message is received, any other ongoing procedure (except for another Reset procedure) on the same E2 interface related to ongoing RIC Services shall be aborted.

8.3.2.3 Unsuccessful operation

Void.

8.3.2.4 Abnormal conditions

Void.

8.3.3 Error Indication

8.3.3.1 General

The Error Indication procedure is initiated by either the E2 Node or the Near-RT RIC to report detected errors in one incoming message, provided they cannot be reported by an appropriate failure message.

This procedure shall be initiated by the E2 Node or the Near-RT RIC.

If the error situation arises due to reception of a message utilizing RIC Service signalling, then the Error Indication procedure uses RIC Service signalling. Otherwise, the procedure uses E2 Support Function signalling.

8.3.3.2 Successful operation

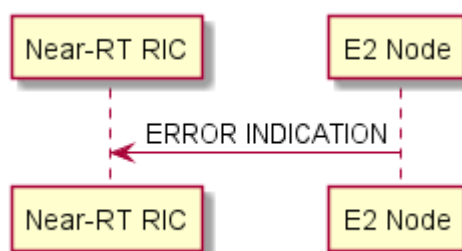


Figure 8.3.3.2-1: Error Indication, (E2 Node initiated) successful operation

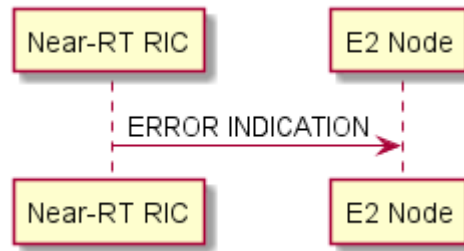


Figure 8.3.3.2-2: Error Indication, (Near-RT RIC Initiated) successful operation

When the conditions defined in clause 10 are fulfilled, the Error Indication procedure shall be initiated by an ERROR INDICATION message sent from the node detecting the error situation.

The ERROR INDICATION message shall contain at least either the *Cause IE* or the *Criticality Diagnostics IE* and may include *RAN Function ID IE* and *RIC Request ID IE*.

8.3.3.3 Unsuccessful operation

Not applicable.

8.3.3.4 Abnormal conditions

Not applicable.

8.3.4 RIC Service Update procedure

8.3.4.1 General

The purpose of the RIC Service Update procedure is to update application level RIC Service related data needed for E2 Node and Near-RT RIC to interoperate correctly over the E2 interface.

This procedure shall be initiated by the E2 Node.

This procedure uses E2 Support Function signalling.

8.3.4.2 Successful operation

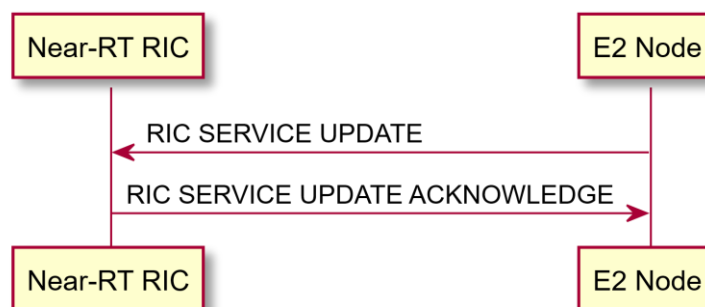


Figure 8.3.4.2-1: RIC Service Update procedure, successful operation

An E2 Node initiates the procedure by sending a RIC SERVICE UPDATE message to the Near-RT RIC.

If the E2 Node has taken into operational use one or more RAN Functions supporting RIC Services, the RIC SERVICE UPDATE message shall include the *RAN Functions Added List IE*.

If the E2 Node has modified one or more RAN Functions supporting RIC Services, the RIC SERVICE UPDATE message shall include the *RAN Functions Modified List IE*.

If the E2 Node has removed from operational use one or more RAN Functions supporting RIC Services, the RIC SERVICE UPDATE message shall include the *RAN Functions Deleted List IE*.

Upon reception of a RIC SERVICE UPDATE message, Near-RT RIC shall update the application level data for E2 Node as follows:

- If the *RAN Function Added List IE* is contained in the RIC SERVICE UPDATE message, Near-RT RIC shall add each listed accepted RAN Function according to the information in the *RAN Function ID IE* and *RAN Function Definition IE* and store the corresponding *RAN Function Revision IE*.
- If the *RAN Function Modified List IE* is contained in the RIC SERVICE UPDATE message, Near-RT RIC shall modify accepted information of supported RAN Functions according to the information in the *RAN Function Definition IE* and update the corresponding *RAN Function Revision IE*.
- If the *RAN Function Deleted List IE* is contained in the RIC SERVICE UPDATE message, Near-RT RIC shall delete information of RAN Function indicated by the *RAN Function ID IE* along with the corresponding *RAN Function Revision IE*.

These changes may be processed in the Near-RT-RIC and may be used when issuing RIC SUBSCRIPTION REQUEST and RIC CONTROL to provide valid *RAN Function ID IE*.

If at least one RAN Function update request present in the RIC SERVICE UPDATE message is successful, then the Near-RT RIC shall send the RIC SERVICE UPDATE ACKNOWLEDGE message to the initiating E2 Node with:

- *RAN Functions Accepted List IE* indicating accepted requests to add, modify, and/or delete the corresponding RAN Function information.
- If required, the *RAN Functions Rejected List IE* indicating rejected requests to add, modify, and/or delete the corresponding RAN Function information.
- If a RAN Function whose *RAN Function Definition IE* received in the *RAN Functions Added List IE* contains one or more RIC assistance services that the Near-RT RIC cannot support, but the RAN Function itself remains acceptable, the Near-RT RIC may, instead of rejecting the entire RAN Function, include it in the *RAN Functions Accepted with Modification List IE* of the RIC SERVICE UPDATE ACKNOWLEDGE message, where the revised definition by the *Revised RAN Function Definition IE* excludes the unsupported RIC assistance services while retaining the rest of the services.

If the Near-RT RIC receives a RIC SERVICE UPDATE message without any IE except for *Message Type IE*, then the Near-RT RIC shall reply with RIC SERVICE UPDATE ACKNOWLEDGE message without any IE except for *Message Type IE*, and shall not perform any updates to the existing application level data.

8.3.4.3 Unsuccessful operation

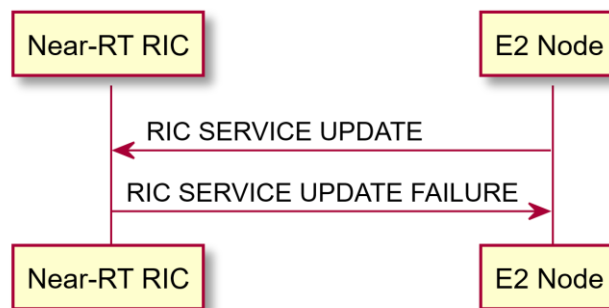


Figure 8.3.4.3-1: RIC Service Update procedure, unsuccessful operation

If the Near-RT RIC cannot accept the update it shall respond with a RIC SERVICE UPDATE FAILURE message with an appropriate cause value.

If the RIC SERVICE UPDATE FAILURE message includes the *Time To Wait IE*, the E2 Node shall wait at least for the indicated time before reinitiating the RIC Service Update procedure towards the same Near-RT RIC. Both nodes shall continue to operate the E2 with their existing RIC Service data.

8.3.4.4 Abnormal conditions

Void.

8.3.4A RIC Service Query procedure

8.3.4A.1 General

The purpose of the RIC Service Query procedure is to ensure alignment between Near-RT RIC and E2 Node concerning application level RIC Service related data needed for E2 Node and Near-RT RIC to interoperate correctly over the E2 interface.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses E2 Support Function signalling.

8.3.4A.2 Successful operation

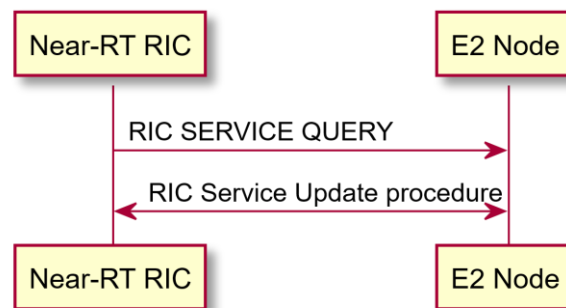


Figure 8.3.4A.2-1: RIC Service Query procedure, successful operation

The Near-RT RIC initiates the procedure by sending a RIC SERVICE QUERY message to the E2 Node.

Upon reception of the RIC SERVICE QUERY message the E2 Node shall initiate the RIC Service Update procedure according to the following considerations:

- If the *RAN Function Accepted List IE* is not present in the RIC SERVICE QUERY message, the E2 Node shall send the RIC SERVICE UPDATE message with the complete list of supported RAN Functions in the *RAN Function Added List IE*.
- If the *RAN Function Accepted List IE* is present in the RIC SERVICE QUERY message and aligns with the list of supported RAN Functions at the E2 Node, the E2 Node shall send the RIC SERVICE UPDATE message without the *RAN Function Added List IE*, *RAN Function Modified List IE* and *RAN Function Deleted List IE*.
- If the *RAN Function Accepted List IE* is present in the RIC SERVICE QUERY message and the list of RAN Functions in the *RAN Function Accepted List IE* does not align with the list of supported RAN Functions at the E2 node, the E2 Node shall send the RIC SERVICE UPDATE message with the *RAN Function Added List IE*, *RAN Function Modified List IE* and/or *RAN Function Deleted List IE* to ensure realignment of RAN Functions between the E2 Node and the Near-RT RIC.

The Near-RT RIC completes the RIC Service Update procedure as described in clause 8.3.4.

8.3.4A.3 Unsuccessful operation

Void.

8.3.4A.4 Abnormal conditions

Void.

8.3.5 E2 Node Configuration Update procedure

8.3.5.1 General

The purpose of the E2 Node Configuration Update procedure is to update application level E2 Node configuration data needed for E2 Node and Near-RT RIC to interoperate correctly over the E2 interface and to support E2 Node initiated TNL association removal.

This procedure shall be initiated by the E2 Node.

This procedure uses E2 Support Function signalling.

8.3.5.2 Successful operation

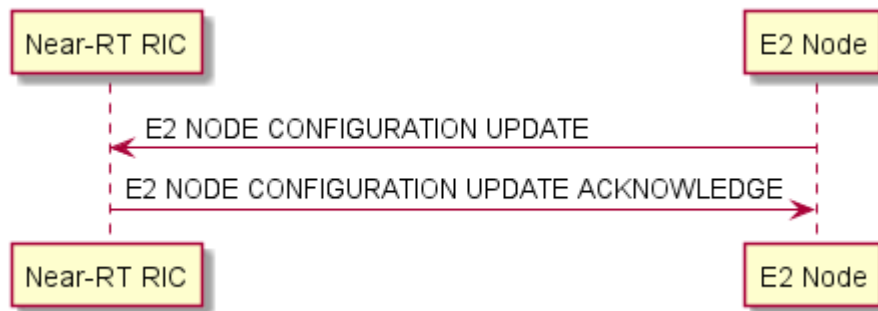


Figure 8.3.5.2-1: E2 Node Configuration Update procedure, successful operation

An E2 Node initiates the procedure by sending a E2 NODE CONFIGURATION UPDATE message to the Near-RT RIC. The message shall include an appropriate set of up-to-date E2 Node-related configuration data that the E2 Node has just taken into operational use.

Upon reception of the E2 NODE CONFIGURATION UPDATE message, Near-RT RIC shall update the application level data for the E2 Node as follows:

Update of E2 Node configuration information in Near-RT RIC:

- If *E2 Node Component Configuration Addition List* IE is contained in the E2 NODE CONFIGURATION UPDATE message, Near-RT RIC shall add the E2 Node Component Configuration information accordingly.
- If *E2 Node Component Configuration Update List* IE is contained in the E2 NODE CONFIGURATION UPDATE message, Near-RT RIC shall modify the E2 Node Component Configuration information accordingly.
- If *E2 Node Component Configuration Removal List* IE is contained in the E2 NODE CONFIGURATION UPDATE message, Near-RT RIC shall remove the E2 Node Component Configuration information accordingly.

If *Global E2 Node ID* IE is contained in the E2 NODE CONFIGURATION UPDATE message for a newly established SCTP association, the Near-RT RIC shall associate the TNL association with the related E2 Node.

If the E2 NODE CONFIGURATION UPDATE message includes *E2 Node TNL Association To Remove List* IE, and the *Endpoint IP address* IE and the *Port Number* IE for both TNL endpoints of the TNL association(s) are included in the *E2 Node TNL Association To Remove List* IE, the Near-RT RIC shall, if supported, consider that the TNL association(s) indicated by both received TNL endpoints will be removed by the E2 Node.

If the E2 NODE CONFIGURATION UPDATE message includes *E2 Node TNL Association To Remove List* IE, and the *Endpoint IP address* IE, or the *Endpoint IP address* IE and the *Port Number* IE for one or both of the TNL endpoints is included in the *E2 Node TNL Association To Remove List* IE in E2 NODE CONFIGURATION UPDATE message, the Near-RT RIC shall, if supported, consider that the TNL association(s) indicated by the received endpoint IP address(es) will be removed by the E2 Node.

After successful update of requested information, Near-RT RIC shall reply with the E2 NODE CONFIGURATION UPDATE ACKNOWLEDGE message to inform the initiating E2 Node that the requested update of application level data was performed successfully.

If the Near-RT RIC receives an E2 NODE CONFIGURATION UPDATE message without any IE except for *Message Type* IE and *Transaction ID* IE, the Near-RT RIC shall reply with the E2 NODE CONFIGURATION UPDATE ACKNOWLEDGE message without performing any updates to the existing configuration.

8.3.5.3 Unsuccessful operation

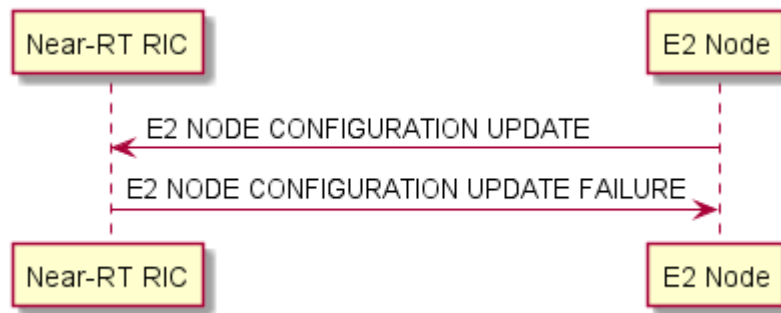


Figure 8.3.5.3-1: E2 Node Configuration Update procedure, unsuccessful operation

If Near-RT RIC cannot accept the E2 NODE CONFIGURATION UPDATE message it shall respond with the E2 NODE CONFIGURATION UPDATE FAILURE message with an appropriate cause value.

If the E2 NODE CONFIGURATION UPDATE FAILURE message includes the *Time To Wait* IE the E2 Node shall wait at least for the indicated time before reinitiating the E2 Node Configuration Update procedure towards the same Near-RT RIC.

If the Near-RT RIC receives an E2 NODE CONFIGURATION UPDATE message containing an *E2 Node Component Configuration Update Item* IE for an E2 Node component that was not previously declared by an *E2 Node Component Configuration Addition Item* IE then the Near-RT RIC shall indicate to the E2 Node that the update failed with appropriate cause value.

If the E2 Node Configuration Update procedure failure occurs, the Near-RT RIC and E2 Node shall continue to operate with their existing configuration data.

8.3.5.4 Abnormal conditions

Void.

8.3.6 E2 Connection Update procedure

8.3.6.1 General

The purpose of the E2 Connection Update procedure is to allow the Near-RT RIC to update the TNL information associated with the E2 interface connection between the E2 Node and Near-RT RIC.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses E2 Support Function signalling.

8.3.6.2 Successful operation

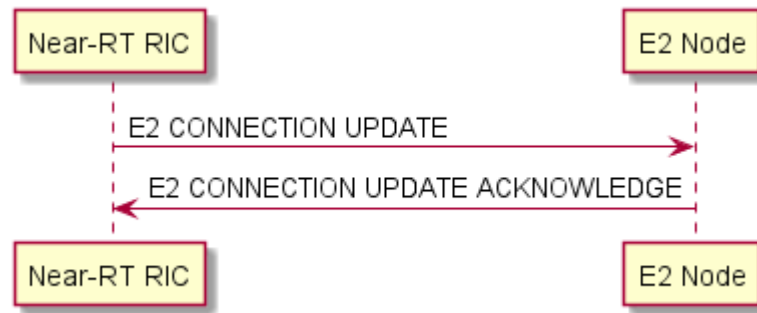


Figure 8.3.6.2-1: E2 Connection Update procedure, successful operation

The Near-RT RIC initiates the procedure by sending a E2 CONNECTION UPDATE message to the E2 Node. The message shall include an appropriate set of up-to-date E2 interface connection data that the E2 Node shall take into account when modifying the E2 interface connection.

Upon reception of a E2 CONNECTION UPDATE message, the E2 Node shall update as follows:

- If *E2 Connection To Add List* IE is contained in the E2 CONNECTION UPDATE message, then the E2 Node shall, if supported, use the information to establish additional TNL Association(s) and configure for use for RIC services and/or E2 support functions according to the *TNL Association Usage* IE in the message.
- If *E2 Connection To Modify List* IE is contained in the E2 CONNECTION UPDATE message, then the E2 Node shall, if supported, use the information to modify the existing usage for RIC services and/or E2 support functions, according to the *TNL Association Usage* IE in the message.
- If *E2 Connection To Remove List* IE is contained in the E2 CONNECTION UPDATE message, then the E2 Node shall, if supported, use the information to remove the existing connection(s). If only one connection remains after successful removal of other connections, the E2 Node shall use this remaining connection for all the RIC services and E2 support functions.

After successful update of E2 interface connection(s), the E2 Node shall reply with the E2 CONNECTION UPDATE ACKNOWLEDGE message to inform the initiating Near-RT RIC that the requested E2 connection update was performed successfully.

If the E2 Node receives a E2 CONNECTION UPDATE message without any IE except for *Message Type* IE and *Transaction ID* IE, the E2 Node shall reply with the E2 CONNECTION ACKNOWLEDGE message without performing any updates to the existing connections.

E2 NODE CONFIGURATION UPDATE procedure shall be the first E2AP procedure triggered on an additional TNLA of an already setup E2 interface instance after the TNL association has become operational, and the Near-RT RIC shall associate the TNLA to the E2 interface instance using the included *Global E2 Node ID*.

An empty E2 NODE CONFIGURATION UPDATE message (i.e. without any IE expect for *Message Type* IE and *Transaction ID* IE) shall be sent by the Near-RT RIC as the first E2AP procedure on the new TNLA, if the E2 Node does not have any Configuration to be updated to Near-RT RIC.

8.3.6.3 Unsuccessful operation

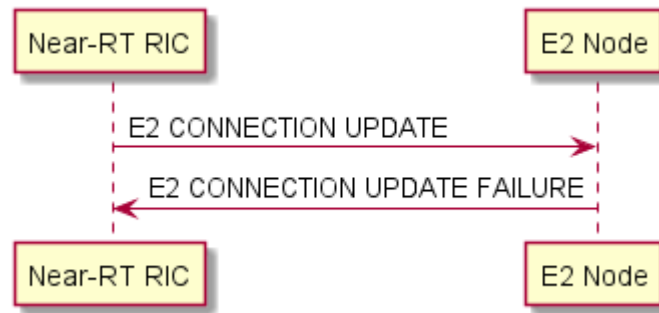


Figure 8.3.6.3-1: E2 Connection Update procedure, unsuccessful operation

If the E2 Node cannot accept the update, it shall respond with a E2 CONNECTION UPDATE FAILURE message with an appropriate cause value.

If the E2 CONNECTION UPDATE FAILURE message includes the *Time To Wait* IE, the Near-RT RIC shall wait at least for the indicated time before reinitiating the E2 Connection Update procedure towards the same E2 Node. Both nodes shall continue to operate with their existing connection(s).

8.3.6.4 Abnormal conditions

Void.

8.3.7 E2 Removal procedure

8.3.7.1 General

The purpose of the E2 removal procedure is to remove the E2 signalling connection between the Near-RT RIC and the E2 node in a controlled manner.

This procedure shall be initiated by the E2 Node or the Near-RT RIC.

This procedure uses E2 Support Function signalling.

8.3.7.2 Successful operation

This procedure may be initiated by either Near-RT RIC or E2 Node.

Successful E2 Removal, E2 Node initiated

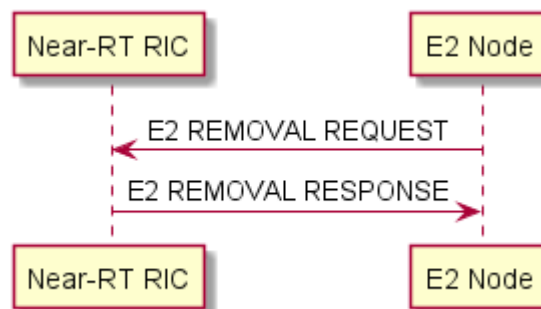


Figure 8.3.7.2-1: E2 Removal, successful operation (E2 Node Initiated)

The E2 Node shall initiate the procedure by sending the E2 REMOVAL REQUEST message to the Near-RT RIC.

Upon reception of the E2 REMOVAL REQUEST message, the Near-RT RIC shall reply with the E2 REMOVAL RESPONSE message.

After receiving the E2 REMOVAL RESPONSE message, the E2 Node shall initiate removal of the TNL association towards the Near-RT RIC, and shall release all resources associated with that E2 signalling connection.

The Near-RT RIC shall then release all resources associated with that E2 signalling connection and erase all application level data.

Successful E2 Removal, Near-RT RIC initiated

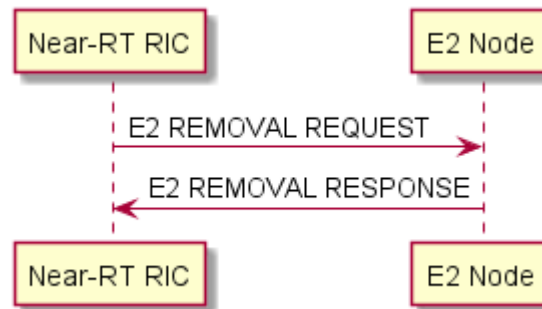


Figure 8.3.7.2-2: E2 Removal, successful operation (Near-RT RIC Initiated)

The Near-RT RIC shall initiate the procedure by sending the E2 REMOVAL REQUEST message to the E2 node.

Upon reception of the E2 REMOVAL REQUEST message, the E2 node shall reply with the E2 REMOVAL RESPONSE message.

After receiving the E2 REMOVAL RESPONSE message, the Near-RT RIC may initiate removal of the TNL association towards the E2 node, and shall release all resources associated with that E2 signalling connection and erase all application level data.

The E2 node shall then release all resources associated with that E2 signalling connection.

Interactions with other procedures:

If the E2 REMOVAL REQUEST message is received, any other ongoing procedure on the same E2 interface related to ongoing RIC Services shall be aborted.

8.3.7.3 Unsuccessful operation

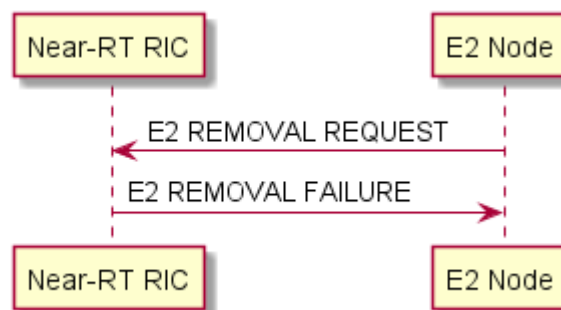


Figure 8.3.7.3-1: E2 Removal procedure (E2 Node Initiated), unsuccessful operation

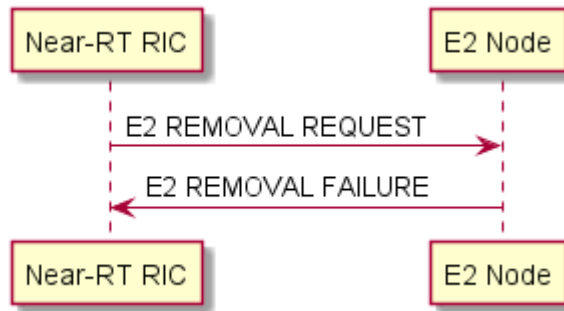


Figure 8.3.7.3-2: E2 Removal procedure (Near-RT RIC Initiated), unsuccessful operation

If the E2 Node cannot accept the E2 REMOVAL REQUEST it shall respond with E2 REMOVAL FAILURE message with an appropriate cause value.

If the Near-RT RIC cannot accept the E2 REMOVAL REQUEST it shall respond with E2 REMOVAL FAILURE message with an appropriate cause value.

8.3.7.4 Abnormal conditions

Void.

8.3.8 E2 Resource Status Procedure

8.3.8.1 General

This procedure is used to report the E2 Node Resource information.

This procedure shall be initiated by the Near-RT RIC.

This procedure uses E2 Support Function signalling.

8.3.8.2 Successful operation

```

@startuml
skin rose
skinparam ParticipantPadding 5
skinparam BoxPadding 10
skinparam lifelineStrategy solid

participant "Near-RT RIC" as near
participant "E2 Node" as ran
near->>ran: E2 RESOURCE STATUS REQUEST
ran-->>near: E2 RESOURCE STATUS RESPONSE
@enduml

```

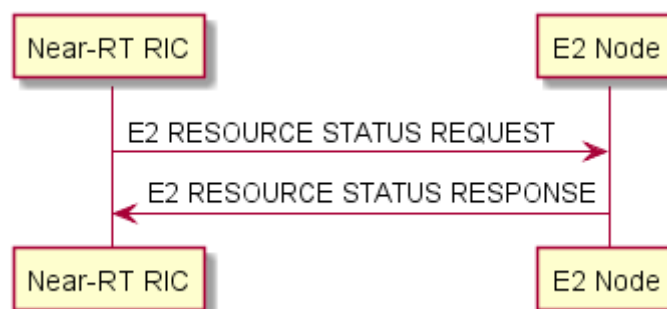


Figure 8.3.8.2-1: E2 Resource Status procedure, successful operation

The Near-RT RIC initiates the procedure by sending an E2 RESOURCE STATUS REQUEST message to the E2 Node.

Upon reception of the E2 RESOURCE STATUS REQUEST message the E2 Node shall process the message according to the following considerations:

The E2 Node:

- Shall initiate the requested load measurement according to the parameters given in the request in case the *Registration Request* IE set to "start"; or
- Shall stop all load measurements and terminate the reporting in case the *Registration Request* IE is set to "stop"; or
- Shall add the load measurement indicated in the request to the load measurements initiated before for the given measurement IDs, in case the *Registration Request* IE is set to "add". If load measurements are already initiated for the requested load measurement, this information shall be ignored.

If the E2 Node is capable to provide all requested resource status information, it shall initiate the measurement as requested by the Near-RT RIC, and respond with the E2 RESOURCE STATUS RESPONSE message

Interaction with other procedures

When starting a measurement, the *Report Characteristics* IE in the E2 RESOURCE STATUS REQUEST indicates the type of objects the E2 Node shall perform load measurements on. The E2 Node shall include in the E2 RESOURCE STATUS UPDATE message:

- the *TNL Capacity Indicator* IE, if the first bit, "TNL Capacity Ind Periodic" of the *Report Characteristics* IE included in the E2 RESOURCE STATUS REQUEST message is set to 1;
- the *Hardware Load Indicator* IE, if the second bit, "HW Load Ind Periodic" of the *Report Characteristics* IE included in the E2 RESOURCE STATUS REQUEST message is set to 1;

If the *Reporting Periodicity* IE in the E2 RESOURCE STATUS REQUEST is present, this indicates the periodicity for the reporting of periodic measurements. The E2 Node shall report once, unless otherwise requested within the *Reporting Periodicity* IE.

8.3.8.3 Unsuccessful operation

```
@startuml
skin rose
skinparam ParticipantPadding 5
skinparam BoxPadding 10
skinparam lifelineStrategy solid

participant "Near-RT RIC" as near
participant "E2 Node" as ran
near->>ran: E2 RESOURCE STATUS REQUEST
ran-->>near: E2 RESOURCE STATUS FAILURE
@enduml
```

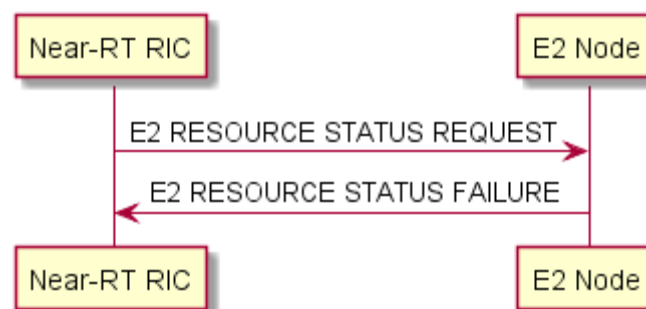


Figure 8.3.8.2-2: E2 Resource Status procedure, unsuccessful operation

If any of the requested load measurements cannot be initiated, the E2 Node shall send the E2 RESOURCE STATUS FAILURE message with an [appropriate cause value](#).

8.3.8.4 Abnormal conditions

If the Near-RT RIC does not receive either the E2 RESOURCE STATUS RESPONSE message or the E2 RESOURCE STATUS FAILURE message, the Near-RT RIC may reinitiate the E2 Resource Status Reporting Initiation procedure towards the same E2 Node, provided that the content of the new E2 RESOURCE STATUS REQUEST message is identical to the content of the previously unacknowledged E2 RESOURCE STATUS REQUEST message with the same Transaction ID.

If the E2 Node receives an E2 RESOURCE STATUS REQUEST message which includes the *Registration Request* IE set to "start" and the *RIC Load Measurement ID* IE corresponding to an existing on-going measurement reporting, for which a different Transaction ID is used, then the E2 Node shall initiate an E2 RESOURCE STATUS FAILURE message with an appropriate cause value.

If the E2 Node receives an E2 RESOURCE STATUS REQUEST message which includes the *Registration Request* IE set to "add" or "stop" and if the E2 Node Load Measurement ID value received in the E2 RESOURCE STATUS REQUEST message is not used, the E2 Node shall initiate an E2 RESOURCE STATUS FAILURE message with an appropriate cause value.

If the *Report Characteristics* IE bitmap is set to "0" (all bits are set to "0") in the E2 RESOURCE STATUS REQUEST message then the E2 Node shall initiate an E2 RESOURCE STATUS FAILURE message with an appropriate cause value.

8.3.9 E2 Resource Status Update procedure

8.3.9.1 General

The purpose of the E2 Resource Status Update procedure is to inform the Near-RT RIC of the E2 Node resource status.

This procedure shall be initiated by the E2 Node.

8.3.9.2 Successful operation

```
@startuml
skin rose
skinparam ParticipantPadding 5
skinparam BoxPadding 10
skinparam lifelineStrategy solid
participant "Near-RT RIC" as near
participant "E2 Node" as ran
ran->>near: E2 RESOURCE STATUS UPDATE
@enduml
```

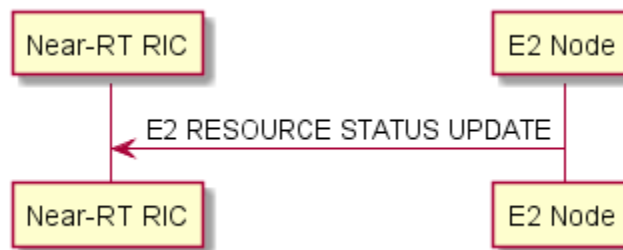


Figure 8.3.9.2-1: E2 Resource Status Update procedure, successful operation

The E2 Node shall report the results of the admitted measurements in the E2 RESOURCE STATUS UPDATE message. The admitted measurements are the measurements that were successfully initiated during the preceding E2 Resource Status Reporting Initiation procedure.

8.3.9.3 Unsuccessful operation

Void.

8.3.9.4 Abnormal conditions

Void.

9 Elements for E2AP communication

9.0 General

Clauses 9.1 and 9.2 describe the structure of the messages and information elements required for the E2AP protocol in tabular format. Clause 9.3 provides the corresponding ASN.1 definition.

The following attributes are used for the tabular description of the messages and information elements: Presence, Range Criticality and Assigned Criticality. Their definition and use can be found in 3GPP TS 36.413 [24].

NOTE: The messages have been defined in accordance with the guidelines specified in 3GPP TR 25.921 [i.2].

9.1 Message functional definition and content

9.1.1 Messages for RIC Functional procedures

9.1.1.1 RIC SUBSCRIPTION REQUEST

This message is sent by the Near-RT RIC to an E2 Node to create a new RIC Subscription in the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Subscription Details	M				YES	reject
>RIC Event Trigger Definition	M		9.2.9		-	
>Sequence of Actions		1.. <maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>RIC Action Type	M		9.2.11		-	
>>RIC Action Definition	O		9.2.12		-	
>>RIC Subsequent Action	O		9.2.13		-	
>>RIC Action Execution Order	O		9.2.35	Used to define a specific execution order.	-	
RIC Subscription Start Time	O		9.2.34		YES	reject
RIC Subscription End Time	O		9.2.34		YES	reject

Range bound	Explanation
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.1.2 RIC SUBSCRIPTION RESPONSE

This message is sent by the E2 Node to accept the request from the Near-RT RIC to create a new RIC Subscription in the E2 Node.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Actions Admitted List		1.. <maxofRICActionID>			YES	reject
>RIC Action ID	M		9.2.10		-	
RIC Actions Not Admitted List		0.. <maxofRICActionID>			YES	reject
>RIC Action ID	M		9.2.10		-	
>Cause	M		9.2.1		-	

Range bound	Explanation
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.1.3 RIC SUBSCRIPTION FAILURE

This message is sent by the E2 Node to inform the Near-RT RIC that the request to create a new RIC Subscription in the E2 Node failed.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
Cause	M		9.2.1		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.4 RIC SUBSCRIPTION DELETE REQUEST

This message is sent by the Near-RT RIC to an E2 Node to request the deletion of an existing Subscription in the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject

9.1.1.5 RIC SUBSCRIPTION DELETE RESPONSE

This message is sent by the E2 Node to accept the request from a Near-RT RIC to delete an existing RIC Subscription in the E2 Node

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject

9.1.1.6 RIC SUBSCRIPTION DELETE FAILURE

This message is sent by the E2 Node to inform the Near-RT RIC that the request to delete an existing RIC Subscription in the E2 Node failed.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
Cause	M		9.2.1		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.6A RIC SUBSCRIPTION DELETE REQUIRED

This message is sent by the E2 Node to request deletion of the existing RIC Subscriptions in the E2 Node.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
List of RIC Subscriptions To Be Removed		1.. <maxofRICrequestID>			EACH	ignore
>RIC Request ID	M		9.2.7		-	-
>RAN Function ID	M		9.2.8		-	-
>Cause	M		9.2.1		-	-

Range bound	Explanation
maxofRICrequestID	Maximum no. of RIC subscription requests supported by Near-RT RIC toward an E2 Node. Value is <1024>.

9.1.1.7 RIC INDICATION

This message is sent by an E2 Node to transfer Report and Insert RIC Service Action information to a Near-RT RIC.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Action ID	M		9.2.10		YES	reject
RIC Indication SN	O		9.2.14		YES	reject
RIC Indication Type	M		9.2.15		YES	reject
RIC Indication Header	M		9.2.17		YES	reject
RIC Indication Message	M		9.2.16		YES	reject
RIC Call process ID	O		9.2.18		YES	reject

9.1.1.8 RIC CONTROL REQUEST

This message is sent by a Near-RT RIC to an E2 Node to initiate or resume a control function logic.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Call Process ID	O		9.2.18		YES	reject
RIC Control Header	M		9.2.20		YES	reject
RIC Control Message	M		9.2.19		YES	reject
RIC Control Ack Request	O		9.2.21		YES	reject

9.1.1.9 RIC CONTROL ACKNOWLEDGE

This message is sent by the E2 Node to inform the Near-RT RIC that the RIC CONTROL REQUEST message was received and to provide information on the outcome of the request.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Call process ID	O		9.2.18		YES	reject
RIC Control Outcome	O		9.2.25		YES	reject

9.1.1.10 RIC CONTROL FAILURE

This message is sent by the E2 Node to inform the Near-RT RIC that the RIC CONTROL REQUEST message has failed to be executed.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Call process ID	O		9.2.18		YES	reject
Cause	M		9.2.1		YES	ignore
RIC Control Outcome	O		9.2.25		YES	Reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.11 RIC SUBSCRIPTION MODIFICATION REQUEST

This message is sent by the Near-RT RIC to an E2 Node to modify an existing Subscription in the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Event Trigger Definition to be Modified	O		9.2.9		YES	ignore
RIC Actions to be Removed List		0..1			YES	ignore
>Action to be Removed Item IEs		1..<maxofRICactionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
RIC Actions to be Modified List		0..1			YES	ignore
>Action to be Modified Item IEs		1..<maxofRICactionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>RIC Action Definition	O		9.2.12		-	
>>RIC Action Execution Order	O		9.2.35			
>>RIC Subsequent Action	O		9.2.13		-	
RIC Actions to be Added List		0..1			YES	ignore
>Action to be Added Item IEs		1..<maxofRICactionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>RIC Action Type	M		9.2.11		-	
>>RIC Action Definition	M		9.2.12		-	
>>RIC Action Execution Order	M		9.2.35			
>>RIC Subsequent Action	O		9.2.13		-	

Range bound	Explanation
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.1.12 RIC SUBSCRIPTION MODIFICATION RESPONSE

This message is sent by the E2 Node to accept the request from the Near-RT RIC to modify an existing E2 subscription in the E2 Node.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Actions Removed List		0..1			YES	ignore
>Action Removed Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
RIC Actions Failed to be Removed List		0..1			YES	ignore
>Action Failed to be Removed Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>Cause	M		9.2.1		-	
RIC Actions Modified List		0..1			YES	ignore
>Action Modified Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
RIC Actions Failed to be Modified List		0..1			YES	ignore
>Action Failed to be Modified Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>Cause	M		9.2.1		-	
RIC Actions Added List		0..1			YES	ignore
>Action Added Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
RIC Actions Failed to be Added List		0..1			YES	ignore
>Action Failed to be Added Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>Cause	M		9.2.1		-	

Range bound	Explanation
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.1.13 RIC SUBSCRIPTION MODIFICATION FAILURE

This message is sent by the E2 Node to inform the Near-RT RIC that the request to modify an existing E2 subscription in the E2 Node failed.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
Cause	M		9.2.1		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.14 RIC SUBSCRIPTION MODIFICATION REQUIRED

This message is sent by the E2 Node to request the Near-RT RIC to modify an existing E2 subscription in the E2 Node.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Actions Required to be Modified List		0..1			YES	ignore
>Action Required to be Modified Item IEs					EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>RIC Time to Wait before subsequent action	M		ENUMERATED (1ms, 2ms, 5ms, 10ms, 20ms, 30ms, 40ms, 50ms, 100ms, 200ms, 500ms, 1s, 2s, 5s, 10s, 20s, 60s, ...)		-	
RIC Actions Required to be Removed List		0..1			YES	ignore
>Action Required to be Removed Item IEs		1..<maxofRICactionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>Cause	M		9.2.1		-	

Range bound	Explanation
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.1.15 RIC SUBSCRIPTION MODIFICATION CONFIRM

This message is sent by the Near-RT RIC to accept the request from the E2 Node to modify an existing E2 subscription in the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Actions Confirmed for Modification List		0..1			YES	ignore
>RIC Action Confirmed for Modification Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
RIC Actions Refused to be Modified List		0..1			YES	ignore
>Action Refused to be Modified Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>Cause	M		9.2.1		-	
RIC Actions Confirmed for Removal List		0..1			YES	ignore
>Action Confirmed for Removal Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
RIC Actions Refused to be Removed List		0..1			YES	ignore
>Action Refused to be Removed Item IEs		1..<maxofRICActionID>			EACH	ignore
>>RIC Action ID	M		9.2.10		-	
>>Cause	M		9.2.1		-	

Range bound	Explanation
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.1.16 RIC SUBSCRIPTION MODIFICATION REFUSE

This message is sent by the Near-RT RIC to deny the request from the E2 Node to modify an existing E2 subscription in the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
Cause	M		9.2.1		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.17 RIC QUERY REQUEST

This message is sent by the Near-RT RIC to an E2 Node to request RAN and/or UE related information from the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Query Header	M		9.2.36		YES	reject
RIC Query Definition	M		9.2.37		YES	reject

9.1.1.18 RIC QUERY RESPONSE

This message is sent by the E2 Node to Near-RT RIC in response to RAN and/or UE related information requested by Near-RT RIC.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
RIC Query Outcome	M		9.2.38		YES	reject

9.1.1.19 RIC QUERY FAILURE

This message is sent by the E2 Node to inform the Near-RT RIC that the requested RAN and/or UE related Information has failed.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function ID	M		9.2.8		YES	reject
Cause	M		9.2.1		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.20 RIC SUBSCRIPTION AUDIT REQUEST

This message is sent by the Near-RT RIC to an E2 Node to initiate the RIC Subscription Audit procedure in the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7	RIC Request ID for RIC Subscription Audit procedure.	YES	reject
RIC Subscription Audit Flag	O		9.2.39		YES	reject
RIC Subscription Audit List	O				YES	reject
>Sequence of RIC Subscriptions		1.. <maxofRICsubscriptions>			EACH	ignore
>>RIC Request ID	M		9.2.7	RIC Request ID for previously established RIC Subscription.	-	
>>RAN Function ID	M		9.2.8		-	

Range bound	Explanation
maxofRICsubscriptions	Maximum no. of RIC Subscriptions to be provided by Near-RT RIC. Value is 2147483648.

9.1.1.21 RIC SUBSCRIPTION AUDIT RESPONSE

This message is sent by the E2 Node to accept the request from the Near-RT RIC to initiate a RIC Subscription Audit in the E2 Node.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7	RIC Request ID for RIC Subscription Audit procedure.	YES	reject
RIC Subscription Confirmed List	O				YES	reject
>Sequence of RIC Subscriptions		1.. <maxofRICsubscriptions>			EACH	ignore
>>RIC Request ID	M		9.2.7	RIC Request ID for previously established RIC Subscription.	-	
>>RAN Function ID	M		9.2.8		-	
>>RIC Action Admitted List	M	1.. <maxofRICactionID>				
>>>RIC Action ID	M		9.2.10			
RIC Subscription Unknown List	O				YES	reject
>Sequence of RIC Subscriptions		1.. <maxofRICsubscriptions>			EACH	ignore
>>RIC Request ID	M		9.2.7	RIC Request ID for previously established RIC Subscription.	-	
>>RAN Function ID	M		9.2.8		-	
RIC Subscription Missing List	O				YES	reject
>Sequence of RIC Subscriptions		1.. <maxofRICsubscriptions>			EACH	ignore
>>RIC Request ID	M		9.2.7	RIC Request ID for previously established RIC Subscription.	-	
>>RAN Function ID	M		9.2.8		-	
>>RIC Action Admitted List	M	1.. <maxofRICactionID>				
>>>RIC Action ID	M		9.2.10			

Range bound	Explanation
maxofRICsubscriptions	Maximum no. of RIC Subscriptions to be provided by Near-RT RIC. Value is 2147483648.
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.1.22 RIC SUBSCRIPTION AUDIT FAILURE

This message is sent by the E2 Node to inform the Near-RT RIC that the request to perform the RIC Subscription Audit in the E2 Node failed.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
Cause	M		9.2.1		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.23 RIC SERVICE LOAD STATUS REQUEST

This message is sent by the Near-RT RIC to an E2 Node to request load reporting for one or more RIC Services.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Load Measurement ID	M		9.2.40	Allocated by Near-RT RIC	YES	reject
E2 Node Load Measurement ID	C- ifRegistrati onRequest StoporAdd		9.2.40	Allocated by E2 Node	YES	ignore
Registration Request	M		ENUMERAT ED (start, stop, add, ...)	Type of request for which the RIC Service Load status is required.	YES	ignore
RAN Function Load Request List	C- ifRegistrati onRequest StartorAdd	1.. <maxofRANfunct ionID>		List of RAN functions requested to provide load information	EACH	reject
>RAN Function ID	M		9.2.8	Id of the declared Function	-	
>RAN Function Load Request	O		9.2.43	Used to request reporting of overall RAN Function load information	-	
>RIC Service Load Request	O		9.2.44	Used to request reporting of overall RIC Service load information	-	
>RIC Subscription Load Request List		0.. <maxofRICreq uestID>			EACH	reject
>>RIC Request ID	M		9.2.7		-	
>>RIC Subscription Load Request	O		9.2.43	Used to request reporting of overall RIC Subscription load information	-	
>>RIC Action Load Request List		0.. <maxofRICact ionID>			EACH	reject
>>>RIC Action ID	M		9.2.10		-	
>>>RIC Action Load Request	M		9.2.43	Used to request reporting of overall RIC Action load information	-	

Reporting Periodicity	O		ENUMERATED (500ms, 1000ms, 2000ms, 5000ms, 10000ms, ...)	Periodicity that can be used for reporting of indicated measurements. Also used as the averaging window length for all measurement object if supported. This IE is ignored if the Registration Request IE is set to "add".	YES	ignore
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Condition	Explanation
ifRegistrationRequestStoporAdd	This IE shall be present if the <i>Registration Request</i> IE is set to the value "stop" or "add".
ifRegistrationRequestStartorAdd	This IE shall be present if the <i>Registration Request</i> IE is set to the value "start" or "add".

Range bound	Explanation
maxofRANfunctionID	Maximum no. of RAN Functions supported by E2 Node. Value is 256.
maxofRICrequestID	Maximum no. of RIC subscription requests supported by Near-RT RIC toward an E2 Node. Value is <1024>.
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.2.24 RIC SERVICE LOAD STATUS RESPONSE

This message is sent by the E2 Node to the Near-RT RIC to accept the request for load reporting for one or more RIC Services.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Load Measurement ID	M		9.2.40	Allocated by Near-RT RIC	YES	reject
E2 Node Load Measurement ID	M		9.2.40	Allocated by E2 Node	YES	ignore
RAN Function Load Confirm List	C- ifRegistrati onRequest StartorAdd	1.. <maxofRANfunc tionID>		List of RAN functions	EACH	reject
>RAN Function ID	M		9.2.8	Id of the declared Function	-	
>RAN Function Load Confirm	O		9.2.45	Used to confirm RAN Function level load reporting	-	
>RIC Service Load Confirm	O		9.2.46	Used to confirm RIC Service load reporting	-	
>RIC Subscription Load Confirm List		0.. <maxofRICreq uestID>			EACH	reject
>>RIC Request ID	M		9.2.7		-	
>>RIC Subscription Load Confirm	O		9.2.45	Used to confirm RIC Subscription level load reporting	-	
>>RIC Action List		0.. <maxofRICact ionID>			EACH	reject
>>>RIC Action ID	M		9.2.10		-	
>>>RIC Action Load Confirm	M		9.2.45	Used to confirm RIC Action level load reporting	-	

Condition	Explanation
ifRegistrationRequestStartorAdd	This IE shall be present if the <i>Registration Request</i> IE in RIC SERVICE LOAD STATUS REQUEST was set to the value "start" or "add".

9.1.2.25 RIC SERVICE LOAD STATUS FAILURE

This message is sent by the E2 Node to inform the Near-RT RIC that the request for load reporting for one or more RIC Services failed.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Load Measurement ID	M		9.2.40	Allocated by Near-RT RIC	YES	reject
E2 Node Load Measurement ID	M		9.2.40	Allocated by E2 Node	YES	ignore
Cause	M		9.2.1		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.26 RIC SERVICE LOAD UPDATE

This message is sent by an E2 Node to a Near-RT RIC to transfer RIC Service load status information.

Direction: E2 Node → Near-RT RIC

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Load Measurement ID	M		9.2.40	Allocated by Near-RT RIC	YES	reject
E2 Node Load Measurement ID	M		9.2.40	Allocated by E2 Node	YES	ignore
RAN Function Load List		1.. <maxofRANfunctionID>		List of RAN functions with load information	EACH	reject
>RAN Function ID	M		9.2.8	Id of the declared Function	-	
>RAN Function Load Information	O		9.2.41	Used to provide overall RAN Function load information	-	
>RIC Service Load Information	O		9.2.42	Used to provide overall RIC Service load information	-	
>RIC Subscription Load List		0.. <maxofRICrequestID>			EACH	reject
>>RIC Request ID	M		9.2.7		-	
>>RIC Subscription Load Information	O		9.2.41	Used to provide overall RIC Subscription load information	-	
>>RIC Action Load List		0.. <maxofRICActionID>			EACH	reject
>>>RIC Action ID	M		9.2.10		-	
>>>RIC Action Load Information	M		9.2.41	Used to provide overall RIC Action load information	-	

Range bound	Explanation
maxofRANfunctionID	Maximum no. of RAN Functions supported by E2 Node. Value is 256.
maxofRICrequestID	Maximum no. of RIC subscription requests supported by Near-RT RIC toward an E2 Node. Value is <1024>.
maxofRICActionID	Maximum no. of Actions to be requested by Near-RT RIC. Value is 16.

9.1.1.27 RIC SUBSCRIPTION STATE CONTROL REQUEST

This message is sent by the Near-RT RIC to an E2 Node to suspend and/or resume one or more RIC Subscriptions.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function State Control List		1.. <maxofRANfunctionID>		List of RAN functions with RIC subscriptions	EACH	reject
>RAN Function ID	M		9.2.8	Id of the declared Function	-	
>RIC Subscription To Be Suspended List		0.. <maxofRICrequestID>			EACH	reject
>>RIC Request ID	M		9.2.7	RIC Request ID of RIC Subscription to be suspended	-	
>>RIC Action List		0.. <maxofRICActionID>		If present, then only the nominated RIC Actions are to be suspended	EACH	reject
>>>RIC Action ID	M		9.2.10		-	
>RIC Subscription To Be Resumed List		0.. <maxofRICrequestID>			EACH	reject
>>RIC Request ID	M		9.2.7	RIC Request ID of RIC Subscription to be resumed	-	
>>RIC Action List		0.. <maxofRICActionID>		If present, then only the nominated RIC Actions are to be resumed	EACH	reject
>>>RIC Action ID	M		9.2.10		-	

Range bound	Explanation
maxofRANfunctionID	Maximum no. of RAN Functions supported by E2 Node. Value is 256.
maxofRICrequestID	Maximum no. of RIC subscription requests supported by Near-RT RIC toward an E2 Node. Value is <1024>.

9.1.1.28 RIC SUBSCRIPTION STATE CONTROL RESPONSE

This message is sent by the E2 Node to the Near-RT RIC to accept the request to suspend and/or resume one or more RIC Subscriptions.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RAN Function State Confirm List		1.. <maxofRANfunctionID>		List of RAN functions with RIC subscriptions	EACH	reject
>RAN Function ID	M		9.2.8	Id of the declared Function	-	
>RIC Subscription Suspended List		0.. <maxofRICrequestID>			EACH	reject
>>RIC Request ID	M		9.2.7	RIC Request ID of suspended RIC Subscription	-	
>>RIC Action List		0.. <maxofRICActionID>			EACH	reject
>>>RIC Action ID	M		9.2.10		-	
>RIC Subscription Resumed List		0.. <maxofRICrequestID>			EACH	reject
>>RIC Request ID	M		9.2.7	RIC Request ID of resumed RIC Subscription	-	
>>RIC Action List		0.. <maxofRICActionID>			EACH	reject
>>>RIC Action ID	M		9.2.10		-	

9.1.1.29 RIC SUBSCRIPTION STATE CONTROL FAILURE

This message is sent by the E2 Node to inform the Near-RT RIC that the request to suspend and/or resume one or more RIC Subscriptions in the E2 Node failed.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
Cause	M		9.2.1		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.30 RIC ASSISTANCE REQUEST

This message is sent by the E2 Node to Near-RT RIC to request an assistance service.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RIC Assistance Header	M		9.2.47		YES	reject
RIC Assistance Message	M		9.2.48		YES	reject
RIC Assistance Update	O		9.2.49		YES	Ignore
RIC Assistance Update Number	O		9.2.50		YES	Ignore

9.1.1.31 RIC ASSISTANCE RESPONSE

This message is sent by the Near-RT RIC to the E2 Node to provide the requested assistance service.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RIC Assistance Header	M		9.2.47		YES	reject
RIC Assistance Outcome	M		9.2.51		YES	ignore

9.1.1.32 RIC ASSISTANCE FAILURE

This message is sent by the Near-RT RIC to inform the E2 Node that the requested RIC Assistance service failed.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
Cause	M		9.2.1		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.1.33 RIC ASSISTANCE INDICATION

This message is sent by the Near-RT RIC to the E2 Node to provide an updated response to the requested assistance service.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	reject
RIC Assistance SN	M		9.2.52		YES	ignore
RIC Assistance Header	M		9.2.47		YES	reject
RIC Assistance Outcome	M		9.2.51		YES	ignore

9.1.1.34 RIC ASSISTANCE HALT

This message is sent by the E2 Node to the Near-RT RIC to halt further updates to the requested assistance service.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
RIC Request ID	M		9.2.7		YES	ignore

9.1.2 Messages for Global Procedures

9.1.2.1 ERROR INDICATION

This message is used to indicate that some error has been detected in the E2 Node or Near-RT RIC.

Direction: E2 Node → Near-RT RIC or Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	ignore
Transaction ID	O		9.2.33	Required if <i>RIC Request ID</i> IE is not present.	YES	reject
RIC Request ID	O		9.2.7	Required if <i>Transaction ID</i> IE is not present.	YES	reject
RAN Function ID	O		9.2.8		YES	reject
Cause	O		9.2.1		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.2 E2 SETUP REQUEST

This message is sent by an E2 Node to a Near-RT RIC to transfer the initialization information.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33	.	YES	reject
Global E2 Node ID	M		9.2.6		YES	reject
RAN Functions Added List		1		List of RAN functions in E2 node.	YES	reject
>RAN Function item		1.. <maxofRANfunctionID>				
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>RAN Function Definition	M		9.2.23	Definition of Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	
>>RAN Function OID	M		9.2.31	Object identifier of corresponding E2SM.	-	
E2 Node Component Configuration Addition List		1		List of E2 Node component configuration information.	YES	reject
>E2 Node Component Configuration Addition Item		1.. <maxofE2nodeComponents>			EACH	reject
>>E2 Node Component Interface Type	M		9.2.26	E2 Node component interface type.	-	
>>E2 Node Component ID	O		9.2.32	E2 Node Component Identifier.	-	
>>E2 Node Component Configuration	M		9.2.27	Contents depends on component interface type.	-	

Range bound	Explanation
maxofRANfunctionID	Maximum no. of RAN Functions supported by E2 Node. Value is 256.
maxofE2nodeComponents	Maximum no. of E2 Node components supported by E2 Node. Value is 1024

9.1.2.3 E2 SETUP RESPONSE

This message is sent by a Near-RT RIC to an E2 Node to transfer the initialization information.

Direction: Near-RT RIC →E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33	.	YES	reject
Global RIC ID	M		9.2.4		YES	reject
RAN Functions Accepted List		0..1				
>RAN Functions ID item		1 .. <maxofRANfunctionID>			YES	Reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	
RAN Functions Rejected List		0..1				
>RAN Functions ID Cause Item		1 .. <maxofRANfunctionID>			YES	reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>Cause	M		9.2.1	Reason for not accepting function.	-	
E2 Node Component Configuration Addition Acknowledge List		1			YES	reject
>E2 Node Component Configuration Addition Acknowledge Item		1.. <maxofE2nodeComponents>			EACH	reject
>>E2 Node Component Interface Type	M		9.2.26	E2 Node component interface type.	-	
>>E2 Node Component ID	M		9.2.32	E2 Node Component Identifier.	-	
>>E2 Node Component Configuration Acknowledge	M		9.2.28	Success or failure with Cause.	-	

RAN Functions Accepted with Modification List		0..1		This list is used only when Near-RT RIC received a RAN Function Definition from the E2 Node that contains RIC assistance services that the Near-RT RIC cannot support, but decided to provide a revised RAN Function definition excluding the unsupported RIC assistance services while retaining the rest of the services. This list shall only be used in this specific scenario and does not apply otherwise.		
>RAN Functions ID item		1 .. <maxofRANfunctionID>			YES	Reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>Revised RAN Function Definition	M		9.2.23	Revised definition of Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	

Range bound	Explanation
maxofRANfunctionID	Maximum no. of RAN Functions supported by E2 Node. Value is 256.
maxofE2nodeComponents	Maximum no. of E2 Node components supported by E2 Node. Value is 1024

9.1.2.4 E2 SETUP FAILURE

This message is sent by the Near-RT RIC to indicate E2 Setup failure.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
Cause	M		9.2.1		YES	ignore
Time To Wait	O		9.2.5		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore
Transport Layer Information	O		9.2.29		YES	ignore

9.1.2.5 RESET REQUEST

This message is sent from a Near-RT RIC to an E2 Node or from an E2 Node to a Near-RT RIC and is used to request the E2 interface between the E2 node and the Near-RT RIC to be reset.

Direction: Near-RT RIC → E2 Node, or E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
Cause	M		9.2.1		YES	ignore

9.1.2.6 RESET RESPONSE

This message is sent by an E2 Node to a Near-RT RIC or from a Near-RT RIC to an E2 Node as a response to a RESET REQUEST message.

Direction: Near-RT RIC → E2 Node, or E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.7 RIC SERVICE UPDATE

This message is sent by an E2 Node to the Near-RT RIC to transfer updated information on RIC Services supported by the E2 Node.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
RAN Functions Added List		0..1		List of added RAN functions in E2 node.		
>RAN Functions Item		1 .. <maxofRANfunctionID>			YES	reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>RAN Function Definition	M		9.2.23	Definition of Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	
>>RAN Function OID	M		9.2.31	Object identifier of corresponding E2SM.	-	
RAN Functions Modified List		0..1		List of Modified RAN functions in E2 node.		
>RAN Functions Item		1 .. <maxofRANfunctionID>			YES	reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>RAN Function Definition	M		9.2.23	Definition of Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	
>>RAN Function OID	M		9.2.31	Object identifier of corresponding E2SM.	-	
RAN Functions Deleted List		0..1		List of deleted RAN functions in E2 node.		
>RAN Functions ID Item		1 .. <maxofRANfunctionID>			YES	reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	

Range bound	Explanation
maxofRANfunctionID	Maximum no. of Functions accepted by Near-RT RIC. Value is 256.

9.1.2.8 RIC SERVICE UPDATE ACKNOWLEDGE

This message is sent by the Near-RT RIC to the E2 Node to acknowledge update of RIC Services supported by the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
RAN Functions Accepted List		0..1		List of Functions accepted by Near-RT RIC.		
>RAN Functions ID Item		1 .. <maxofRANfunctionID>			YES	reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	
RAN Functions Rejected List		0..1		List of Functions not accepted by Near-RT RIC.		
>RAN Functions Cause Item		1 .. <maxofRANfunctionID>			YES	reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>Cause	M		9.2.1	Reason for not accepting function.	-	
RAN Functions Accepted with Modification List		0..1		This list is used only when Near-RT RIC received a RAN Function Definition from the E2 Node that contains RIC assistance services that the Near-RT RIC cannot support, but decided to provide a revised RAN Function definition excluding the unsupported RIC assistance services while retaining the rest of the services. This list shall only be used in this specific scenario and does not apply otherwise.		
>RAN Functions ID item		1 .. <maxofRANfunctionID>			YES	reject

>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>Revised RAN Function Definition	M		9.2.23	Revised definition of Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	

Range bound	Explanation
maxofRANfunctionID	Maximum no. of Functions accepted by Near-RT RIC. Value is 256.

9.1.2.9 RIC SERVICE UPDATE FAILURE

This message is sent by the Near-RT RIC to the E2 Node to indicate RIC SERVICE Update Failure.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
Cause	M		9.2.1	Reason for failure.	YES	reject
Time To Wait	O		9.2.5		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.10 RIC SERVICE QUERY

This message is sent by a Near-RT RIC to an E2 Node to request a E2 Node initiated RIC Service Update procedure.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33	.	YES	reject
RAN Functions Accepted List		0..1		Complete list of Functions previously accepted by Near-RT RIC.		
>RAN Functions ID Item		1 .. <maxofRANfunctionID>			YES	reject
>>RAN Function ID	M		9.2.8	Id of the declared Function.	-	
>>RAN Function Revision	M		9.2.24	Revision counter.	-	

Range bound	Explanation
maxofRANfunctionID	Maximum no. of Functions accepted by Near-RT RIC. Value is 256.

9.1.2.11 E2 NODE CONFIGURATION UPDATE

This message is sent by an E2 Node to the Near-RT RIC to transfer updated information on the E2 Node Configuration information.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
Global E2 Node ID	O		9.2.6	Required when sent as first message on new TNL association.	YES	reject
E2 Node Component Configuration Addition List		0..1			YES	reject
>E2 Node Component Configuration Addition Item		1.. <maxofE2nodeComponents>			EACH	reject
>>E2 Node Component Interface Type	M		9.2.26	E2 Node component interface type.	-	
>>E2 Node Component ID	M		9.2.32	E2 Node Component Identifier.	-	
>>E2 Node Component Configuration	M		9.2.27	Contents depends on component type.	-	
E2 Node Component Configuration Update List		0..1			YES	reject
>E2 Node Component Configuration Update Item		1.. <maxofE2nodeComponents>			EACH	reject
>>E2 Node Component Interface Type	M		9.2.26	E2 Node component interface type.	-	
>>E2 Node Component ID	M		9.2.32	E2 Node Component Identifier.	-	
>>E2 Node Component Configuration	M		9.2.27	Contents depends on component type.	-	
E2 Node Component Configuration Removal List		0..1			YES	reject
>E2 Node Component Configuration Removal Item		1.. <maxofE2nodeComponents>			EACH	reject
>>E2 Node Component Interface Type	M		9.2.26	E2 Node component interface type.	-	
>>E2 Node Component ID	M		9.2.32	E2 Node Component Identifier.	-	
E2 Node TNL Association To Remove List		0..1			YES	reject
>E2 Node TNL Association To Remove Item IEs		1.. <maxofTNLA>			EACH	reject
>> Transport Layer Information	M		9.2.29	Transport Layer Address of the E2 node.	-	-
>> Transport Layer Information Near-RT RIC	O		9.2.29	Transport Layer Address of the Near-RT RIC.	-	-

Range bound	Explanation
maxofE2nodeComponents	Maximum no. of E2 Node components supported by E2 Node. Value is 1024.
maxofTNLA	Maximum no. of TNL Associations supported by E2 Node. Value is 32.

9.1.2.12 E2 NODE CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by Near-RT RIC to E2 Node to acknowledge update of E2 Node Configuration supported by the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33	.	YES	reject
E2 Node Component Configuration Addition Acknowledge List		0..1			YES	reject
>E2 Node Component Configuration Addition Acknowledge Item		1.. <maxofE2node Components>			EACH	reject
>>E2 Node Component Interface Type	M		9.2.26	E2 Node component interface type.	-	
>>E2 Node Component ID	M		9.2.32	E2 Node Component Identifier.	-	
>>E2 Node Component Configuration Acknowledge	M		9.2.28	Success or failure with Cause.	-	
E2 Node Component Configuration Update Acknowledge List		0..1			YES	reject
>E2 Node Component Configuration Update Acknowledge Item		1.. <maxofE2node Components>			EACH	reject
>>E2 Node Component Interface Type	M		9.2.26	E2 Node component interface type.	-	
>>E2 Node Component ID	O		9.2.32	E2 Node Component Identifier.	-	
>>E2 Node Component Configuration Update Acknowledge	M		9.2.28	Success or failure with Cause.	-	
E2 Node Component Configuration Removal Acknowledge List		0..1			YES	reject
>E2 Node Component Configuration Removal Acknowledge Item		1.. <maxofE2node Components>			EACH	reject
>>E2 Node Component Interface Type	M		9.2.26	E2 Node component interface type.	-	
>>E2 Node Component ID	M		9.2.32	E2 Node Component Identifier.	-	
>>E2 Node Component Configuration Acknowledge	M		9.2.28	Success or failure with Cause.	-	

Range bound	Explanation
maxofE2nodeComponents	Maximum no. of E2 Node components supported by E2 Node. Value is 1024.

9.1.2.13 E2 NODE CONFIGURATION UPDATE FAILURE

This message is sent by Near-RT RIC to E2 Node to indicate E2 Node Configuration Update Failure.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33	.	YES	reject
Cause	M		9.2.1	Cause	YES	reject
Time To Wait	O		9.2.5		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.14 E2 CONNECTION UPDATE

This message is sent by Near-RT RIC to E2 Node to initiate update of E2 Connection supported by the E2 Node.

Direction: Near-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
E2 Connection To Add List		0..1			YES	ignore
>E2 Connection to Add Item IEs		1.. <maxofTNLA>			EACH	ignore
>>Transport Layer Information	M		9.2.29	Transport layer address and port number of Near-RT RIC.		
>>TNL Association Usage	M		9.2.30	Indicates how E2 connection is to be used.		
E2 Connection To Remove List		0..1			YES	ignore
>E2 Connection to Remove Item IEs		1.. <maxofTNLA>			EACH	ignore
>>Transport Layer Information	M		9.2.29	Transport layer address and port number of Near-RT RIC.		
E2 Connection To Modify List		0..1			YES	ignore
>E2 Connection to Modify Item IEs		1.. <maxofTNLA>			EACH	ignore
>>Transport Layer Information	M		9.2.29	Transport layer address and port number of Near-RT RIC.		
>>TNL Association Usage	M		9.2.30	Indicates how E2 connection is to be used.		

Range bound	Explanation
maxofTNLA	Maximum no. of TNL Associations supported by E2 Node. Value is 32.

9.1.2.15 E2 CONNECTION UPDATE ACKNOWLEDGE

This message is sent by E2 Node to the Near-RT RIC to acknowledge update of E2 Connection supported by the E2 Node.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
E2 Connection Setup List		0..1			YES	ignore
>E2 Connection Setup Item IEs		1.. <maxofTNLA>			EACH	ignore
>>Transport Layer Information	M		9.2.29	Transport layer address and port number of Near-RT RIC.		
>>TNL Association Usage	M		9.2.30	Indicates how E2 connection is to be used.		
E2 Connection Failed to Setup List		0..1			YES	ignore
>E2 Connection failed to setup Item IEs		1.. <maxofTNLA>			EACH	ignore
>>Transport Layer Information	M		9.2.29	Transport layer address and port number of Near-RT RIC.		
>>Cause	M		9.2.1			

Range bound	Explanation
maxofTNLA	Maximum no. of TNL Associations supported by E2 Node. Value is 32.

9.1.2.16 E2 CONNECTION UPDATE FAILURE

This message is sent by E2 Node to the Near-RT RIC to inform failure of the requested E2 Connection updates.

Direction: E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
Cause	M		9.2.1		YES	reject
Time To Wait	O		9.2.5		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.17 E2 REMOVAL REQUEST

This message is sent by either the E2 Node or the Near-RT RIC to initiate the removal of the E2 signalling connection and the related resources.

Direction: Near-RT RIC → E2 Node, or E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject

9.1.2.18 E2 REMOVAL RESPONSE

This message is sent by either the E2 Node or the Near-RT RIC to acknowledge the initiation of removal of the E2 signalling connection and the related resources.

Direction: Near-RT RIC → E2 Node, or E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.19 E2 REMOVAL FAILURE

This message is sent by either the E2 Node or the Near-RT RIC to indicate that removing the E2 signalling connection and the related resources cannot be accepted.

Direction: Near-RT RIC → E2 Node, or E2 Node → Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
Cause	M		9.2.1		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.20 E2 RESOURCE STATUS REQUEST

This message is sent by the Near-RT RIC to an E2 Node to initiate the requested measurement according to the parameters given in the message.

Direction: Near RT RIC → E2 Node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33		YES	reject
RIC Load Measurement ID	M		9.2.40	Allocated by Near-RT RIC	YES	reject
E2 Node Load Measurement ID	C- ifRegistrati onRequest StoporAdd		9.2.40	Allocated by E2 Node	YES	ignore
Registration Request	M		ENUMERAT ED (start, stop, add, ...)	Type of request for which the resource status is required.	YES	ignore
Report Characteristics	C- ifRegistrati onRequest Start		BIT STRING (SIZE(8))	Each position in the bitmap indicates measurement object the E2 Node is requested to report. First Bit = TNL Capacity Ind Periodic, Second Bit = HW Load Ind Periodic. Other bits shall be ignored by the E2 Node.	YES	ignore
Reporting Periodicity	O		ENUMERAT ED (500ms, 1000ms, 2000ms, 5000ms,100 00ms, ...)	Periodicity that can be used for reporting of indicated measurements. Also used as the averaging window length for all measurement objects if supported. This IE is ignored if the <i>Registration Request</i> IE is set to "add".	YES	ignore

Condition	Explanation
ifRegistrationRequestStoporAdd	This IE shall be present if the <i>Registration Request</i> IE is set to the value "stop" or "add".
ifRegistrationRequestStart	This IE shall be present if the <i>Registration Request</i> IE is set to the value "start".

9.1.2.21 E2 RESOURCE STATUS RESPONSE

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33	.	YES	reject
RIC Load Measurement ID	M		9.2.40	Allocated by Near-RT RIC	YES	reject
E2 Node Load Measurement ID	M		9.2.40	Allocated by E2 Node	YES	reject
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.22 E2 RESOURCE STATUS FAILURE

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33	.	YES	reject
RIC Load Measurement ID	M		9.2.40	Allocated by Near-RT RIC	YES	reject
E2 Node Load Measurement ID	M		9.2.40	Allocated by E2 Node	YES	reject
Cause	M		9.2.1		YES	ignore
Criticality Diagnostics	O		9.2.2		YES	ignore

9.1.2.23 E2 RESOURCE STATUS UPDATE

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.3		YES	reject
Transaction ID	M		9.2.33	.	YES	reject
RIC Load Measurement ID	M		9.2.40	Allocated by Near-RT RIC	YES	reject
E2 Node Load Measurement ID	M		9.2.40	Allocated by E2 Node	YES	reject
TNL Capacity Indicator	O		9.2.54		YES	ignore
Hardware Load Indicator	O		9.2.55		YES	ignore

9.2 Information Element definitions

9.2.0 General

When specifying information elements which are to be represented by bit strings, if not otherwise specifically stated in the semantics description of the concerned IE or elsewhere, the following principle applies with regards to the ordering of bits:

- The first bit (leftmost bit) contains the Most Significant Bit (MSB);
- The last bit (rightmost bit) contains the Least Significant Bit (LSB);
- When importing bit strings from other specifications, the first bit of the bit string contains the first bit of the concerned information.

9.2.1 Cause

The purpose of the *Cause* IE is to indicate the reason for a particular event for the E2AP protocol.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
CHOICE Cause Group	M			
>RIC services				
>>RIC Request	O		ENUMERATED (RAN Function ID invalid, Action not supported, Excessive actions, Duplicate action, Duplicate Event Trigger, Function resource limit, RIC Request ID unknown, Inconsistent Action/subsequent Action sequence, Control message invalid, RIC Call process ID invalid, Control timer expired, Control failed to execute, System not ready, unspecified, ..., RIC Subscription End Time expired, RIC Subscription Time invalid, Duplicate RIC Request ID, Event Trigger not supported, Requested Information Unavailable, Invalid Information Request)	
>>RIC Service	O		ENUMERATED RAN Function not supported, Excessive functions, RIC resource limit,...)	
>>E2 Node	O		ENUMERATED (E2 node component unknown, ...)	
>Transport Layer				
>>Transport Layer Cause	M		ENUMERATED (Unspecified, Transport Resource Unavailable, ...)	
>Protocol				
>>Protocol Cause	M		ENUMERATED (Transfer Syntax Error, Abstract Syntax Error (Reject), Abstract Syntax Error (Ignore and Notify), Message not Compatible with Receiver State, Semantic Error, Abstract Syntax Error (Falsely Constructed Message), Unspecified, ...)	
>Misc				
>>Miscellaneous Cause	M		ENUMERATED (Control Processing Overload, Hardware Failure, O&M Intervention, Unspecified, ...)	
>Service layer				
>>Service layer Cause	M		9.2.53	

The meaning of the different cause values is described in the following table. In general, "not supported" cause values indicate that the related capability is missing. On the other hand, "not available" cause values indicate that the related capability is present, but insufficient resources are available to perform the requested action.

RIC Request cause	Meaning
Unspecified	Sent for RIC service cause when none of the specified cause values applies.
RAN Function ID invalid	Requested function Id invalid or not known by E2 Node.
Action not supported	Requested Action not supported by RAN function.
Excessive actions	Excessive number of actions requested for RAN Function.
Duplicate action	Same action requested more than once in same subscription request.
Duplicate Event Trigger	Subscription request has same event trigger as previously accepted subscription request.
Function resource limit	RAN function has reached resource limit.
RIC Request ID unknown	RIC Request ID sent to Near-RT RIC is unknown.
Inconsistent Action/subsequent Action sequence	RAN Function has detected inconsistent sequence of requested Action and Subsequent Action.
Control message invalid	RAN Function has detected invalid RIC CONTROL REQUEST message.
RIC Call process ID invalid	RAN function has detected invalid RIC Call Process ID in RIC CONTROL REQUEST.
Control timer expired	RIC Control Request received by E2 Node after the associated RIC Time to Wait timer had expired.
Control failed to execute	Requested control procedure initiated by RIC Control Request failed to be executed in the E2 Node.
System not ready	RAN Function is not ready to receive RIC Subscription or RIC Control message.
RIC Subscription End Time expired	RIC SUBSCRIPTION DELETE REQUIRED is triggered to inform Near-RT RIC that end time has expired.
RIC Subscription Time invalid	E2 Node received RIC SUBSCRIPTION REQUEST containing an invalid RIC Subscription Start Time and/or RIC Subscription End Time.
Duplicate RIC Request ID	E2 node does not support handling of same RIC Request ID as previously accepted subscription request.
Event Trigger not supported	Requested event trigger definition or modification - not supported by RAN function.
Requested Information Unavailable	Information requested by Near-RT RIC is not available at E2 Node.
Invalid Information Request	Information requested by Near-RT RIC is invalid.

RIC Service cause	Meaning
RAN Function not supported	The RAN Function described by E2 Node is not supported by Near-RT RIC.
Excessive functions	RIC has reached a limit on the number of declared RAN functions.
RIC resource limit	RIC has reached a resource limit.

E2 Node configuration cause	Meaning
E2 Node component unknown	The received message refers to an unknown E2 Node component.

Transport Layer cause	Meaning
Unspecified	Sent when none of the cause values below applies but still the cause is Transport Network Layer related.
Transport Resource Unavailable	The required transport resources are not available.

Protocol cause	Meaning
Transfer Syntax Error	The received message included a transfer syntax error.
Abstract Syntax Error (Reject)	The received message included an abstract syntax error and the concerning criticality indicated "reject".
Abstract Syntax Error (Ignore And Notify)	The received message included an abstract syntax error and the concerning criticality indicated "ignore and notify".
Message Not Compatible With Receiver State	The received message was not compatible with the receiver state.
Semantic Error	The received message included a semantic error.
Abstract Syntax Error (Falsely Constructed Message)	The received message contained IEs or IE groups in wrong order or with too many occurrences.
Unspecified	Sent when none of the above cause values applies but still the cause is Protocol related.

Miscellaneous cause	Meaning
Control Processing Overload	Control processing overload.
Hardware Failure	Action related to hardware failure.
O&M Intervention	The action is due to O&M intervention.
Unspecified Failure	Sent when none of the above cause values applies and the cause is not related to any of the categories Radio Network Layer, Transport Network Layer, NAS or Protocol.

Service Layer cause	Meaning
Service Layer Cause	Used as a transparent container transporting the service layer specific cause information

9.2.2 Criticality Diagnostics

The *Criticality Diagnostics* IE is sent by the E2 Node or the Near-RT RIC when parts of a received message have not been comprehended, or were missing, or if the message contained logical errors. When applicable, it contains information about which IEs were not comprehended or were missing.

For further details on how to use the *Criticality Diagnostics* IE, (see clause 10). The conditions for inclusion of the *Transaction ID* IE are described in clause 10.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Procedure Code	O		INTEGER (0..255)	Procedure Code is to be used if Criticality Diagnostics is part of Error Indication procedure, and not within the response message of the same procedure that caused the error.
Triggering Message	O		ENUMERATED (initiating message, successful outcome, unsuccessful outcome)	The Triggering Message is used only if the Criticality Diagnostics is part of Error Indication procedure.
Procedure Criticality	O		ENUMERATED (reject, ignore, notify)	This Procedure Criticality is used for reporting the Criticality of the Triggering message (Procedure).
RIC Request ID	O		9.2.7	
Information Element Criticality Diagnostics		0 .. <maxnoof Errors>		
>IE Criticality	M		ENUMERATED (reject, ignore, notify)	The IE Criticality is used for reporting the criticality of the triggering IE. The value 'ignore' shall not be used.
>IE ID	M		INTEGER (0..65535)	The IE ID of the not understood or missing IE.
>Type of Error	M		ENUMERATED (not understood, missing, ...)	

Range bound	Explanation
maxnoofErrors	Maximum no. of IE errors allowed to be reported with a single message. The value for maxnoofErrors is 256.

9.2.3 Message Type

The *Message Type* IE uniquely identifies the message being sent. It is mandatory for all messages.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Message Type				
>Procedure Code	M		INTEGER (0..255)	
>Type of Message	M		CHOICE (Initiating Message, Successful Outcome, Unsuccessful Outcome, ...)	

9.2.4 Global RIC ID

This IE is used to globally identify the Near-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		3GPP TS 38.423 clause 9.2.2.4	
Near-RT RIC ID	M		BIT STRING (SIZE(20))	

9.2.5 Time to wait

This IE defines the minimum allowed waiting times.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Time to wait	M		ENUMERATED(1s, 2s, 5s, 10s, 20s, 60s)	

9.2.6 Global E2 Node ID

This IE is used to globally identify an E2 node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE	M			
>gNB				To be used when E2 Node supports gNB mode or both gNB and en-gNB modes.
>>Global gNB ID	M		3GPP TS 38.423 clause 9.2.2.1	
>>Global en-gNB ID	O		3GPP TS 36.423 clause 9.2.112	Required when E2 node also supports NR with en-gNB mode.
>>gNB-CU-UP ID	O		3GPP TS 37.483 clause 9.3.1.15	Required when E2 Node supports only gNB-CU-UP functionality.
>>gNB-DU ID	O		3GPP TS 38.473 clause 9.3.1.9	Required when E2 Node supports only gNB-DU functionality.
>en-gNB				To be used when E2 Node supports en-gNB mode only.
>>Global en-gNB ID	M		3GPP TS 36.423 clause 9.2.112	
>>en-gNB-CU-UP ID	O		3GPP TS 37.483 clause 9.3.1.15	Required when E2 Node supports only gNB-CU-UP functionality.
>>en-gNB-DU ID	O		3GPP TS 38.473 clause 9.3.1.9	Required when E2 Node supports only gNB-DU functionality.
>ng-eNB				To be used when E2 Node supports ng-eNB mode or both ng-eNB and eNB modes.
>>Global ng-eNB ID	M		3GPP TS 38.423 clause 9.2.2.2	
>>Global eNB ID	O		3GPP TS 36.423 clause 9.2.22	Required when E2 Node also supports E-UTRA with eNB mode.
>>ng-eNB-DU ID	O		3GPP TS 37.473 clause 9.3.1.9	Required when E2 node supports only ng-eNB DU functionality.
>eNB				To be used when E2 Node supports eNB mode only.
>>Global eNB ID	M		3GPP TS 36.423 clause 9.2.22	

9.2.7 RIC Request ID

This information element indicates the RIC Request ID , and shall be unique for a given E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Requestor ID	M		INTEGER (0.. 65535)	
RIC Instance ID	M		INTEGER (0..65535)	

9.2.8 RAN Function ID

This information element indicates the RAN Function ID, and shall be unique within a given E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Function ID	M		INTEGER (0..4095)	Value 0 reserved for Near-RT RIC internal usage.

9.2.9 RIC Event Trigger Definition

This information element indicates the RIC event trigger description used by the RIC Subscription procedure.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Event Trigger Definition	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.10 RIC Action ID

This information element indicates the Action ID number for a RIC Service Action, and shall be unique within the given RIC Request ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Action ID	M		INTEGER (0..255)	

9.2.11 RIC Action Type

This IE defines the type of RIC Service Action to be executed.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Action Type	M		ENUMERATED (Insert, Report, Policy, ...)	

9.2.12 RIC Action Definition

This information element provides parameters to be used when executed a Report, Insert or Policy RIC Service Actions .

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Action Definition	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.13 RIC Subsequent Action

This IE defines the subsequent action to be taken after completing a particular RIC Service Action and shall be present when RIC Action Type set to Insert.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Subsequent Action Type	M		ENUMERATED (Continue, Halt, ...)	
RIC Time to Wait	M		ENUMERATED (1ms, 2ms, 5ms, 10ms, 20ms, 30ms, 40ms, 50ms, 100ms, 200ms, 500ms, 1s, 2s, 5s, 10s, 20s, 60s, ...)	

9.2.14 RIC Indication Sequence Number (SN)

This information element indicates the Indication Sequence Number (SN).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Indication SN	M		INTEGER (0..65535)	

9.2.15 RIC Indication Type

This IE defines the Indication Type.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Indication Type	M		ENUMERATED (Insert, Report, ...)	

9.2.16 RIC Indication message

This information element carries the RIC indication message used for Insert and Report RIC Service Actions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Indication message	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.17 RIC Indication header

This information element carries the RIC indication header used for Insert and Report RIC Service Actions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Indication header	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.18 RIC Call Process ID

This information element carries the RIC Call Process ID used for the Insert and Control RIC Service Actions. The RIC Call Process ID shall be unique within a given RAN Function on a given E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Call Process ID	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.19 RIC Control message

This information element carries the RIC Control message.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Control Message	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.20 RIC Control header

This information element carries the RIC Control Header.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Control header	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.21 RIC Control Ack Request

This IE defines whether and when the RIC CONTROL ACKNOWLEDGE message shall be sent by the E2 Node as described in the below table.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Control Ack Request	M		ENUMERATED (NoAck, Ack, ...)	

The meaning of the different values is described in the following table.

RIC Service cause	Meaning
NoAck	Optional RIC Control Acknowledgement is not required.
Ack	Optional RIC Control Acknowledgement is required.

9.2.22 Void

9.2.23 RAN Function Definition

This information element carries the RAN Function Definition.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Function Definition	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.24 RAN Function Revision

This information element carries the RAN Function Revision.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Function Revision	M		INTEGER (0..4095)	

9.2.25 RIC Control Outcome

This information element carries the RIC Control Outcome.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Control Outcome	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.26 E2 Node Component Interface Type

This IE is used to identify an E2 node component type.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E2 node component interface type	M		ENUMERATED (ng, xn, e1, f1, w1, s1, x2, ...)	

9.2.27 E2 Node Component Configuration

This IE is used to carry the E2 Node component configuration update information of a specific E2 Node component.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SEQUENCE	M			
>E2 Node Component Request Part	M		OCTET STRING	Contents depend on component type and used to carry new or updated component configuration. See the table below.
>E2 Node Component Response Part	M		OCTET STRING	Contents depend on component type and used to carry new or updated component configuration. See the table below.

NOTE: E2 node may generate the content of this IE using information derived from prior messages sent or received over the corresponding 3GPP Interfaces. For the E2 SETUP REQUEST message, the E2 node may derive the content of this IE from the prior messages pertaining to the Request part and the Response part of the Component Addition list in the following table and/or the Request part and the Response part of the Component Update list in the following table.

In all cases the information is a data structure defined by the applicable 3GPP specification as specified in the following table.

E2 Node component message content	Component Addition list		Component Update list	
	Request part	Response part	Request part	Response part
gNB case				
>NG (AMF Name)	NG SETUP REQUEST, 3GPP TS 38.413 [19] clause 9.2.6.1	NG SETUP RESPONSE, 3GPP TS 38.413 [19] clause 9.2.6.2	RAN CONFIGURATION UPDATE, 3GPP TS 38.413 [19] clause 9.2.6.4 Or AMF CONFIGURATION UPDATE, 3GPP TS 38.413 [19] clause 9.2.6.7	RAN CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 38.413 [19] clause 9.2.6.5 Or AMF CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 38.413 [19] clause 9.2.6.8
>Xn (Neighbour Global NG-RAN Node ID)	XN SETUP REQUEST, 3GPP TS 38.423 [20] clause 9.1.3.1	XN SETUP RESPONSE, 3GPP TS 38.423 [20] clause 9.1.3.2	NG-RAN NODE CONFIGURATION UPDATE, 3GPP TS 38.423 [20] clause 9.1.3.4	NG-RAN NODE CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 38.423 [20] clause 9.1.3.5
>E1 (gNB-CU-UP ID)	GNB-CU-UP E1 SETUP REQUEST, 3GPP TS 37.483 [21] clause 9.2.1.4 Or GNB-CU-CP E1 SETUP REQUEST, 3GPP TS 37.483 [21] clause 9.2.1.7	GNB-CU-UP E1 SETUP RESPONSE, 3GPP TS 37.483 [21] clause 9.2.1.5 Or GNB-CU-CP E1 SETUP RESPONSE, 3GPP TS 37.483 [21] clause 9.2.1.8	GNB-CU-UP CONFIGURATION UPDATE, 3GPP TS 37.483 [21] clause 9.2.1.10 Or GNB-CU-CP CONFIGURATION UPDATE, 3GPP TS 37.483 [21] clause 9.2.1.13	GNB-CU-UP CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 37.483 [21] clause 9.2.1.11 Or GNB-CU-CP CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 37.483 [21] clause 9.2.1.14
>F1 (gNB-DU ID)	F1 SETUP REQUEST, 3GPP TS 38.473 [22] clause 9.2.1.4	F1 SETUP RESPONSE, 3GPP TS 38.473 [22] clause 9.2.1.5	GNB-DU CONFIGURATION UPDATE, 3GPP TS 38.473 [22] clause 9.2.1.7 Or GNB-CU CONFIGURATION UPDATE, 3GPP TS 38.473 [22] clause 9.2.1.10	GNB-DU CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 38.473 [22] clause 9.2.1.8 Or GNB-CU CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 38.473 [22] clause 9.2.1.11
>X2 (Neighbour Global eNB ID)	EN-DC X2 SETUP REQUEST, 3GPP TS 36.423 [25] clause 9.1.2.31	EN-DC X2 SETUP RESPONSE, 3GPP TS 36.423 [25] clause 9.1.2.32	EN-DC CONFIGURATION UPDATE, 3GPP TS 36.423 [25] clause 9.1.2.34	EN-DC CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 36.423 [25] clause 9.1.2.35
eNB case				

E2 Node component message content	Component Addition list		Component Update list	
	Request part	Response part	Request part	Response part
>NG (AMF Name)	NG SETUP REQUEST, 3GPP TS 38.413 [19] clause 9.2.6.1	NG SETUP RESPONSE, 3GPP TS 38.413 [19] clause 9.2.6.2	RAN CONFIGURATION UPDATE, 3GPP TS 38.413 [19] clause 9.2.6.4 Or AMF CONFIGURATION UPDATE, 3GPP TS 38.413 [19] clause 9.2.6.7	RAN CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 38.413 [19] clause 9.2.6.5 Or AMF CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 38.413 [19] clause 9.2.6.8
>Xn (Neighbour Global NG-RAN Node ID)	XN SETUP REQUEST, 3GPP TS 38.423 [20] clause 9.1.3.1	XN SETUP RESPONSE, 3GPP TS 38.423 [20] clause 9.1.3.2	NG-RAN NODE CONFIGURATION UPDATE, 3GPP TS 38.423 [20] clause 9.1.3.4	NG-RAN NODE CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 38.423 [20] clause 9.1.3.5
>W1 (ng-eNB-DU ID)	W1 SETUP REQUEST, 3GPP TS 37.473 [23] clause 9.2.1.4	W1 SETUP RESPONSE, 3GPP TS 37.473 [23] clause 9.2.1.5	NG-ENB-DU CONFIGURATION UPDATE, 3GPP TS 37.473 [23] clause 9.2.1.7 Or NG-ENB-CU CONFIGURATION UPDATE, 3GPP TS 37.473 [23] clause 9.2.1.10	NG-ENB-DU CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 37.473 [23] clause 9.2.1.8 Or NG-ENB-CU CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 37.473 [23] clause 9.2.1.11
>S1 (MME Name)	S1 SETUP REQUEST, 3GPP TS 36.413 [24] clause 9.1.8.4	S1 SETUP RESPONSE, 3GPP TS 36.413 [24] clause 9.1.8.5	ENB CONFIGURATION UPDATE, 3GPP TS 36.413 [24] clause 9.1.8.7 Or MME CONFIGURATION UPDATE, 3GPP TS 36.413 [24] clause 9.1.8.10	ENB CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 36.413 [24] clause 9.1.8.8 Or MME CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 36.413 [24] clause 9.1.8.11
>X2 (when neighbour is eNB) (Neighbour Global eNB ID)	X2 SETUP REQUEST, 3GPP TS 36.423 [25] clause 9.1.2.3	X2 SETUP RESPONSE, 3GPP TS 36.423 [25] clause 9.1.2.4	ENB CONFIGURATION UPDATE, 3GPP TS 36.423 [25] clause 9.1.2.8	ENB CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 36.423 [25] clause 9.1.2.9

E2 Node component message content	Component Addition list		Component Update list	
	Request part	Response part	Request part	Response part
>X2 (when neighbour is en-gNB) (Neighbour Global eNB ID)	EN-DC X2 SETUP REQUEST, 3GPP TS 36.423 [25] clause 9.1.2.31	EN-DC X2 SETUP RESPONSE, 3GPP TS 36.423 [25] clause 9.1.2.32	EN-DC CONFIGURATION UPDATE, 3GPP TS 36.423 [25] clause 9.1.2.34	EN-DC CONFIGURATION UPDATE ACKNOWLEDGE, 3GPP TS 36.423 [25] clause 9.1.2.35

9.2.28 E2 Node Component Configuration Acknowledge

This IE is used to carry the E2 Node component configuration update acknowledge of a specific E2 Node component.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Outcome	M		ENUMERATED (success, failure,...)	
Cause	O		9.2.1	Cause for failure.

9.2.29 Transport Layer Information

This information element provides Near-RT RIC address and optionally port number to be used by an E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transport Layer Address	M		BIT STRING (SIZE(1..160,...))	To be passed to transport layer without interpretation.
Transport Layer Port	O		BIT STRING (SIZE(16))	To be passed to transport layer without interpretation.

9.2.30 TNL Association Usage

This information element provides TNL association usage.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TNL Association Usage	M		ENUMERATED (ric service, support functions, both,..)	Indicates whether E2 connection to be used for RIC services only, or E2 support functions only, or both.

9.2.31 RAN Function OID

This information element carries the RAN Function OID and shall uniquely refer to a specific E2 Service Model (E2SM).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Function Service Model OID	M		PrintableString(SIZE(1..1000,...))	Object Identifier of the specific RAN Function definition. Formatted as per OID.

9.2.32 E2 Node Component ID

This IE is used to locally identify an E2 node component.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE <i>E2 node component interface type</i>	M			
>NG				
>>AMF name	M		3GPP TS 38.413 [19] clause 9.3.3.21	Serving AMF.
>Xn				
>>Global NG-RAN Node ID	M		3GPP TS 38.423 [20] clause 9.2.2.3	Neighbour gNB or ng-eNB.
>E1				
>>gNB-CU-UP ID	M		3GPP TS 37.483 [21] clause 9.3.1.15	
>F1				
>>gNB-DU ID	M		3GPP TS 38.473 [22] clause 9.3.1.9	
>W1				
>>ng-eNB-DU ID	M		3GPP TS 37.473 [23] clause 9.3.1.9	
>S1				
>>MME name	M		3GPP TS 36.413 [24], clause 9.1.8.5	Serving MME.
>X2				
>>Global eNB ID	O		3GPP TS 36.423 [25] clause 9.2.22	Neighbour eNB.
>>Global en-gNB ID	O		3GPP TS 36.423 [25] clause 9.2.112	Neighbour en-gNB.

9.2.33 Transaction ID

The *Transaction ID* IE uniquely identifies a procedure among all ongoing parallel procedures of the same type initiated by the same protocol peer. Messages belonging to the same procedure shall use the same Transaction ID. The Transaction ID is determined by the initiating peer of a procedure.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transaction ID	M		INTEGER (0..255, ...)	

9.2.34 RIC Subscription Time

The *RIC Subscription Time* IE is used to set the start and end time of a RIC Subscription.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Subscription time	M		OCTET STRING (SIZE(8))	Encoded using the 64-bit timestamp format as defined in clause 6 of IETF RFC 5905 [26].

9.2.35 RIC Action Execution Order

This IE is used to modify the default RIC service action execution order.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Action Execution Order	M		INTEGER (0..255, ...)	0 used to indicate "Any-order" 1..255 Used to enforce a specific execution order.

9.2.36 RIC Query Header

This information element carries the RIC Query Header.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Query Header	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.37 RIC Query Definition

This information element carries the RIC Query Definition.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Query Definition	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.38 RIC Query Outcome

This information element carries the RIC Query Outcome.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Query Outcome	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.39 RIC Subscription Audit Flag

The purpose of the *RIC Subscription Audit Flag* IE is to modify to E2 Node response to a RIC SUBSCRIPTION AUDIT REQUEST message.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
Listed Records Only	O		ENUMERATED (true, ...)	If TRUE then E2 Node shall consider the listed records only.

9.2.40 Load Measurement ID

This information element carries the Load Measurement ID allocated by Near-RT RIC and E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Load measurement ID	M		INTEGER (1..4095,...)	

9.2.41 RIC Load Information

This information element is used to encode the load information from the E2 Node. At least one of the listed IE shall be present.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Load Status	O		ENUMERATED (overload, not overload, ...)	
Load Estimate	O		INTEGER (0..100)	Load estimation in percentage of capacity

9.2.42 RIC Service Load Information

This information element carries the RIC Service Load Information for one or more RIC services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Service Report Load Information	O		9.2.41	Used to carry load information on REPORT service
RIC Service Insert Load Information	O		9.2.41	Used to carry load information on INSERT service
RIC Service Control Load Information	O		9.2.41	Used to carry load information on CONTROL service
RIC Service Policy Load Information	O		9.2.41	Used to carry load information on POLICY service
RIC Service Query Load Information	O		9.2.41	Used to carry load information on QUERY service

9.2.43 RIC Load Request

This information element requests reporting for the coreresponding RIC Load Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Load Request	M		ENUMERATED (true, ...)	If TRUE then the corresponding load information reporting is requested

9.2.44 RIC Service Load Request

This information element requests reporting for the coreresponding RIC Load Information for one or more specific RIC services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Service Report Load Request	O		9.2.43	Used to request reporting of load information on REPORT service
RIC Service Insert Load Request	O		9.2.43	Used to request reporting of load information on INSERT service
RIC Service Control Load Request	O		9.2.43	Used to request reporting of load information on CONTROL service
RIC Service Policy Load Request	O		9.2.43	Used to request reporting of load information on POLICY service
RIC Service Query Load Request	O		9.2.43	Used to request reporting of load information on QUERY service

9.2.45 RIC Load Confirm

This information element cofirms reporting for the coreresponding RIC Load Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Load Confirm	M		ENUMERATED (true, ...)	If TRUE then the corresponding load information reporting is confirmed

9.2.46 RIC Service Load Confirm

This information element confirms reporting for the coreresponding RIC Load Information for one or more specific RIC services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Service Report Load Confirm	O		9.2.45	Used to confirm reporting of load information on REPORT service
RIC Service Insert Load Confirm	O		9.2.45	Used to confirm reporting of load information on INSERT service
RIC Service Control Load Confirm	O		9.2.45	Used to confirm reporting of load information on CONTROL service
RIC Service Policy Load Confirm	O		9.2.45	Used to confirm reporting of load information on POLICY service
RIC Service Query Load Confirm	O		9.2.45	Used to confirm reporting of load information on QUERY service

9.2.47 RIC Assistance Header

This information element carries the RIC Assistance Header used to indicate a specific RIC Assistance service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Assistance Header	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.48 RIC Assistance Message

This information element carries the RIC Assistance Message used to request a specific RIC Assistance service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Assistance Message	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.49 RIC Assistance Update

This information element indicates if updates are requested for a RIC Assistance service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Assistance Update	M		ENUMERATED (true, ...)	If TRUE, then the requested RIC Assistance service is to be updated

9.2.50 RIC Assistance Update Number

This information element indicates the maximum number of updates for a requested RIC Assistance service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Assistance Update Number	M		INTEGER (0..65535)	

9.2.51 RIC Assistance Outcome

This information element carries the RIC Assistance Outcome used to provide a specific RIC Assistance service.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Assistance Outcome	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.52 RIC Assistance SN

This information element indicates the RIC Assistance SN (Sequence Number).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Assistance SN	M		INTEGER (0..65535)	

9.2.53 Service Level Cause

This information element carries the Service Level Cause message.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Service Level Cause	M		OCTET STRING	Defined in RAN Function specific E2 Service Model, see O-RAN WG3.TS.E2SM [3].

9.2.54 TNL Capacity Indicator

This IE is defined in [3] clause 6.2.3.38.

9.2.55 Hardware Load Indicator

This IE is defined in [3] clause 6.2.3.39.

9.3 Message and Information Element Abstract Syntax (with ASN.1)

9.3.1 General

E2AP ASN.1 definition conforms to Recommendation ITU-T X.691 [15], Recommendation ITU-T X.680 [16] and Recommendation ITU-T X.681 [17].

Sub clause 9.3 presents the Abstract Syntax of the E2AP protocol with ASN.1. In case there is contradiction between the ASN.1 definition in this sub clause and the tabular format in sub clause 9.1 and 9.2, the ASN.1 shall take precedence, except for the definition of conditions for the presence of conditional elements, in which the tabular format shall take precedence.

The ASN.1 definition specifies the structure and content of E2AP messages. E2AP messages can contain any IEs specified in the object set definitions for that message without the order or number of occurrences being restricted by ASN.1. However, for this version of the standard, a sending entity shall construct an E2AP message according to the PDU definitions module and with the following additional rules:

- IEs shall be ordered (in an IE container) in the order they appear in object set definitions.
- Object set definitions specify how many times IEs may appear. An IE shall appear exactly once if the presence field in an object has value "mandatory". An IE may appear at most once if the presence field in an object has value "optional" or "conditional". If in a tabular format there is multiplicity specified for an IE (i.e., an IE list) then in the corresponding ASN.1 definition the list definition is separated into two parts. The first part defines an IE container list where the list elements reside. The second part defines list elements. The IE container list appears as an IE of its own. For this version of the standard an IE container list may contain only one kind of list elements.

NOTE: In the above "IE" means an IE in the object set with an explicit ID. If one IE needs to appear more than once in one object set, then the different occurrences will have different IE IDs.

If an E2AP message that is not constructed as defined above is received, this shall be considered as Abstract Syntax Error, and the message shall be handled as defined for Abstract Syntax Error in clause 10.

9.3.2 Usage of private message mechanism for non-standard use

The private message mechanism for non-standard use is not supported with E2AP.

9.3.3 Elementary Procedure definitions

```
-- ASN1START
-- *****
--
-- Elementary Procedure definitions
-- Derived from 3GPP 38.413 NGAP
-- *****

E2AP-PDU-Descriptions {
iso(1) identified-organization(3) dod(6) internet(1) private(4) enterprise(1) 53148 e2(1) version2 (2)
e2ap(1) e2ap-PDU-Descriptions (0) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- *****
--
-- IE parameter types from other modules.
--
-- *****

IMPORTS
    Criticality,
    ProcedureCode
FROM E2AP-CommonDataTypes

    E2connectionUpdate,
    E2connectionUpdateAcknowledge,
    E2connectionUpdateFailure,
    E2nodeConfigurationUpdate,
    E2nodeConfigurationUpdateAcknowledge,
    E2nodeConfigurationUpdateFailure,
    E2ResourceStatusRequest,
    E2ResourceStatusResponse,
    E2ResourceStatusFailure,
    E2ResourceStatusUpdate,
    E2RemovalRequest,
    E2RemovalResponse,
    E2RemovalFailure,
    E2setupFailure,
    E2setupRequest,
    E2setupResponse,
    ErrorIndication,
```

```

ResetRequest,
ResetResponse,
RICassistanceFailure,
RICassistanceIndication,
RICassistanceHalt,
RICassistanceRequest,
RICassistanceResponse,
RICcontrolAcknowledge,
RICcontrolFailure,
RICcontrolRequest,
RICindication,
RICserviceLoadStatusRequest,
RICserviceLoadStatusResponse,
RICserviceLoadStatusFailure,
RICserviceLoadUpdate,
RICserviceQuery,
RICserviceUpdate,
RICserviceUpdateAcknowledge,
RICserviceUpdateFailure,
RICsubscriptionFailure,
RICsubscriptionRequest,
RICsubscriptionResponse,
RICsubscriptionAuditFailure,
RICsubscriptionAuditRequest,
RICsubscriptionAuditResponse,
RICsubscriptionDeleteFailure,
RICsubscriptionDeleteRequest,
RICsubscriptionDeleteResponse,
RICsubscriptionDeleteRequired,
RICsubscriptionModificationRequest,
RICsubscriptionModificationResponse,
RICsubscriptionModificationFailure,
RICsubscriptionModificationRequired,
RICsubscriptionModificationConfirm,
RICsubscriptionModificationRefuse,
RICsubscriptionStateControlRequest,
RICsubscriptionStateControlResponse,
RICsubscriptionStateControlFailure,
RICqueryRequest,
RICqueryResponse,
RICqueryFailure
FROM E2AP-PDU-Contents

id-E2connectionUpdate,
id-E2nodeConfigurationUpdate,
id-E2removal,
id-E2resourceStatusReportingInitiation,
id-E2resourceStatusUpdate,
id-E2setup,
id-ErrorIndication,
id-Reset,
id-RICassistance,
id-RICassistanceIndication,
id-RICassistanceHalt,
id-RICcontrol,
id-RICindication,
id-RICserviceLoadStatus,
id-RICserviceLoadUpdate,
id-RICserviceQuery,
id-RICserviceUpdate,
id-RICsubscription,
id-RICsubscriptionAudit,
id-RICsubscriptionDelete,
id-RICsubscriptionDeleteRequired,
id-RICsubscriptionModification,
id-RICsubscriptionModificationRequired,
id-RICsubscriptionStateControl,
id-RICquery
FROM E2AP-Constants;

```

```

-- *****
--
-- Interface Elementary Procedure Class
--
-- *****

```

```

E2AP-ELEMENTARY-PROCEDURE ::= CLASS {
    &InitiatingMessage                                ,
    &SuccessfulOutcome                                OPTIONAL ,

```

```

        &UnsuccessfulOutcome      OPTIONAL ,
        &procedureCode            ProcedureCode UNIQUE ,
        &criticality               Criticality   DEFAULT ignore
    }

WITH SYNTAX {
    INITIATING MESSAGE      &InitiatingMessage
    [SUCCESSFUL OUTCOME     &SuccessfulOutcome]
    [UNSUCCESSFUL OUTCOME   &UnsuccessfulOutcome]
    PROCEDURE CODE          &procedureCode
    [CRITICALITY            &criticality]
}

-- *****
--
-- Interface PDU Definition
--
-- *****

E2AP-PDU ::= CHOICE {
    initiatingMessage      InitiatingMessage,
    successfulOutcome      SuccessfulOutcome,
    unsuccessfulOutcome     UnsuccessfulOutcome,
    ...
}

InitiatingMessage ::= SEQUENCE {
    procedureCode      E2AP-ELEMENTARY-PROCEDURE.&procedureCode      ({E2AP-ELEMENTARY-PROCEDURES}),
    criticality        E2AP-ELEMENTARY-PROCEDURE.&criticality        ({E2AP-ELEMENTARY-
PROCEDURES}{@procedureCode}),
    value              E2AP-ELEMENTARY-PROCEDURE.&InitiatingMessage  ({E2AP-ELEMENTARY-
PROCEDURES}{@procedureCode})
}

SuccessfulOutcome ::= SEQUENCE {
    procedureCode      E2AP-ELEMENTARY-PROCEDURE.&procedureCode      ({E2AP-ELEMENTARY-PROCEDURES}),
    criticality        E2AP-ELEMENTARY-PROCEDURE.&criticality        ({E2AP-ELEMENTARY-
PROCEDURES}{@procedureCode}),
    value              E2AP-ELEMENTARY-PROCEDURE.&SuccessfulOutcome  ({E2AP-ELEMENTARY-
PROCEDURES}{@procedureCode})
}

UnsuccessfulOutcome ::= SEQUENCE {
    procedureCode      E2AP-ELEMENTARY-PROCEDURE.&procedureCode      ({E2AP-ELEMENTARY-PROCEDURES}),
    criticality        E2AP-ELEMENTARY-PROCEDURE.&criticality        ({E2AP-ELEMENTARY-
PROCEDURES}{@procedureCode}),
    value              E2AP-ELEMENTARY-PROCEDURE.&UnsuccessfulOutcome ({E2AP-ELEMENTARY-
PROCEDURES}{@procedureCode})
}

-- *****
--
-- Interface Elementary Procedure List
--
-- *****

E2AP-ELEMENTARY-PROCEDURES E2AP-ELEMENTARY-PROCEDURE ::= {
    E2AP-ELEMENTARY-PROCEDURES-CLASS-1      |
    E2AP-ELEMENTARY-PROCEDURES-CLASS-2,
    ...
}

E2AP-ELEMENTARY-PROCEDURES-CLASS-1 E2AP-ELEMENTARY-PROCEDURE ::= {
    ricSubscription                |
    ricSubscriptionAudit           |
    ricSubscriptionDelete          |
    ricSubscriptionModification    |
    ricSubscriptionModificationRequired |
    ricSubscriptionStateControl    |
    ricQuery                      |
    ricServiceUpdate               |
    ricAssistance                  |
    ricControl                     |
    ricServiceLoadStatus           |
    e2setup                        |
    e2nodeConfigurationUpdate      |
    e2connectionUpdate             |
    reset                          |
    e2removal                      |
}

```

```

    e2resourceStatusReportingInitiation,
    ...
}

E2AP-ELEMENTARY-PROCEDURES-CLASS-2 E2AP-ELEMENTARY-PROCEDURE ::= {
    ricAssistanceIndication          |
    ricAssistanceHalt                |
    ricIndication                    |
    ricServiceQuery                  |
    ricServiceLoadUpdate              |
    errorIndication                  |
    ricSubscriptionDeleteRequired     |
    e2resourceStatusUpdate,          |
    ...
}

-- *****
--
-- Interface Elementary Procedures
--
-- *****

-- New for v01.01
e2connectionUpdate E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E2connectionUpdate
    SUCCESSFUL OUTCOME      E2connectionUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    E2connectionUpdateFailure
    PROCEDURE CODE          id-E2connectionUpdate
    CRITICALITY              reject
}

e2nodeConfigurationUpdate E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E2nodeConfigurationUpdate
    SUCCESSFUL OUTCOME      E2nodeConfigurationUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    E2nodeConfigurationUpdateFailure
    PROCEDURE CODE          id-E2nodeConfigurationUpdate
    CRITICALITY              reject
}

-- New for v02.01
e2removal E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E2RemovalRequest
    SUCCESSFUL OUTCOME      E2RemovalResponse
    UNSUCCESSFUL OUTCOME    E2RemovalFailure
    PROCEDURE CODE          id-E2removal
    CRITICALITY              reject
}

e2setup E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E2setupRequest
    SUCCESSFUL OUTCOME      E2setupResponse
    UNSUCCESSFUL OUTCOME    E2setupFailure
    PROCEDURE CODE          id-E2setup
    CRITICALITY              reject
}

errorIndication E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ErrorIndication
    PROCEDURE CODE          id-ErrorIndication
    CRITICALITY              ignore
}

e2resourceStatusReportingInitiation E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E2ResourceStatusRequest
    SUCCESSFUL OUTCOME      E2ResourceStatusResponse
    UNSUCCESSFUL OUTCOME    E2ResourceStatusFailure
    PROCEDURE CODE          id-E2resourceStatusReportingInitiation
    CRITICALITY              reject
}

e2resourceStatusUpdate E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      E2ResourceStatusUpdate
    PROCEDURE CODE          id-E2resourceStatusUpdate
    CRITICALITY              ignore
}

reset E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      ResetRequest
    SUCCESSFUL OUTCOME      ResetResponse
}

```

```

PROCEDURE CODE      id-Reset
CRITICALITY         reject
}

ricAssistance E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICassistanceRequest
    SUCCESSFUL OUTCOME       RICassistanceResponse
    UNSUCCESSFUL OUTCOME    RICassistanceFailure
    PROCEDURE CODE          id-RIcAssistance
    CRITICALITY              reject
}

ricAssistanceIndication E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICassistanceIndication
    PROCEDURE CODE          id-RIcAssistanceIndication
    CRITICALITY              reject
}

ricAssistanceHalt E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICassistanceHalt
    PROCEDURE CODE          id-RIcAssistanceHalt
    CRITICALITY              reject
}

ricControl E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICcontrolRequest
    SUCCESSFUL OUTCOME       RICcontrolAcknowledge
    UNSUCCESSFUL OUTCOME    RICcontrolFailure
    PROCEDURE CODE          id-RIcControl
    CRITICALITY              reject
}

ricIndication E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICindication
    PROCEDURE CODE          id-RIcIndication
    CRITICALITY              ignore
}

ricServiceLoadStatus E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICserviceLoadStatusRequest
    SUCCESSFUL OUTCOME       RICserviceLoadStatusResponse
    UNSUCCESSFUL OUTCOME    RICserviceLoadStatusFailure
    PROCEDURE CODE          id-RIcServiceLoadStatus
    CRITICALITY              ignore
}

ricServiceLoadUpdate E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICserviceLoadUpdate
    PROCEDURE CODE          id-RIcServiceLoadUpdate
    CRITICALITY              ignore
}

ricServiceQuery E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICserviceQuery
    PROCEDURE CODE          id-RIcServiceQuery
    CRITICALITY              ignore
}

ricServiceUpdate E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICserviceUpdate
    SUCCESSFUL OUTCOME       RICserviceUpdateAcknowledge
    UNSUCCESSFUL OUTCOME    RICserviceUpdateFailure
    PROCEDURE CODE          id-RIcServiceUpdate
    CRITICALITY              reject
}

ricSubscription E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICsubscriptionRequest
    SUCCESSFUL OUTCOME       RICsubscriptionResponse
    UNSUCCESSFUL OUTCOME    RICsubscriptionFailure
    PROCEDURE CODE          id-RIcSubscription
    CRITICALITY              reject
}

ricSubscriptionAudit E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE      RICsubscriptionAuditRequest
    SUCCESSFUL OUTCOME       RICsubscriptionAuditResponse
    UNSUCCESSFUL OUTCOME    RICsubscriptionAuditFailure
    PROCEDURE CODE          id-RIcSubscriptionAudit

```



```

    CRITICALITY                reject
}

ricSubscriptionDelete E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE        RICsubscriptionDeleteRequest
    SUCCESSFUL OUTCOME        RICsubscriptionDeleteResponse
    UNSUCCESSFUL OUTCOME      RICsubscriptionDeleteFailure
    PROCEDURE CODE            id-RICsubscriptionDelete
    CRITICALITY                reject
}

ricSubscriptionDeleteRequired E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE        RICsubscriptionDeleteRequired
    PROCEDURE CODE            id-RICsubscriptionDeleteRequired
    CRITICALITY                ignore
}

ricSubscriptionModification E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE        RICsubscriptionModificationRequest
    SUCCESSFUL OUTCOME        RICsubscriptionModificationResponse
    UNSUCCESSFUL OUTCOME      RICsubscriptionModificationFailure
    PROCEDURE CODE            id-RICsubscriptionModification
    CRITICALITY                reject
}

ricSubscriptionModificationRequired E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE        RICsubscriptionModificationRequired
    SUCCESSFUL OUTCOME        RICsubscriptionModificationConfirm
    UNSUCCESSFUL OUTCOME      RICsubscriptionModificationRefuse
    PROCEDURE CODE            id-RICsubscriptionModificationRequired
    CRITICALITY                reject
}

ricSubscriptionStateControl E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE        RICsubscriptionStateControlRequest
    SUCCESSFUL OUTCOME        RICsubscriptionStateControlResponse
    UNSUCCESSFUL OUTCOME      RICsubscriptionStateControlFailure
    PROCEDURE CODE            id-RICsubscriptionStateControl
    CRITICALITY                reject
}

ricQuery E2AP-ELEMENTARY-PROCEDURE ::= {
    INITIATING MESSAGE        RICqueryRequest
    SUCCESSFUL OUTCOME        RICqueryResponse
    UNSUCCESSFUL OUTCOME      RICqueryFailure
    PROCEDURE CODE            id-RICquery
    CRITICALITY                reject
}

END
-- ASN1STOP

```

9.3.4 PDU definitions

```

-- ASN1START
-- *****
--
-- PDU definitions for E2AP
-- Derived from 3GPP 38.413 (NGAP)
--
-- *****

E2AP-PDU-Contents {
iso(1) identified-organization(3) dod(6) internet(1) private(4) enterprise(1) 53148 e2(1) version2 (2)
e2ap(1) e2ap-PDU-Contents (1) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- *****
--
-- IE parameter types from other modules.
--
-- *****

IMPORTS

```

```
HardwareLoadIndicator,
TNLCapacityIndicator
FROM E2SM-COMMON-IES
```

```
Cause,
CriticalityDiagnostics,
E2nodeComponentConfiguration,
E2nodeComponentConfigurationAck,
E2nodeComponentID,
E2nodeComponentInterfaceType,
GlobalE2node-ID,
GlobalRIC-ID,
LoadMeasurementID,
RANfunctionDefinition,
RANfunctionID,
RANfunctionOID,
RANfunctionRevision,
RegistrationRequest,
RegistrationRequest-E2,
ReportCharacteristics,
ReportingPeriodicity,
ReportingPeriodicity-E2,
RICactionDefinition,
RICactionExecutionOrder,
RICactionID,
RICactionType,
RICassistanceHeader,
RICassistanceMessage,
RICassistanceOutcome,
RICassistanceSN,
RICassistanceUpdate,
RICassistanceUpdateNumber,
RICcallProcessID,
RICcontrolAckRequest,
RICcontrolHeader,
RICcontrolMessage,
RICcontrolOutcome,
RICeventTriggerDefinition,
RICindicationHeader,
RICindicationMessage,
RICindicationSN,
RICindicationType,
RICloadConfirm,
RICloadInformation,
RICloadRequest,
RICqueryHeader,
RICqueryDefinition,
RICqueryOutcome,
RICrequestID,
RICserviceLoadConfirm,
RICserviceLoadInformation,
RICserviceLoadRequest,
RICsubsequentAction,
RICsubscriptionTime,
RICsubscriptionAuditFlag,
RICtimeToWait,
TimeToWait,
TNLinformation,
TNLusage,
TransactionID
FROM E2AP-IES
```

```
ProtocolIE-Container{},
ProtocolIE-ContainerList{},
ProtocolIE-SingleContainer{},
E2AP-PROTOCOL-IES,
E2AP-PROTOCOL-IES-PAIR
FROM E2AP-Containers
```

```
id-Cause,
id-CriticalityDiagnostics,
id-E2connectionSetup,
id-E2connectionSetupFailed,
id-E2connectionSetupFailed-Item,
id-E2connectionFailed-Item,
id-E2connectionUpdate-Item,
id-E2connectionUpdateAdd,
```

id-E2connectionUpdateModify,
 id-E2connectionUpdateRemove,
 id-E2connectionUpdateRemove-Item,
 id-E2nodeComponentConfigAddition,
 id-E2nodeComponentConfigAddition-Item,
 id-E2nodeComponentConfigAdditionAck,
 id-E2nodeComponentConfigAdditionAck-Item,
 id-E2nodeComponentConfigRemoval,
 id-E2nodeComponentConfigRemoval-Item,
 id-E2nodeComponentConfigRemovalAck,
 id-E2nodeComponentConfigRemovalAck-Item,
 id-E2nodeComponentConfigUpdate,
 id-E2nodeComponentConfigUpdate-Item,
 id-E2nodeComponentConfigUpdateAck,
 id-E2nodeComponentConfigUpdateAck-Item,
 id-E2nodeLoadMeasurementID,
 id-E2nodeTNLassociationRemoval,
 id-E2nodeTNLassociationRemoval-Item,
 id-E2resourceStatusReportingInitiation,
 id-E2resourceStatusUpdate,
 id-GlobalE2node-ID,
 id-GlobalRIC-ID,
 id-HardwareLoadIndicator,
 id-RANfunctionID,
 id-RANfunctionID-Item,
 id-RANfunctionIEcause-Item,
 id-RANfunction-Item,
 id-RANfunctionLoad-Item,
 id-RANfunctionLoad-List,
 id-RANfunctionLoadConfirm-Item,
 id-RANfunctionLoadConfirm-List,
 id-RANfunctionLoadRequest-Item,
 id-RANfunctionLoadRequest-List,
 id-RANfunctionsAccepted,
 id-RANfunctionsAcceptModified-List,
 id-RANfunction-AcceptModified-Item,
 id-RANfunctionsAdded,
 id-RANfunctionsDeleted,
 id-RANfunctionsModified,
 id-RANfunctionsRejected,
 id-RANfunctionStateConfirm-Item,
 id-RANfunctionStateConfirm-List,
 id-RANfunctionStateControl-Item,
 id-RANfunctionStateControl-List,
 id-RegistrationRequest,
 id-RegistratioRequest-E2,
 id-ReportingPeriodicity,
 id-ReportingPeriodicity-E2,
 id-ReportCharacteristics,
 id-RICaction-Admitted-Item,
 id-RICactionID,
 id-RICaction-NotAdmitted-Item,
 id-RICactions-Admitted,
 id-RICactions-NotAdmitted,
 id-RICaction-ToBeSetup-Item,
 id-RICactionsToBeRemovedForModification-List,
 id-RICaction-ToBeRemovedForModification-Item,
 id-RICactionsToBeModifiedForModification-List,
 id-RICaction-ToBeModifiedForModification-Item,
 id-RICactionsToBeAddedForModification-List,
 id-RICaction-ToBeAddedForModification-Item,
 id-RICactionsRemovedForModification-List,
 id-RICaction-RemovedForModification-Item,
 id-RICactionsFailedToBeRemovedForModification-List,
 id-RICaction-FailedToBeRemovedForModification-Item,
 id-RICactionsModifiedForModification-List,
 id-RICaction-ModifiedForModification-Item,
 id-RICactionsFailedToBeModifiedForModification-List,
 id-RICaction-FailedToBeModifiedForModification-Item,
 id-RICactionsAddedForModification-List,
 id-RICaction-AddedForModification-Item,
 id-RICactionsFailedToBeAddedForModification-List,
 id-RICaction-FailedToBeAddedForModification-Item,
 id-RICactionsRequiredToBeModified-List,
 id-RICaction-RequiredToBeModified-Item,
 id-RICactionsRequiredToBeRemoved-List,
 id-RICaction-RequiredToBeRemoved-Item,
 id-RICactionsConfirmedForModification-List,
 id-RICaction-ConfirmedForModification-Item,

```

id-RIActionList-Item,
id-RIActionLoad-Item,
id-RIActionLoadConfirm-Item,
id-RIActionLoadRequest-Item,
id-RIActionsRefusedToBeModified-List,
id-RIAction-RefusedToBeModified-Item,
id-RIActionsConfirmedForRemoval-List,
id-RIAction-ConfirmedForRemoval-Item,
id-RIActionsRefusedToBeRemoved-List,
id-RIAction-RefusedToBeRemoved-Item,
id-RICAssistanceHeader,
id-RICAssistanceMessage,
id-RICAssistanceOutcome,
id-RICAssistanceSN,
id-RICAssistanceUpdate,
id-RICAssistanceUpdateNumber,
id-RICcallProcessID,
id-RICcontrolAckRequest,
id-RICcontrolHeader,
id-RICcontrolMessage,
id-RICcontrolOutcome,
id-RIceventTriggerDefinitionToBeModified,
id-RICindicationHeader,
id-RICindicationMessage,
id-RICindicationSN,
id-RICindicationType,
id-RICloadMeasurementID,
id-RICrequestID,
id-RICserviceQuery,
id-RICsubscriptionAuditFlag,
id-RICsubscriptionAuditList,
id-RICsubscriptionAudit-Item,
id-RICsubscriptionAuditAction-Item,
id-RICsubscriptionAuditConfirmedList,
id-RICsubscriptionAuditMissingList,
id-RICsubscriptionAuditUnkownList,
id-RICsubscriptionDetails,
id-RICsubscriptionList-Item,
id-RICsubscriptionLoad-Item,
id-RICsubscriptionLoadConfirm-Item,
id-RICsubscriptionLoadRequest-Item,
id-RICsubscriptionToBeRemoved,
id-RICsubscription-withCause-Item,
id-RICsubscriptionStartTime,
id-RICsubscriptionEndTime,
id-RICqueryHeader,
id-RICqueryDefinition,
id-RICqueryOutcome,
id-TimeToWait,
id-TNLCapacityIndicator,
id-TNLInformation,
id-TransactionID,

maxofE2nodeComponents,
maxofRANfunctionID,
maxofRIActionID,
maxofRICrequestID,
maxofRICsubscriptions,
maxofTNLA
FROM E2AP-Constants;

```

```

-- *****
--
-- MESSAGES FOR RIC FUNCTIONAL PROCEDURES
--
-- *****
--
-- *****
--
-- RIC Subscription Elementary Procedure
--
-- *****
-- *****
--
-- RIC SUBSCRIPTION REQUEST
--
-- *****
RICsubscriptionRequest ::= SEQUENCE {

```

```

    protocolIEs          ProtocolIE-Container    {{RICsubscriptionRequest-IEs}},
    ...
}

RICsubscriptionRequest-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcrequestID          CRITICALITY reject    TYPE RICrequestID          PRESENCE
mandatory } |
    { ID id-RANfunctionID        CRITICALITY reject    TYPE RANfunctionID        PRESENCE
mandatory } |
    { ID id-RICsubscriptionDetails CRITICALITY reject    TYPE RICsubscriptionDetails PRESENCE
mandatory },
    ...,
    { ID id-RICsubscriptionStartTime CRITICALITY reject    TYPE RICsubscriptionTime    PRESENCE
optional } |
    { ID id-RICsubscriptionEndTime   CRITICALITY reject    TYPE RICsubscriptionTime    PRESENCE
optional }
}

RICsubscriptionDetails ::= SEQUENCE {
    ricEventTriggerDefinition    RICeventTriggerDefinition,
    ricAction-ToBeSetup-List     RICactions-ToBeSetup-List,
    ...
}

RICactions-ToBeSetup-List ::= SEQUENCE (SIZE(1..maxofRICactionID)) OF ProtocolIE-SingleContainer {
{RICaction-ToBeSetup-ItemIEs} }

RICaction-ToBeSetup-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIaction-ToBeSetup-Item CRITICALITY ignore    TYPE RICaction-ToBeSetup-Item PRESENCE
mandatory },
    ...
}

RICaction-ToBeSetup-Item ::= SEQUENCE {
    ricActionID                RICactionID,
    ricActionType              RICactionType,
    ricActionDefinition        RICactionDefinition OPTIONAL,
    ricSubsequentAction        RICsubsequentAction OPTIONAL,
    ...,
    ricActionExecutionOrder    RICactionExecutionOrder OPTIONAL -- New in E2APv03.00
}

-- *****
--
-- RIC SUBSCRIPTION RESPONSE
--
-- *****
RICsubscriptionResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container{{RICsubscriptionResponse-IEs}},
    ...
}

RICsubscriptionResponse-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcrequestID          CRITICALITY reject    TYPE RICrequestID          PRESENCE
mandatory } |
    { ID id-RANfunctionID        CRITICALITY reject    TYPE RANfunctionID        PRESENCE
mandatory } |
    { ID id-RIactions-Admitted    CRITICALITY reject    TYPE RICaction-Admitted-List PRESENCE
mandatory } |
    { ID id-RIactions-NotAdmitted CRITICALITY reject    TYPE RICaction-NotAdmitted-List PRESENCE
optional },
    ...
}

RICaction-Admitted-List ::= SEQUENCE (SIZE(1..maxofRICactionID)) OF ProtocolIE-SingleContainer{{RICaction-
Admitted-ItemIEs}}

RICaction-Admitted-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIaction-Admitted-Item CRITICALITY ignore    TYPE RICaction-Admitted-Item PRESENCE
mandatory },
    ...
}

RICaction-Admitted-Item ::= SEQUENCE {
    ricActionID                RICactionID,

```

```

...
}

RICAction-NotAdmitted-List ::= SEQUENCE (SIZE(0..maxofRICActionID)) OF ProtocolIE-SingleContainer {
{RICAction-NotAdmitted-ItemIEs} }

RICAction-NotAdmitted-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIAction-NotAdmitted-Item CRITICALITY ignore TYPE RICAction-NotAdmitted-Item PRESENCE
mandatory },
    ...
}

RICAction-NotAdmitted-Item ::= SEQUENCE {
    ricActionID          RICActionID,
    cause                Cause,
    ...
}

-- *****
--
-- RIC SUBSCRIPTION FAILURE
--
-- *****
RICSubscriptionFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICSubscriptionFailure-IEs}},
    ...
}

RICSubscriptionFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID          CRITICALITY reject TYPE RIRequestID          PRESENCE
mandatory }|
    { ID id-RANfunctionID        CRITICALITY reject TYPE RANfunctionID        PRESENCE
mandatory }|
    { ID id-Cause                CRITICALITY reject TYPE Cause                PRESENCE mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE
optional },
    ...
}

-- *****
--
-- RIC Subscription Audit Elementary Procedure
--
-- *****
-- *****
-- RIC SUBSCRIPTION AUDIT REQUEST
--
-- *****
RICSubscriptionAuditRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICSubscriptionAuditRequest-IEs}},
    ...
}

RICSubscriptionAuditRequest-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID          CRITICALITY reject TYPE RIRequestID          PRESENCE
mandatory }|
    { ID id-RISubscriptionAuditFlag CRITICALITY reject TYPE RISubscriptionAuditFlag PRESENCE
optional }|
    { ID id-RISubscriptionAuditList CRITICALITY reject TYPE RISubscriptionAuditList PRESENCE
optional },
    ...
}

RICSubscriptionAuditList ::= SEQUENCE (SIZE(1..maxofRICsubscriptions)) OF ProtocolIE-SingleContainer {
{RICSubscriptionAudit-ItemIEs} }

RICSubscriptionAudit-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RISubscriptionAudit-Item CRITICALITY ignore TYPE RISubscriptionAudit-Item PRESENCE
mandatory },
    ...
}

RICSubscriptionAudit-Item ::= SEQUENCE {
    ricRequestID          RIRequestID,
    ranFunctionID         RANfunctionID,
    ...
}

```

```
-- *****
--
-- RIC SUBSCRIPTION AUDIT RESPONSE
--
-- *****
RICsubscriptionAuditResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container({RICsubscriptionAuditResponse-IEs}),
    ...
}

RICsubscriptionAuditResponse-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID          CRITICALITY reject          TYPE RIRequestID
      PRESENCE mandatory } |
    { ID id-RICsubscriptionAuditConfirmedList CRITICALITY reject          TYPE
RICsubscriptionAuditActionList PRESENCE mandatory } |
    { ID id-RICsubscriptionAuditUnkownList    CRITICALITY reject          TYPE RICsubscriptionAuditList
      PRESENCE optional } |
    { ID id-RICsubscriptionAuditMissingList   CRITICALITY reject          TYPE
RICsubscriptionAuditActionList PRESENCE optional },
    ...
}

RICsubscriptionAuditActionList ::= SEQUENCE (SIZE(1..maxofRICsubscriptions)) OF ProtocolIE-SingleContainer
{ {RICsubscriptionAuditAction-ItemIEs} }

RICsubscriptionAuditAction-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICsubscriptionAuditAction-Item CRITICALITY ignore TYPE RICsubscriptionAuditAction-Item
      PRESENCE mandatory },
    ...
}

RICsubscriptionAuditAction-Item ::= SEQUENCE {
    ricRequestID          RIRequestID,
    ranFunctionID          RANfunctionID,
    ricAction-Admitted-List RICAction-Admitted-List,
    ...
}

-- *****
--
-- RIC SUBSCRIPTION AUDIT FAILURE
--
-- *****
RICsubscriptionAuditFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    ({RICsubscriptionAuditFailure-IEs}),
    ...
}

RICsubscriptionAuditFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID          CRITICALITY reject TYPE RIRequestID          PRESENCE
mandatory } |
    { ID id-Cause                CRITICALITY reject TYPE Cause                PRESENCE
mandatory } |
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE
optional },
    ...
}

-- *****
--
-- RIC Subscription Delete Elementary Procedure
--
-- *****
--
-- RIC SUBSCRIPTION DELETE REQUEST
--
-- *****
RICsubscriptionDeleteRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    ({RICsubscriptionDeleteRequest-IEs}),
    ...
}
```

```

RICsubscriptionDeleteRequest-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID                CRITICALITY reject  TYPE RIRequestID                PRESENCE
mandatory    }|
    { ID id-RANfunctionID              CRITICALITY reject  TYPE RANfunctionID              PRESENCE
mandatory    },
    ...
}

-- *****
--
-- RIC SUBSCRIPTION DELETE RESPONSE
--
-- *****
RICsubscriptionDeleteResponse ::= SEQUENCE {
    protocolIEs                ProtocolIE-Container    {{RICsubscriptionDeleteResponse-IEs}},
    ...
}

RICsubscriptionDeleteResponse-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID                CRITICALITY reject  TYPE RIRequestID                PRESENCE
mandatory    }|
    { ID id-RANfunctionID              CRITICALITY reject  TYPE RANfunctionID              PRESENCE
mandatory    },
    ...
}

-- *****
--
-- RIC SUBSCRIPTION DELETE FAILURE
--
-- *****
RICsubscriptionDeleteFailure ::= SEQUENCE {
    protocolIEs                ProtocolIE-Container    {{RICsubscriptionDeleteFailure-IEs}},
    ...
}

RICsubscriptionDeleteFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID                CRITICALITY reject  TYPE RIRequestID                PRESENCE
mandatory    }|
    { ID id-RANfunctionID              CRITICALITY reject  TYPE RANfunctionID              PRESENCE
mandatory    }|
    { ID id-Cause                      CRITICALITY ignore  TYPE Cause                      PRESENCE
mandatory    }|
    { ID id-CriticalityDiagnostics      CRITICALITY ignore  TYPE CriticalityDiagnostics      PRESENCE
optional    },
    ...
}

-- *****
--
-- RIC Subscription Delete Required Elementary Procedure
--
-- *****
--
-- RIC SUBSCRIPTION DELETE REQUIRED
--
-- *****
RICsubscriptionDeleteRequired ::= SEQUENCE {
    protocolIEs                ProtocolIE-Container    {{RICsubscriptionDeleteRequired-IEs}},
    ...
}

RICsubscriptionDeleteRequired-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RISubscriptionToBeRemoved    CRITICALITY ignore  TYPE RISubscription-List-withCause
PRESENCE mandatory    },
    ...
}

RICsubscription-List-withCause ::= SEQUENCE (SIZE(1..maxofRIRequestID)) OF ProtocolIE-SingleContainer {
{{RICsubscription-withCause-ItemIEs} }

RICsubscription-withCause-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RISubscription-withCause-Item CRITICALITY ignore  TYPE RISubscription-withCause-Item
PRESENCE mandatory    },
    ...
}

RICsubscription-withCause-Item ::= SEQUENCE {

```



```

    ricRequestID          RICrequestID,
    ranFunctionID         RANfunctionID,
    cause                 Cause,
    ...
}

-- *****
--
-- RIC Subscription Modification Elementary Procedure
--
-- *****
-- *****
--
-- RIC SUBSCRIPTION MODIFICATION REQUEST
--
-- *****
RICsubscriptionModificationRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICsubscriptionModificationRequest-IEs}},
    ...
}

RICsubscriptionModificationRequest-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICrequestID          CRITICALITY reject  TYPE RICrequestID
      PRESENCE mandatory}|
    { ID id-RANfunctionID        CRITICALITY reject  TYPE RANfunctionID
      PRESENCE mandatory}|
    { ID id-RICeventTriggerDefinitionToBeModified CRITICALITY ignore  TYPE RICeventTriggerDefinition
      PRESENCE optional}|
    { ID id-RICactionsToBeRemovedForModification-List CRITICALITY ignore  TYPE RICactions-
      ToBeRemovedForModification-List PRESENCE optional}|
    { ID id-RICactionsToBeModifiedForModification-List CRITICALITY ignore  TYPE RICactions-
      ToBeModifiedForModification-List PRESENCE optional}|
    { ID id-RICactionsToBeAddedForModification-List CRITICALITY ignore  TYPE RICactions-
      ToBeAddedForModification-List PRESENCE optional},
    ...
}

RICactions-ToBeRemovedForModification-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-
SingleContainer {{RICaction-ToBeRemovedForModification-ItemIEs} }

RICaction-ToBeRemovedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICaction-ToBeRemovedForModification-Item CRITICALITY ignore  TYPE RICaction-
      ToBeRemovedForModification-Item PRESENCE mandatory },
    ...
}

RICaction-ToBeRemovedForModification-Item ::= SEQUENCE {
    ricActionID          RICactionID,
    ...
}

RICactions-ToBeModifiedForModification-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-
SingleContainer {{RICaction-ToBeModifiedForModification-ItemIEs} }

RICaction-ToBeModifiedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICaction-ToBeModifiedForModification-Item CRITICALITY ignore  TYPE RICaction-
      ToBeModifiedForModification-Item PRESENCE mandatory },
    ...
}

RICaction-ToBeModifiedForModification-Item ::= SEQUENCE {
    ricActionID          RICactionID,
    ricActionDefinition  RICactionDefinition    OPTIONAL,
    ricActionExecutionOrder RICactionExecutionOrder OPTIONAL,
    ricSubsequentAction  RICsubsequentAction    OPTIONAL,
    ...
}

RICactions-ToBeAddedForModification-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-
SingleContainer {{RICaction-ToBeAddedForModification-ItemIEs} }

RICaction-ToBeAddedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICaction-ToBeAddedForModification-Item CRITICALITY ignore  TYPE RICaction-
      ToBeAddedForModification-Item PRESENCE mandatory },
    ...
}

RICaction-ToBeAddedForModification-Item ::= SEQUENCE {

```

```

ricActionID          RICactionID,
ricActionType        RICactionType,
ricActionDefinition   RICactionDefinition,
ricActionExecutionOrder RICactionExecutionOrder,
ricSubsequentAction   RICsubsequentAction   OPTIONAL,
...
}
-- *****
--
-- RIC SUBSCRIPTION MODIFICATION RESPONSE
--
-- *****
RICsubscriptionModificationResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICsubscriptionModificationResponse-IEs}},
    ...
}

RICsubscriptionModificationResponse-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcrequestID          CRITICALITY reject  TYPE RIcrequestID
      PRESENCE mandatory}}|
    { ID id-RANfunctionID        CRITICALITY reject  TYPE RANfunctionID
      PRESENCE mandatory}}|
    { ID id-RIcActionsRemovedForModification-List CRITICALITY ignore  TYPE RIcActions-
      RemovedForModification-List PRESENCE optional}}|
    { ID id-RIcActionsFailedToBeRemovedForModification-List CRITICALITY ignore  TYPE RIcActions-
      FailedToBeRemovedForModification-List PRESENCE optional}}|
    { ID id-RIcActionsModifiedForModification-List CRITICALITY ignore  TYPE RIcActions-
      ModifiedForModification-List PRESENCE optional}}|
    { ID id-RIcActionsFailedToBeModifiedForModification-List CRITICALITY ignore  TYPE RIcActions-
      FailedToBeModifiedForModification-List PRESENCE optional}}|
    { ID id-RIcActionsAddedForModification-List CRITICALITY ignore  TYPE RIcActions-
      AddedForModification-List PRESENCE optional}}|
    { ID id-RIcActionsFailedToBeAddedForModification-List CRITICALITY ignore  TYPE RIcActions-
      FailedToBeAddedForModification-List PRESENCE optional}},
    ...
}

RIcActions-RemovedForModification-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-
SingleContainer {{RIcAction-RemovedForModification-ItemIEs} }

RIcAction-RemovedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcAction-RemovedForModification-Item CRITICALITY ignore  TYPE RIcAction-
      RemovedForModification-Item PRESENCE mandatory },
    ...
}

RIcAction-RemovedForModification-Item ::= SEQUENCE {
    ricActionID          RICactionID,
    ...
}

RIcActions-FailedToBeRemovedForModification-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-
SingleContainer {{RIcAction-FailedToBeRemovedForModification-ItemIEs} }

RIcAction-FailedToBeRemovedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcAction-FailedToBeRemovedForModification-Item CRITICALITY ignore  TYPE RIcAction-
      FailedToBeRemovedForModification-Item PRESENCE mandatory },
    ...
}

RIcAction-FailedToBeRemovedForModification-Item ::= SEQUENCE {
    ricActionID          RICactionID,
    cause                Cause,
    ...
}

RIcActions-ModifiedForModification-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-
SingleContainer {{RIcAction-ModifiedForModification-ItemIEs} }

RIcAction-ModifiedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcAction-ModifiedForModification-Item CRITICALITY ignore  TYPE RIcAction-
      ModifiedForModification-Item PRESENCE mandatory },
    ...
}

RIcAction-ModifiedForModification-Item ::= SEQUENCE {
    ricActionID          RICactionID,
    ...
}

```

```
RICActions-FailedToBeModifiedForModification-List ::= SEQUENCE (SIZE(0..maxofRICActionID)) OF ProtocolIE-
SingleContainer {{RICAction-FailedToBeModifiedForModification-ItemIEs} }
```

```
RICAction-FailedToBeModifiedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIAction-FailedToBeModifiedForModification-Item CRITICALITY ignore TYPE RICAction-
FailedToBeModifiedForModification-Item PRESENCE mandatory },
    ...
}
```

```
RICAction-FailedToBeModifiedForModification-Item ::= SEQUENCE {
    ricActionID RICActionID,
    cause Cause,
    ...
}
```

```
RICActions-AddedForModification-List ::= SEQUENCE (SIZE(0..maxofRICActionID)) OF ProtocolIE-
SingleContainer {{RICAction-AddedForModification-ItemIEs} }
```

```
RICAction-AddedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIAction-AddedForModification-Item CRITICALITY ignore TYPE RICAction-AddedForModification-
Item PRESENCE mandatory },
    ...
}
```

```
RICAction-AddedForModification-Item ::= SEQUENCE {
    ricActionID RICActionID,
    ...
}
```

```
RICActions-FailedToBeAddedForModification-List ::= SEQUENCE (SIZE(0..maxofRICActionID)) OF ProtocolIE-
SingleContainer {{RICAction-FailedToBeAddedForModification-ItemIEs} }
```

```
RICAction-FailedToBeAddedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIAction-FailedToBeAddedForModification-Item CRITICALITY ignore TYPE RICAction-
FailedToBeAddedForModification-Item PRESENCE mandatory },
    ...
}
```

```
RICAction-FailedToBeAddedForModification-Item ::= SEQUENCE {
    ricActionID RICActionID,
    cause Cause,
    ...
}
```

```
-- *****
```

```
--
-- RIC SUBSCRIPTION MODIFICATION FAILURE
--
```

```
-- *****
```

```
RICsubscriptionModificationFailure ::= SEQUENCE {
    protocolIEs ProtocolIE-Container {{RICsubscriptionModificationFailure-IEs}},
    ...
}
```

```
RICsubscriptionModificationFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcrequestID CRITICALITY reject TYPE RICcrequestID PRESENCE
mandatory}|
    { ID id-RANfunctionID CRITICALITY reject TYPE RANfunctionID PRESENCE
mandatory}|
    { ID id-Cause CRITICALITY reject TYPE Cause PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE
optional},
    ...
}
```

```
-- *****
```

```
--
-- RIC Subscription Modification Required Elementary Procedure
--
```

```
-- *****
```

```
-- *****
```

```
--
```

```
-- RIC SUBSCRIPTION MODIFICATION REQUIRED
--
```

```
-- *****
```

```
RICsubscriptionModificationRequired ::= SEQUENCE {
    protocolIEs ProtocolIE-Container {{RICsubscriptionModificationRequired-IEs}},
    ...
}
```

```

RICsubscriptionModificationRequired-IEs E2AP-PROTOCOL-IES ::= {
  { ID id-RIRequestID                                CRITICALITY reject  TYPE RIRequestID
    PRESENCE mandatory}}|
  { ID id-RANfunctionID                              CRITICALITY reject  TYPE RANfunctionID
    PRESENCE mandatory}}|
  { ID id-RIActionsRequiredToBeModified-List CRITICALITY ignore  TYPE RIActions-RequiredToBeModified-
List    PRESENCE optional}}|
  { ID id-RIActionsRequiredToBeRemoved-List CRITICALITY ignore  TYPE RIActions-RequiredToBeRemoved-
List    PRESENCE optional}},
  ...
}

RIActions-RequiredToBeModified-List ::= SEQUENCE (SIZE(0..maxofRICActionID)) OF ProtocolIE-
SingleContainer {{RICAction-RequiredToBeModified-ItemIEs} }

RICAction-RequiredToBeModified-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-RIAction-RequiredToBeModified-Item CRITICALITY ignore  TYPE RICAction-RequiredToBeModified-
Item    PRESENCE mandatory },
  ...
}

RICAction-RequiredToBeModified-Item ::= SEQUENCE {
  ricActionID          RICActionID,
  ricTimeToWait        RICtimeToWait,
  ...
}

RIActions-RequiredToBeRemoved-List ::= SEQUENCE (SIZE(0..maxofRICActionID)) OF ProtocolIE-SingleContainer
{{RICAction-RequiredToBeRemoved-ItemIEs} }

RICAction-RequiredToBeRemoved-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-RIAction-RequiredToBeRemoved-Item CRITICALITY ignore  TYPE RICAction-RequiredToBeRemoved-
Item    PRESENCE mandatory },
  ...
}

RICAction-RequiredToBeRemoved-Item ::= SEQUENCE {
  ricActionID          RICActionID,
  cause                Cause,
  ...
}

-- *****
--
-- RIC SUBSCRIPTION MODIFICATION CONFIRM
--
-- *****
RICsubscriptionModificationConfirm ::= SEQUENCE {
  protocolIEs          ProtocolIE-Container    {{RICsubscriptionModificationConfirm-IEs}},
  ...
}

RICsubscriptionModificationConfirm-IEs E2AP-PROTOCOL-IES ::= {
  { ID id-RIRequestID                                CRITICALITY reject  TYPE RIRequestID
    PRESENCE mandatory}}|
  { ID id-RANfunctionID                              CRITICALITY reject  TYPE RANfunctionID
    PRESENCE mandatory}}|
  { ID id-RIActionsConfirmedForModification-List CRITICALITY ignore  TYPE RIActions-
ConfirmedForModification-List    PRESENCE optional}}|
  { ID id-RIActionsRefusedToBeModified-List CRITICALITY ignore  TYPE RIActions-
RefusedToBeModified-List    PRESENCE optional}}|
  { ID id-RIActionsConfirmedForRemoval-List CRITICALITY ignore  TYPE RIActions-
ConfirmedForRemoval-List    PRESENCE optional}}|
  { ID id-RIActionsRefusedToBeRemoved-List CRITICALITY ignore  TYPE RIActions-
RefusedToBeRemoved-List    PRESENCE optional}},
  ...
}

RIActions-ConfirmedForModification-List ::= SEQUENCE (SIZE(0..maxofRICActionID)) OF ProtocolIE-
SingleContainer {{RICAction-ConfirmedForModification-ItemIEs} }

RICAction-ConfirmedForModification-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-RIAction-ConfirmedForModification-Item CRITICALITY ignore  TYPE RICAction-
ConfirmedForModification-Item    PRESENCE mandatory },
  ...
}

RICAction-ConfirmedForModification-Item ::= SEQUENCE {

```

```

        ricActionID                RICactionID,
        ...
    }
    RICactions-RefusedToBeModified-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-SingleContainer
    {{RICaction-RefusedToBeModified-ItemIEs} }

    RICaction-RefusedToBeModified-ItemIEs E2AP-PROTOCOL-IES ::= {
        { ID id-RIAction-RefusedToBeModified-Item                CRITICALITY ignore TYPE RICaction-
        RefusedToBeModified-Item PRESENCE mandatory },
        ...
    }

    RICaction-RefusedToBeModified-Item ::= SEQUENCE {
        ricActionID                RICactionID,
        cause                       Cause,
        ...
    }

    RICactions-ConfirmedForRemoval-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-SingleContainer
    {{RICaction-ConfirmedForRemoval-ItemIEs} }

    RICaction-ConfirmedForRemoval-ItemIEs E2AP-PROTOCOL-IES ::= {
        { ID id-RIAction-ConfirmedForRemoval-Item                CRITICALITY ignore TYPE RICaction-
        ConfirmedForRemoval-Item PRESENCE mandatory },
        ...
    }

    RICaction-ConfirmedForRemoval-Item ::= SEQUENCE {
        ricActionID                RICactionID,
        ...
    }

    RICactions-RefusedToBeRemoved-List ::= SEQUENCE (SIZE(0..maxofRICactionID)) OF ProtocolIE-SingleContainer
    {{RICaction-RefusedToBeRemoved-ItemIEs} }

    RICaction-RefusedToBeRemoved-ItemIEs E2AP-PROTOCOL-IES ::= {
        { ID id-RIAction-RefusedToBeRemoved-Item                CRITICALITY ignore TYPE RICaction-
        RefusedToBeRemoved-Item PRESENCE mandatory },
        ...
    }

    RICaction-RefusedToBeRemoved-Item ::= SEQUENCE {
        ricActionID                RICactionID,
        cause                       Cause,
        ...
    }

-- *****
--
-- RIC SUBSCRIPTION MODIFICATION REFUSE
--
-- *****
RICsubscriptionModificationRefuse ::= SEQUENCE {
    protocolIEs                ProtocolIE-Container    {{RICsubscriptionModificationRefuse-IEs}},
    ...
}

RICsubscriptionModificationRefuse-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID                CRITICALITY reject TYPE RIRequestID                PRESENCE
    mandatory}|
    { ID id-RANfunctionID                CRITICALITY reject TYPE RANfunctionID                PRESENCE
    mandatory}|
    { ID id-Cause                       CRITICALITY reject TYPE Cause                PRESENCE mandatory}|
    { ID id-CriticalityDiagnostics        CRITICALITY ignore TYPE CriticalityDiagnostics        PRESENCE
    optional},
    ...
}

-- *****
--
-- RIC Subscription State Control Elementary Procedure
--
-- *****
-- *****
--
-- RIC SUBSCRIPTION STATE CONTROL REQUEST
--
-- *****

```

```

RICsubscriptionStateControlRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICsubscriptionStateControlRequest-
IES}},
    ...
}

RICsubscriptionStateControlRequest-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcrequestID          CRITICALITY reject  TYPE RIcrequestID          PRESENCE
mandatory    }|
    { ID id-RANfunctionStateControl-List  CRITICALITY reject  TYPE RANfunctionStateControl-List PRESENCE
mandatory    },
    ...
}

RANfunctionStateControl-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-SingleContainer {
{RANfunctionStateControl-ItemIEs} }

RANfunctionStateControl-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RANfunctionStateControl-Item  CRITICALITY reject  TYPE RANfunctionStateControl-Item
PRESENCE mandatory },
    ...
}

RANfunctionStateControl-Item ::= SEQUENCE {
    ranFunctionID          RANfunctionID,
    ricSubscriptionToBeSuspended-list  RICsubscriptionToBeSuspended-List  OPTIONAL,
    ricSubscriptionToBeResumed-list    RICsubscriptionToBeResumed-List    OPTIONAL,
    ...
}

RICsubscriptionToBeSuspended-List ::= SEQUENCE (SIZE(1..maxofRICrequestID)) OF ProtocolIE-SingleContainer
{ {RICsubscriptionList-ItemIEs} }

RICsubscriptionToBeResumed-List ::= SEQUENCE (SIZE(1..maxofRICrequestID)) OF ProtocolIE-SingleContainer {
{RICsubscriptionList-ItemIEs} }

RICsubscriptionList-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcsubscriptionList-Item  CRITICALITY reject  TYPE RICsubscriptionList-Item          PRESENCE
mandatory    },
    ...
}

RICsubscriptionList-Item ::= SEQUENCE {
    ricRequestID          RICrequestID,
    ricAction-list        RICAction-List  OPTIONAL,
    ...
}

RICAction-List ::= SEQUENCE (SIZE(1..maxofRICActionID)) OF ProtocolIE-SingleContainer { {RICAction-
ItemIEs} }

RICAction-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcActionList-Item  CRITICALITY reject  TYPE RICActionList-Item          PRESENCE mandatory },
    ...
}

RICActionList-Item ::= SEQUENCE {
    ricActionID          RICActionID,
    ...
}

-- *****
--
-- RIC SUBSCRIPTION STATE CONTROL RESPONSE
--
-- *****
RICsubscriptionStateControlResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICsubscriptionStateControlResponse-IEs}},
    ...
}

RICsubscriptionStateControlResponse-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIcrequestID          CRITICALITY reject  TYPE RIcrequestID          PRESENCE
mandatory    }|
    { ID id-RANfunctionStateConfirm-List  CRITICALITY reject  TYPE RANfunctionStateConfirm-List PRESENCE
mandatory    },
    ...
}

```

```

RANfunctionStateConfirm-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-SingleContainer {
{RANfunctionStateConfirm-ItemIEs} }

RANfunctionStateConfirm-ItemIEs E2AP-PROTOCOL-IES ::= {
{ ID id-RANfunctionStateConfirm-Item CRITICALITY reject TYPE RANfunctionStateConfirm-Item
PRESENCE mandatory },
...
}

RANfunctionStateConfirm-Item ::= SEQUENCE {
ranFunctionID RANfunctionID,
ricSubscriptionSuspended-list RICsubscriptionSuspended-List OPTIONAL,
ricSubscriptionResumed-list RICsubscriptionResumed-List OPTIONAL,
...
}

RICsubscriptionSuspended-List ::= SEQUENCE (SIZE(1..maxofRICrequestID)) OF ProtocolIE-SingleContainer {
{RICsubscriptionList-ItemIEs} }

RICsubscriptionResumed-List ::= SEQUENCE (SIZE(1..maxofRICrequestID)) OF ProtocolIE-SingleContainer {
{RICsubscriptionList-ItemIEs} }

-- *****
--
-- RIC SUBSCRIPTION STATE CONTROL FAILURE
--
-- *****
RICsubscriptionStateControlFailure ::= SEQUENCE {
protocolIEs ProtocolIE-Container {{RICsubscriptionStateControlFailure-IEs}},
...
}

RICsubscriptionStateControlFailure-IEs E2AP-PROTOCOL-IES ::= {
{ ID id-RIcrequestID CRITICALITY reject TYPE RIcrequestID
PRESENCE mandatory }|
{ ID id-Cause CRITICALITY reject TYPE Cause
PRESENCE mandatory }|
{ ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics
PRESENCE optional },
...
}

-- *****
--
-- RIC Assistance Elementary Procedure
--
-- *****
--
-- RIC ASSISTANCE REQUEST
--
-- *****
RICassistanceRequest ::= SEQUENCE {
protocolIEs ProtocolIE-Container {{RICassistanceRequest-IEs}},
...
}

RICassistanceRequest-IEs E2AP-PROTOCOL-IES ::= {
{ ID id-RIcrequestID CRITICALITY reject TYPE RIcrequestID PRESENCE
mandatory }|
{ ID id-RIcassistanceHeader CRITICALITY reject TYPE RIcassistanceHeader PRESENCE
mandatory }|
{ ID id-RIcassistanceMessage CRITICALITY reject TYPE RIcassistanceMessage PRESENCE
mandatory }|
{ ID id-RIcassistanceUpdate CRITICALITY reject TYPE RIcassistanceUpdate PRESENCE
optional }|
{ ID id-RIcassistanceUpdateNumber CRITICALITY reject TYPE RIcassistanceUpdateNumber PRESENCE
optional },
...
}

-- *****
--
-- RIC ASSISTANCE RESPONSE
--
-- *****

```

```

RICAssistanceResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICAssistanceResponse-IEs}},
    ...
}

RICAssistanceResponse-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID          CRITICALITY reject  TYPE RIRequestID          PRESENCE
mandatory }|
    { ID id-RIAssistanceHeader    CRITICALITY reject  TYPE RIAssistanceHeader    PRESENCE
mandatory }|
    { ID id-RIAssistanceOutcome    CRITICALITY reject  TYPE RIAssistanceOutcome    PRESENCE
mandatory },
    ...
}

-- *****
--
-- RIC ASSISTANCE FAILURE
--
-- *****
RICAssistanceFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICAssistanceFailure-IEs}},
    ...
}

RICAssistanceFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID          CRITICALITY reject  TYPE RIRequestID          PRESENCE
mandatory }|
    { ID id-Cause                CRITICALITY ignore  TYPE Cause                PRESENCE
mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional },
    ...
}

-- *****
--
-- RIC Assistance Indication Elementary Procedure
--
-- *****
--
-- RIC ASSISTANCE INDICATION
--
-- *****
RICAssistanceIndication ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICAssistanceIndication-IEs}},
    ...
}

RICAssistanceIndication-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RIRequestID          CRITICALITY reject  TYPE RIRequestID          PRESENCE
mandatory }|
    { ID id-RIAssistanceSN       CRITICALITY reject  TYPE RIAssistanceSN       PRESENCE
mandatory }|
    { ID id-RIAssistanceHeader    CRITICALITY reject  TYPE RIAssistanceHeader    PRESENCE
mandatory }|
    { ID id-RIAssistanceOutcome    CRITICALITY reject  TYPE RIAssistanceOutcome    PRESENCE
mandatory },
    ...
}

-- *****
--
-- RIC Assistance Halt Elementary Procedure
--
-- *****
--
-- RIC ASSISTANCE HALT
--
-- *****
RICAssistanceHalt ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICAssistanceHalt-IEs}},
    ...
}

RICAssistanceHalt-IEs E2AP-PROTOCOL-IES ::= {

```



```

    { ID id-RCrequestID          CRITICALITY reject  TYPE RCrequestID          PRESENCE
mandatory    },
    ...
}

-- *****
--
-- RIC Indication Elementary Procedure
--
-- *****
-- *****
--
-- RIC INDICATION
--
-- *****
RICindication ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICindication-IEs}},
    ...
}

RICindication-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RCrequestID          CRITICALITY reject  TYPE RCrequestID          PRESENCE
mandatory    }|
    { ID id-RANfunctionID        CRITICALITY reject  TYPE RANfunctionID        PRESENCE
mandatory    }|
    { ID id-RCactionID           CRITICALITY reject  TYPE RCactionID           PRESENCE
mandatory    }|
    { ID id-RCindicationSN        CRITICALITY reject  TYPE RCindicationSN        PRESENCE
optional     }|
    { ID id-RCindicationType      CRITICALITY reject  TYPE RCindicationType      PRESENCE
mandatory    }|
    { ID id-RCindicationHeader    CRITICALITY reject  TYPE RCindicationHeader    PRESENCE
mandatory    }|
    { ID id-RCindicationMessage   CRITICALITY reject  TYPE RCindicationMessage   PRESENCE
mandatory    }|
    { ID id-RCcallProcessID       CRITICALITY reject  TYPE RCcallProcessID       PRESENCE
optional     },
    ...
}

-- *****
--
-- RIC Control Elementary Procedure
--
-- *****
-- *****
--
-- RIC CONTROL REQUEST
--
-- *****
RICcontrolRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICcontrolRequest-IEs}},
    ...
}

RICcontrolRequest-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RCrequestID          CRITICALITY reject  TYPE RCrequestID          PRESENCE
mandatory    }|
    { ID id-RANfunctionID        CRITICALITY reject  TYPE RANfunctionID        PRESENCE
mandatory    }|
    { ID id-RCcallProcessID       CRITICALITY reject  TYPE RCcallProcessID       PRESENCE
optional     }|
    { ID id-RCcontrolHeader       CRITICALITY reject  TYPE RCcontrolHeader       PRESENCE
mandatory    }|
    { ID id-RCcontrolMessage      CRITICALITY reject  TYPE RCcontrolMessage      PRESENCE
mandatory    }|
    { ID id-RCcontrolAckRequest   CRITICALITY reject  TYPE RCcontrolAckRequest   PRESENCE
optional     },
    ...
}

-- *****
--
-- RIC CONTROL ACKNOWLEDGE
--
-- *****
RICcontrolAcknowledge ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICcontrolAcknowledge-IEs}},
    ...
}

```

```

RICcontrolAcknowledge-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICrequestID          CRITICALITY reject  TYPE RICrequestID          PRESENCE
mandatory  }|
    { ID id-RANfunctionID         CRITICALITY reject  TYPE RANfunctionID         PRESENCE
mandatory  }|
    { ID id-RICcallProcessID      CRITICALITY reject  TYPE RICcallProcessID      PRESENCE
optional   }|
    { ID id-RICcontrolOutcome     CRITICALITY reject  TYPE RICcontrolOutcome     PRESENCE
optional   },
    ...
}
-- *****
--
-- RIC CONTROL FAILURE
--
-- *****
RICcontrolFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICcontrolFailure-IEs}},
    ...
}

RICcontrolFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICrequestID          CRITICALITY reject  TYPE RICrequestID          PRESENCE
mandatory  }|
    { ID id-RANfunctionID         CRITICALITY reject  TYPE RANfunctionID         PRESENCE
mandatory  }|
    { ID id-RICcallProcessID      CRITICALITY reject  TYPE RICcallProcessID      PRESENCE
optional   }|
    { ID id-Cause                 CRITICALITY ignore  TYPE Cause                 PRESENCE
mandatory  }|
    { ID id-RICcontrolOutcome     CRITICALITY reject  TYPE RICcontrolOutcome     PRESENCE
optional   },
    ...
    { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional   }
}
-- *****
--
-- RIC QUERY Elementary Procedure
--
-- *****
--
-- RIC QUERY REQUEST
--
-- *****
RICQueryRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICQueryRequest-IEs}},
    ...
}

RICQueryRequest-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICrequestID          CRITICALITY reject  TYPE RICrequestID          PRESENCE
mandatory  }|
    { ID id-RANfunctionID         CRITICALITY reject  TYPE RANfunctionID         PRESENCE
mandatory  }|
    { ID id-RICqueryHeader        CRITICALITY reject  TYPE RICqueryHeader        PRESENCE
mandatory  }|
    { ID id-RICqueryDefinition    CRITICALITY reject  TYPE RICqueryDefinition    PRESENCE
mandatory  },
    ...
}
-- *****
--
-- RIC QUERY RESPONSE
--
-- *****
RICQueryResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICQueryResponse-IEs}},
    ...
}

RICQueryResponse-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICrequestID          CRITICALITY reject  TYPE RICrequestID          PRESENCE
mandatory  }|

```

```

    { ID id-RANfunctionID          CRITICALITY reject  TYPE RANfunctionID          PRESENCE
mandatory    }|
    { ID id-RICQueryOutcome        CRITICALITY reject  TYPE RICQueryOutcome          PRESENCE
mandatory    },
    ...
}
-- *****
--
-- RIC QUERY FAILURE
--
-- *****
RICQueryFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICQueryFailure-IEs}},
    ...
}

RICQueryFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICrequestID          CRITICALITY reject  TYPE RICrequestID          PRESENCE
mandatory    }|
    { ID id-RANfunctionID        CRITICALITY reject  TYPE RANfunctionID        PRESENCE
mandatory    }|
    { ID id-Cause                CRITICALITY ignore   TYPE Cause                PRESENCE
mandatory    }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore   TYPE CriticalityDiagnostics PRESENCE
optional    },
    ...
}

-- *****
--
-- RIC Service Load Status Elementary Procedure
--
-- *****
--
-- RIC SERVICE LOAD STATUS REQUEST
--
-- *****
RICServiceLoadStatusRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICServiceLoadStatusRequest-IEs}},
    ...
}

RICServiceLoadStatusRequest-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICloadMeasurementID    CRITICALITY reject  TYPE LoadMeasurementID    PRESENCE
mandatory    }|
    { ID id-E2nodeLoadMeasurementID CRITICALITY ignore   TYPE LoadMeasurementID    PRESENCE
conditional}|
    -- This IE shall be present if the Registration Request IE is set to the value "stop" or "add". --
    { ID id-RegistrationRequest      CRITICALITY ignore   TYPE RegistrationRequest    PRESENCE
mandatory    }|
    { ID id-RANfunctionLoadRequest-List CRITICALITY reject  TYPE RANfunctionLoadRequest-List PRESENCE
conditional }|
    -- This IE shall be present if the Registration Request IE is set to the value "start" or "add". -
    { ID id-ReportingPeriodicity     CRITICALITY ignore   TYPE ReportingPeriodicity    PRESENCE
optional    },
    ...
}

RANfunctionLoadRequest-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-SingleContainer {
{RANfunctionLoadRequest-ItemIEs} }

RANfunctionLoadRequest-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RANfunctionLoadRequest-Item CRITICALITY reject  TYPE RANfunctionLoadRequest-Item PRESENCE
mandatory    },
    ...
}

RANfunctionLoadRequest-Item ::= SEQUENCE {
    ranFunctionID          RANfunctionID,
    ranFunctionLoadRequest RICloadRequest          OPTIONAL,
    ricServiceLoadRequest  RICserviceLoadRequest    OPTIONAL,
    ricSubscriptionLoadRequest-list RICsubscriptionLoadRequest-List    OPTIONAL,
    ...
}

RICsubscriptionLoadRequest-List ::= SEQUENCE (SIZE(1..maxofRICrequestID)) OF ProtocolIE-SingleContainer {
{RICsubscriptionLoadRequest-ItemIEs} }

```

```

RICsubscriptionLoadRequest-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-RICsubscriptionLoadRequest-Item CRITICALITY reject TYPE RICsubscriptionLoadRequest-ItemIE
    PRESENCE mandatory },
  ...
}

RICsubscriptionLoadRequest-ItemIE ::= SEQUENCE {
  ricRequestID RICrequestID,
  ricSubscriptionLoadRequest RICloadRequest OPTIONAL,
  ricActionLoadRequest-list RICactionLoadRequest-List OPTIONAL,
  ...
}

RICactionLoadRequest-List ::= SEQUENCE (SIZE(1..maxofRICactionID)) OF ProtocolIE-SingleContainer {
  {RICactionLoadRequest-ItemIEs} }

RICactionLoadRequest-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-RICactionLoadRequest-Item CRITICALITY reject TYPE RICactionLoadRequest-Item
    PRESENCE mandatory },
  ...
}

RICactionLoadRequest-Item ::= SEQUENCE {
  ricActionID RICactionID,
  ricActionLoadRequest RICloadRequest,
  ...
}

-- *****
--
-- RIC SERVICE LOAD STATUS RESPONSE
--
-- *****
RICserviceLoadStatusResponse ::= SEQUENCE {
  protocolIEs ProtocolIE-Container {{RICserviceLoadStatusResponse-IEs}},
  ...
}

RICserviceLoadStatusResponse-IEs E2AP-PROTOCOL-IES ::= {
  { ID id-RICloadMeasurementID CRITICALITY reject TYPE LoadMeasurementID PRESENCE
    mandatory }|
  { ID id-E2nodeLoadMeasurementID CRITICALITY ignore TYPE LoadMeasurementID PRESENCE
    mandatory }|
  { ID id-RANfunctionLoadConfirm-List CRITICALITY reject TYPE RANfunctionLoadConfirm-List PRESENCE
    conditional },
  -- This IE shall be present if the Registration Request IE in RIC SERVICE LOAD STATUS REQUEST was set
  to the value "start" or "add". -
  ...
}

RANfunctionLoadConfirm-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-SingleContainer {
  {RANfunctionLoadConfirm-ItemIEs} }

RANfunctionLoadConfirm-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-RANfunctionLoadConfirm-Item CRITICALITY reject TYPE RANfunctionLoadConfirm-Item PRESENCE
    mandatory },
  ...
}

RANfunctionLoadConfirm-Item ::= SEQUENCE {
  ranFunctionID RANfunctionID,
  ranFunctionLoadConfirm RICloadConfirm OPTIONAL,
  ricServiceLoadConfirm RICserviceLoadConfirm OPTIONAL,
  ricSubscriptionLoadConfirm-list RICsubscriptionLoadConfirm-List OPTIONAL,
  ...
}

RICsubscriptionLoadConfirm-List ::= SEQUENCE (SIZE(1..maxofRICrequestID)) OF ProtocolIE-SingleContainer {
  {RICsubscriptionLoadConfirm-ItemIEs} }

RICsubscriptionLoadConfirm-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-RICsubscriptionLoadConfirm-Item CRITICALITY reject TYPE RICsubscriptionLoadConfirm-ItemIE
    PRESENCE mandatory },
  ...
}

RICsubscriptionLoadConfirm-ItemIE ::= SEQUENCE {
  ricRequestID RICrequestID,
  ricSubscriptionLoadConfirm RICloadConfirm OPTIONAL,

```

```

    ricActionLoadConfirm-list          RICActionLoadConfirm-List          OPTIONAL,
    ...
}

RICActionLoadConfirm-List ::= SEQUENCE (SIZE(1..maxofRICActionID)) OF ProtocolIE-SingleContainer {
{RICActionLoadConfirm-ItemIEs} }

RICActionLoadConfirm-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICActionLoadConfirm-Item CRITICALITY reject TYPE RICActionLoadConfirm-Item
    PRESENCE mandatory },
    ...
}

RICActionLoadConfirm-Item ::= SEQUENCE {
    ricActionID          RICActionID,
    ricActionLoadConfirm RICloadConfirm,
    ...
}

-- *****
--
-- RIC SERVICE LOAD STATUS FAILURE
--
-- *****
RICServiceLoadStatusFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICServiceLoadStatusFailure-IEs}},
    ...
}

RICServiceLoadStatusFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICloadMeasurementID CRITICALITY reject TYPE LoadMeasurementID PRESENCE
mandatory }|
    { ID id-E2nodeLoadMeasurementID CRITICALITY ignore TYPE LoadMeasurementID PRESENCE
mandatory }|
    { ID id-Cause CRITICALITY ignore TYPE Cause PRESENCE
mandatory }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore TYPE CriticalityDiagnostics PRESENCE
optional },
    ...
}

-- *****
--
-- RIC Service Load Update Elementary Procedure
--
-- *****
--
-- RIC SERVICE LOAD UPDATE
--
-- *****
RICServiceLoadUpdate ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICServiceLoadUpdate-IEs}},
    ...
}

RICServiceLoadUpdate-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-RICloadMeasurementID CRITICALITY reject TYPE LoadMeasurementID PRESENCE
mandatory }|
    { ID id-E2nodeLoadMeasurementID CRITICALITY ignore TYPE LoadMeasurementID PRESENCE
mandatory }|
    { ID id-RANfunctionLoad-List CRITICALITY reject TYPE RANfunctionLoad-List PRESENCE
mandatory },
    ...
}

RANfunctionLoad-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-SingleContainer {
{RANfunctionLoad-ItemIEs} }

RANfunctionLoad-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RANfunctionLoad-Item CRITICALITY reject TYPE RANfunctionLoad-Item PRESENCE
mandatory },
    ...
}

RANfunctionLoad-Item ::= SEQUENCE {
    ranFunctionID          RANfunctionID,
    ranFunctionLoadInformation RICloadInformation OPTIONAL,
    ricServiceLoadInformation RICServiceLoadInformation OPTIONAL,

```

```

    ricSubscriptionLoad-list          RICsubscriptionLoad-List          OPTIONAL,
    ...
}

RICsubscriptionLoad-List ::= SEQUENCE (SIZE(1..maxofRICrequestID)) OF ProtocolIE-SingleContainer {
{RICsubscriptionLoad-ItemIEs} }

RICsubscriptionLoad-ItemIEs          E2AP-PROTOCOL-IES ::= {
    { ID id-RICsubscriptionLoad-Item    CRITICALITY reject    TYPE RICsubscriptionLoad-ItemIE          PRESENCE
mandatory    },
    ...
}

RICsubscriptionLoad-ItemIE ::= SEQUENCE {
    ricRequestID                      RICrequestID,
    ricSubscriptionLoadInformation      RICloadInformation          OPTIONAL,
    ricActionLoad-list                 RICactionLoad-List          OPTIONAL,
    ...
}

RICactionLoad-List ::= SEQUENCE (SIZE(1..maxofRICactionID)) OF ProtocolIE-SingleContainer {
{RICactionLoad-ItemIEs} }

RICactionLoad-ItemIEs          E2AP-PROTOCOL-IES ::= {
    { ID id-RIcactionLoad-Item    CRITICALITY reject    TYPE RICactionLoad-Item          PRESENCE
mandatory    },
    ...
}

RICactionLoad-Item ::= SEQUENCE {
    ricActionID                      RICactionID,
    ricActionLoadInformation          RICloadInformation,
    ...
}

-- *****
--
-- E2 Resource Status Elementary Procedure
--
-- *****
-- *****
--
-- E2 RESOURCE STATUS REQUEST
--
-- *****

E2ResourceStatusRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container          { {E2ResourceStatusRequestIEs} },
    ...
}

E2ResourceStatusRequestIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject    TYPE TransactionID          PRESENCE
mandatory    }|
    { ID id-RIcloadMeasurementID    CRITICALITY reject    TYPE LoadMeasurementID          PRESENCE
mandatory    }|
    { ID id-E2NodeLoadMeasurementID    CRITICALITY ignore    TYPE LoadMeasurementID          PRESENCE
conditional}|
    -- This IE shall be present if the Registration Request IE is set to the value "stop" or "add". --
    { ID id-RegistrationRequest-E2    CRITICALITY ignore    TYPE RegistrationRequest-E2          PRESENCE
mandatory}|
    { ID id-ReportCharacteristics          CRITICALITY ignore    TYPE ReportCharacteristics          PRESENCE
conditional}|
    -- This IE shall be present if the Registration Request IE is set to the value "start". --
    { ID id-ReportingPeriodicity-E2    CRITICALITY ignore    TYPE ReportingPeriodicity-E2          PRESENCE
optional},
    ...
}

-- *****
--
-- E2 RESOURCE STATUS RESPONSE
--
-- *****

E2ResourceStatusResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container          { { E2ResourceStatusResponseIEs} },
    ...
}

```

```

E2ResourceStatusResponseIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE mandatory
    }|
    { ID id-RICLoadMeasurementID    CRITICALITY reject  TYPE LoadMeasurementID        PRESENCE mandatory
    }|
    { ID id-E2NodeLoadMeasurementID CRITICALITY reject  TYPE LoadMeasurementID        PRESENCE
mandatory }|
    { ID id-CriticalityDiagnostics  CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional },
    ...
}

-- *****
--
-- E2 RESOURCE STATUS FAILURE
--
-- *****

E2ResourceStatusFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container      { { E2ResourceStatusFailureIEs} },
    ...
}

E2ResourceStatusFailureIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE mandatory
    }|
    { ID id-RICLoadMeasurementID    CRITICALITY reject  TYPE LoadMeasurementID        PRESENCE mandatory
    }|
    { ID id-E2NodeLoadMeasurementID CRITICALITY reject  TYPE LoadMeasurementID        PRESENCE mandatory
    }|
    { ID id-Cause                  CRITICALITY ignore  TYPE Cause                  PRESENCE mandatory
    }|
    { ID id-CriticalityDiagnostics  CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional },
    ...
}

-- *****
--
-- E2 Resource Status Update Elementary Procedure
--
-- *****

-- *****
--
-- E2 RESOURCE STATUS UPDATE
--
-- *****

E2ResourceStatusUpdate ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container      {{ E2ResourceStatusUpdateIEs}},
    ...
}

E2ResourceStatusUpdateIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory }|
    { ID id-RICLoadMeasurementID    CRITICALITY reject  TYPE LoadMeasurementID        PRESENCE
mandatory }|
    { ID id-E2NodeLoadMeasurementID CRITICALITY reject  TYPE LoadMeasurementID        PRESENCE
mandatory }|
    { ID id-TNLCapacityIndicator    CRITICALITY ignore  TYPE TNLCapacityIndicator    PRESENCE
optional }|
    { ID id-HardwareLoadIndicator    CRITICALITY ignore  TYPE HardwareLoadIndicator    PRESENCE
optional },
    ...
}

-- *****
--
-- MESSAGES FOR GLOBAL PROCEDURES
--
-- *****

```

```
-- *****
--
-- Error Indication Elementary Procedure
--
-- *****
-- *****
--
-- ERROR INDICATION
--
-- *****
ErrorIndication ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{ErrorIndication-IEs}},
    ...
}

ErrorIndication-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
optional    }|
    { ID id-RICrequestID          CRITICALITY reject  TYPE RICrequestID          PRESENCE
optional    }|
    { ID id-RANfunctionID         CRITICALITY reject  TYPE RANfunctionID         PRESENCE
optional    }|
    { ID id-Cause                 CRITICALITY ignore  TYPE Cause                 PRESENCE
optional    }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional    },
    ...
}

-- *****
--
-- E2 Setup Elementary Procedure
--
-- *****
-- *****
--
-- E2 SETUP REQUEST
--
-- *****

E2setupRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    { {E2setupRequestIEs} },
    ...
}

E2setupRequestIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory    }|
    { ID id-GlobalE2node-ID       CRITICALITY reject  TYPE GlobalE2node-ID          PRESENCE
mandatory    }|
    { ID id-RANfunctionsAdded      CRITICALITY reject  TYPE RANfunctions-List        PRESENCE
mandatory    }|
    { ID id-E2nodeComponentConfigAddition CRITICALITY reject  TYPE E2nodeComponentConfigAddition-List
mandatory    },
    ...
}

-- *****
--
-- E2 SETUP RESPONSE
--
-- *****

E2setupResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    { {E2setupResponseIEs} },
    ...
}

E2setupResponseIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory    }|
    { ID id-GlobalRIC-ID          CRITICALITY reject  TYPE GlobalRIC-ID          PRESENCE
mandatory    }|
    { ID id-RANfunctionsAccepted   CRITICALITY reject  TYPE RANfunctionsID-List    PRESENCE
optional    }|
    { ID id-RANfunctionsRejected   CRITICALITY reject  TYPE RANfunctionsIDcause-List
optional    }|
    { ID id-E2nodeComponentConfigAdditionAck CRITICALITY reject  TYPE E2nodeComponentConfigAdditionAck-
List          PRESENCE mandatory }|

```



```

    { ID id-RANfunctionsAcceptModified-List      CRITICALITY reject  TYPE RANfunctions-AcceptModified-List
      PRESENCE optional    },
    ...
}

RANfunctions-AcceptModified-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-SingleContainer
{ {RANfunction-AcceptModified-ItemIEs} }

RANfunction-AcceptModified-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-RANfunction-AcceptModified-Item      CRITICALITY ignore  TYPE RANfunction-AcceptModified-
    Item          PRESENCE mandatory    },
  ...
}

RANfunction-AcceptModified-Item ::= SEQUENCE {
  ranFunctionID          RANfunctionID,
  revRanFunctionDefinition RANfunctionDefinition,
  ranFunctionRevision     RANfunctionRevision,
  ...
}

-- *****
--
-- E2 SETUP FAILURE
--
-- *****

E2setupFailure ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {E2setupFailureIEs} },
  ...
}

E2setupFailureIEs E2AP-PROTOCOL-IES ::= {
  { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory    }|
  { ID id-Cause                  CRITICALITY ignore  TYPE Cause                  PRESENCE
mandatory    }|
  { ID id-TimeToWait             CRITICALITY ignore  TYPE TimeToWait             PRESENCE
optional     }|
  { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional     }|
  { ID id-TNLinformation         CRITICALITY ignore  TYPE TNLinformation         PRESENCE
optional     },
  ...
}

-- *****
--
-- E2 Connection Update Elementary Procedure
--
-- *****
--
-- E2 CONNECTION UPDATE
--
-- *****

E2connectionUpdate ::= SEQUENCE {
  protocolIEs      ProtocolIE-Container      { {E2connectionUpdate-IEs} },
  ...
}

E2connectionUpdate-IEs E2AP-PROTOCOL-IES ::= {
  { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory    }|
  { ID id-E2connectionUpdateAdd  CRITICALITY reject  TYPE E2connectionUpdate-List      PRESENCE
optional     }|
  { ID id-E2connectionUpdateRemove CRITICALITY reject  TYPE E2connectionUpdateRemove-List PRESENCE
optional     }|
  { ID id-E2connectionUpdateModify CRITICALITY reject  TYPE E2connectionUpdate-List      PRESENCE
optional     },
  ...
}

E2connectionUpdate-List ::= SEQUENCE (SIZE(1..maxofTNLA)) OF ProtocolIE-SingleContainer {
{E2connectionUpdate-ItemIEs} }

E2connectionUpdate-ItemIEs E2AP-PROTOCOL-IES ::= {

```

```

    { ID id-E2connectionUpdate-Item          CRITICALITY ignore  TYPE E2connectionUpdate-Item          PRESENCE
mandatory  },
    ...
}

E2connectionUpdate-Item ::= SEQUENCE {
    tnlInformation          TNLinformation,
    tnlUsage                TNLusage,
    ...
}

E2connectionUpdateRemove-List ::= SEQUENCE (SIZE(1..maxofTNLA)) OF ProtocolIE-SingleContainer {
{E2connectionUpdateRemove-ItemIEs} }

E2connectionUpdateRemove-ItemIEs  E2AP-PROTOCOL-IES ::= {
    { ID id-E2connectionUpdateRemove-Item  CRITICALITY ignore  TYPE E2connectionUpdateRemove-Item
    PRESENCE mandatory  },
    ...
}

E2connectionUpdateRemove-Item ::= SEQUENCE {
    tnlInformation          TNLinformation,
    ...
}

-- *****
--
-- E2 CONNECTION UPDATE ACKNOWLEDGE
--
-- *****
E2connectionUpdateAcknowledge ::= SEQUENCE {
    protocolIEs            ProtocolIE-Container    {{E2connectionUpdateAck-IEs}},
    ...
}

E2connectionUpdateAck-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory  }|
    { ID id-E2connectionSetup      CRITICALITY reject  TYPE E2connectionUpdate-List      PRESENCE
optional   }|
    { ID id-E2connectionSetupFailed CRITICALITY reject  TYPE E2connectionSetupFailed-List  PRESENCE
optional   },
    ...
}

E2connectionSetupFailed-List ::= SEQUENCE (SIZE(1..maxofTNLA)) OF ProtocolIE-SingleContainer {
{E2connectionSetupFailed-ItemIEs} }

E2connectionSetupFailed-ItemIEs  E2AP-PROTOCOL-IES ::= {
    { ID id-E2connectionSetupFailed-Item          CRITICALITY ignore  TYPE E2connectionSetupFailed-Item
    PRESENCE mandatory  },
    ...
}

E2connectionSetupFailed-Item ::= SEQUENCE {
    tnlInformation          TNLinformation,
    cause                  Cause,
    ...
}

-- *****
--
-- E2 CONNECTION UPDATE FAILURE
--
-- *****
E2connectionUpdateFailure ::= SEQUENCE {
    protocolIEs            ProtocolIE-Container    {{E2connectionUpdateFailure-IEs}},
    ...
}

E2connectionUpdateFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID
    PRESENCE mandatory  }|
    { ID id-Cause                  CRITICALITY reject  TYPE Cause
    PRESENCE optional    }|
    { ID id-TimeToWait             CRITICALITY ignore  TYPE TimeToWait
    PRESENCE optional    }|

```

```

    { ID id-CriticalityDiagnostics          CRITICALITY ignore  TYPE CriticalityDiagnostics
      PRESENCE optional    },
    ...
  }

-- *****
--
-- E2 Node Configuration Update Elementary Procedure
--
-- *****
-- *****
-- E2 NODE CONFIGURATION UPDATE
--
-- *****
E2nodeConfigurationUpdate ::= SEQUENCE {
  protocolIEs          ProtocolIE-Container    {{E2nodeConfigurationUpdate-IEs}},
  ...
}

E2nodeConfigurationUpdate-IEs E2AP-PROTOCOL-IES ::= {
  { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID
    PRESENCE mandatory    },
  { ID id-GlobalE2node-ID        CRITICALITY reject  TYPE GlobalE2node-ID
    PRESENCE optional    },
  { ID id-E2nodeComponentConfigAddition  CRITICALITY reject  TYPE E2nodeComponentConfigAddition-List
    PRESENCE optional    },
  { ID id-E2nodeComponentConfigUpdate    CRITICALITY reject  TYPE E2nodeComponentConfigUpdate-List
    PRESENCE optional    },
  { ID id-E2nodeComponentConfigRemoval    CRITICALITY reject  TYPE E2nodeComponentConfigRemoval-List
    PRESENCE optional    },
  { ID id-E2nodeTNLassociationRemoval    CRITICALITY reject  TYPE E2nodeTNLassociationRemoval-List
    PRESENCE optional    },
  ...
}

E2nodeComponentConfigAddition-List ::= SEQUENCE (SIZE(1..maxofE2nodeComponents)) OF ProtocolIE-
SingleContainer { {E2nodeComponentConfigAddition-ItemIEs} }

E2nodeComponentConfigAddition-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-E2nodeComponentConfigAddition-Item  CRITICALITY reject  TYPE E2nodeComponentConfigAddition-
Item          PRESENCE mandatory    },
  ...
}

E2nodeComponentConfigAddition-Item ::= SEQUENCE {
  e2nodeComponentInterfaceType    E2nodeComponentInterfaceType,
  e2nodeComponentID                E2nodeComponentID,
  e2nodeComponentConfiguration    E2nodeComponentConfiguration,
  ...
}

E2nodeComponentConfigUpdate-List ::= SEQUENCE (SIZE(1..maxofE2nodeComponents)) OF ProtocolIE-
SingleContainer { {E2nodeComponentConfigUpdate-ItemIEs} }

E2nodeComponentConfigUpdate-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-E2nodeComponentConfigUpdate-Item    CRITICALITY reject  TYPE E2nodeComponentConfigUpdate-Item
    PRESENCE mandatory    },
  ...
}

E2nodeComponentConfigUpdate-Item ::= SEQUENCE {
  e2nodeComponentInterfaceType    E2nodeComponentInterfaceType,
  e2nodeComponentID                E2nodeComponentID,
  e2nodeComponentConfiguration    E2nodeComponentConfiguration,
  ...
}

E2nodeComponentConfigRemoval-List ::= SEQUENCE (SIZE(1..maxofE2nodeComponents)) OF ProtocolIE-
SingleContainer { {E2nodeComponentConfigRemoval-ItemIEs} }

E2nodeComponentConfigRemoval-ItemIEs E2AP-PROTOCOL-IES ::= {
  { ID id-E2nodeComponentConfigRemoval-Item    CRITICALITY reject  TYPE E2nodeComponentConfigRemoval-Item
    PRESENCE mandatory    },
  ...
}

E2nodeComponentConfigRemoval-Item ::= SEQUENCE {
  e2nodeComponentInterfaceType    E2nodeComponentInterfaceType,

```

```

    e2nodeComponentID          E2nodeComponentID,
    ...
}

E2nodeTNLassociationRemoval-List ::= SEQUENCE (SIZE(1..maxofTNLA)) OF ProtocolIE-SingleContainer {
{E2nodeTNLassociationRemoval-ItemIEs} }

E2nodeTNLassociationRemoval-ItemIEs      E2AP-PROTOCOL-IES ::= {
    { ID id-E2nodeTNLassociationRemoval-Item      CRITICALITY reject  TYPE E2nodeTNLassociationRemoval-Item
      PRESENCE mandatory },
    ...
}

E2nodeTNLassociationRemoval-Item ::= SEQUENCE {
    tnlInformation          TNLinformation,
    tnlInformationRIC        TNLinformation,
    ...
}

-- *****
--
-- E2 NODE CONFIGURATION UPDATE ACKNOWLEDGE
--
-- *****
E2nodeConfigurationUpdateAcknowledge ::= SEQUENCE {
    protocolIEs              ProtocolIE-Container    {{E2nodeConfigurationUpdateAcknowledge-IEs}},
    ...
}

E2nodeConfigurationUpdateAcknowledge-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID
      PRESENCE mandatory }|
    { ID id-E2nodeComponentConfigAdditionAck      CRITICALITY reject  TYPE
E2nodeComponentConfigAdditionAck-List      PRESENCE optional }|
    { ID id-E2nodeComponentConfigUpdateAck      CRITICALITY reject  TYPE
E2nodeComponentConfigUpdateAck-List      PRESENCE optional }|
    { ID id-E2nodeComponentConfigRemovalAck      CRITICALITY reject  TYPE
E2nodeComponentConfigRemovalAck-List      PRESENCE optional },
    ...
}

E2nodeComponentConfigAdditionAck-List ::= SEQUENCE (SIZE(1..maxofE2nodeComponents)) OF ProtocolIE-
SingleContainer { {E2nodeComponentConfigAdditionAck-ItemIEs} }

E2nodeComponentConfigAdditionAck-ItemIEs      E2AP-PROTOCOL-IES ::= {
    { ID id-E2nodeComponentConfigAdditionAck-Item      CRITICALITY reject  TYPE
E2nodeComponentConfigAdditionAck-Item      PRESENCE mandatory },
    ...
}

E2nodeComponentConfigAdditionAck-Item ::= SEQUENCE {
    e2nodeComponentInterfaceType      E2nodeComponentInterfaceType,
    e2nodeComponentID                  E2nodeComponentID,
    e2nodeComponentConfigurationAck    E2nodeComponentConfigurationAck,
    ...
}

E2nodeComponentConfigUpdateAck-List ::= SEQUENCE (SIZE(1..maxofE2nodeComponents)) OF ProtocolIE-
SingleContainer { {E2nodeComponentConfigUpdateAck-ItemIEs} }

E2nodeComponentConfigUpdateAck-ItemIEs      E2AP-PROTOCOL-IES ::= {
    { ID id-E2nodeComponentConfigUpdateAck-Item      CRITICALITY reject  TYPE
E2nodeComponentConfigUpdateAck-Item      PRESENCE mandatory },
    ...
}

E2nodeComponentConfigUpdateAck-Item ::= SEQUENCE {
    e2nodeComponentInterfaceType      E2nodeComponentInterfaceType,
    e2nodeComponentID                  E2nodeComponentID,
    e2nodeComponentConfigurationAck    E2nodeComponentConfigurationAck,
    ...
}

E2nodeComponentConfigRemovalAck-List ::= SEQUENCE (SIZE(1..maxofE2nodeComponents)) OF ProtocolIE-
SingleContainer { {E2nodeComponentConfigRemovalAck-ItemIEs} }

E2nodeComponentConfigRemovalAck-ItemIEs      E2AP-PROTOCOL-IES ::= {
    { ID id-E2nodeComponentConfigRemovalAck-Item      CRITICALITY reject  TYPE
E2nodeComponentConfigRemovalAck-Item      PRESENCE mandatory },

```

```

    ...
}

E2nodeComponentConfigRemovalAck-Item ::= SEQUENCE {
    e2nodeComponentInterfaceType      E2nodeComponentInterfaceType,
    e2nodeComponentID                 E2nodeComponentID,
    e2nodeComponentConfigurationAck    E2nodeComponentConfigurationAck,
    ...
}

-- *****
--
-- E2 NODE CONFIGURATION UPDATE FAILURE
--
-- *****
E2nodeConfigurationUpdateFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{E2nodeConfigurationUpdateFailure-IEs}},
    ...
}

E2nodeConfigurationUpdateFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID
      PRESENCE mandatory   }|
    { ID id-Cause                  CRITICALITY ignore  TYPE Cause
      PRESENCE mandatory   }|
    { ID id-TimeToWait             CRITICALITY ignore  TYPE TimeToWait
      PRESENCE optional     }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics
      PRESENCE optional     },
    ...
}

-- *****
--
-- Reset Elementary Procedure
--
-- *****

-- *****
--
-- RESET REQUEST
--
-- *****

ResetRequest ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    { {ResetRequestIEs} },
    ...
}

ResetRequestIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
      mandatory   }|
    { ID id-Cause                  CRITICALITY ignore  TYPE Cause                  PRESENCE
      mandatory   },
    ...
}

-- *****
--
-- RESET RESPONSE
--
-- *****

ResetResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    { {ResetResponseIEs} },
    ...
}

ResetResponseIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
      mandatory   }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
      optional   },
    ...
}

-- *****
--
-- RIC Service Update Elementary Procedure

```

```
--
-- *****
-- *****
--
-- RIC SERVICE UPDATE
--
-- *****
RICServiceUpdate ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICserviceUpdate-IEs}},
    ...
}

RICServiceUpdate-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory    }|
    { ID id-RANfunctionsAdded      CRITICALITY reject  TYPE RANfunctions-List          PRESENCE
optional    }|
    { ID id-RANfunctionsModified   CRITICALITY reject  TYPE RANfunctions-List          PRESENCE
optional    }|
    { ID id-RANfunctionsDeleted    CRITICALITY reject  TYPE RANfunctionsID-List        PRESENCE
optional    },
    ...
}

RANfunctions-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-SingleContainer { {RANfunction-
ItemIEs} }

RANfunction-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RANfunction-Item          CRITICALITY ignore  TYPE RANfunction-Item          PRESENCE
mandatory    },
    ...
}

RANfunction-Item ::= SEQUENCE {
    ranFunctionID          RANfunctionID,
    ranFunctionDefinition   RANfunctionDefinition,
    ranFunctionRevision     RANfunctionRevision,
    ranFunctionOID          RANfunctionOID,
    ...
}

RANfunctionsID-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-
SingleContainer{{RANfunctionID-ItemIEs}}

RANfunctionID-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RANfunctionID-Item          CRITICALITY ignore          TYPE RANfunctionID-Item          PRESENCE
mandatory    },
    ...
}

RANfunctionID-Item ::= SEQUENCE {
    ranFunctionID          RANfunctionID,
    ranFunctionRevision     RANfunctionRevision,
    ...
}

-- *****
-- *****
-- RIC SERVICE UPDATE ACKNOWLEDGE
--
-- *****
RICServiceUpdateAcknowledge ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    {{RICserviceUpdateAcknowledge-IEs}},
    ...
}

RICServiceUpdateAcknowledge-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory    }|
    { ID id-RANfunctionsAccepted    CRITICALITY reject  TYPE RANfunctionsID-List          PRESENCE
optional    }|
    { ID id-RANfunctionsRejected    CRITICALITY reject  TYPE RANfunctionsIDcause-List        PRESENCE
optional    }|
    { ID id-RANfunctionsAcceptModified-List CRITICALITY reject  TYPE RANfunctions-AcceptModified-List
PRESENCE optional    },
    ...
}

```

```

RANfunctionsIDcause-List ::= SEQUENCE (SIZE(1..maxofRANfunctionID)) OF ProtocolIE-SingleContainer {
{RANfunctionIDcause-ItemIEs} }

RANfunctionIDcause-ItemIEs E2AP-PROTOCOL-IES ::= {
    { ID id-RANfunctionIDcause-Item      CRITICALITY ignore  TYPE RANfunctionIDcause-Item      PRESENCE
mandatory    },
    ...
}

RANfunctionIDcause-Item ::= SEQUENCE {
    ranFunctionID      RANfunctionID,
    cause              Cause,
    ...
}

-- *****
--
-- RIC SERVICE UPDATE FAILURE
--
-- *****
RICServiceUpdateFailure ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container    {{RICServiceUpdateFailure-IEs}},
    ...
}

RICServiceUpdateFailure-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID      CRITICALITY reject  TYPE TransactionID      PRESENCE
mandatory    }|
    { ID id-Cause              CRITICALITY reject  TYPE Cause              PRESENCE
mandatory    }|
    { ID id-TimeToWait         CRITICALITY ignore  TYPE TimeToWait         PRESENCE
optional     }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional     },
    ...
}

-- *****
--
-- RIC Service Query Elementary Procedure
--
-- *****
--
-- RIC SERVICE QUERY
--
-- *****
RICServiceQuery ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container    {{RICServiceQuery-IEs}},
    ...
}

RICServiceQuery-IEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID      CRITICALITY reject  TYPE TransactionID      PRESENCE
mandatory    }|
    { ID id-RANfunctionsAccepted CRITICALITY reject  TYPE RANfunctionsID-List      PRESENCE
optional     },
    ...
}

-- *****
--
-- E2 Removal Elementary Procedure
--
-- *****
--
-- E2 REMOVAL REQUEST
--
-- *****
E2RemovalRequest ::= SEQUENCE {
    protocolIEs      ProtocolIE-Container    { {E2RemovalRequestIEs} },
    ...
}

E2RemovalRequestIEs E2AP-PROTOCOL-IES ::= {

```

```

    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory    },
    ...
}

-- *****
--
-- E2 REMOVAL RESPONSE
--
-- *****

E2RemovalResponse ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    { {E2RemovalResponseIEs} },
    ...
}

E2RemovalResponseIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
    PRESENCE mandatory    }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional    },
    ...
}

-- *****
--
-- E2 REMOVAL FAILURE
--
-- *****

E2RemovalFailure ::= SEQUENCE {
    protocolIEs          ProtocolIE-Container    { {E2RemovalFailureIEs} },
    ...
}

E2RemovalFailureIEs E2AP-PROTOCOL-IES ::= {
    { ID id-TransactionID          CRITICALITY reject  TYPE TransactionID          PRESENCE
mandatory    }|
    { ID id-Cause                  CRITICALITY ignore  TYPE Cause                  PRESENCE
mandatory    }|
    { ID id-CriticalityDiagnostics CRITICALITY ignore  TYPE CriticalityDiagnostics PRESENCE
optional    },
    ...
}

}

END
-- ASN1STOP

```

9.3.5 Information Element definitions

```

-- ASN1START
-- *****
-- E2AP
-- Information Element Definitions
--
-- *****

E2AP-IEs {
iso(1) identified-organization(3) dod(6) internet(1) private(4) enterprise(1) 53148 e2(1) version2 (2)
e2ap(1) e2ap-IEs (2)}

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

IMPORTS
    Criticality,
    Presence,
    ProcedureCode,
    ProtocolIE-ID,
    TriggeringMessage
FROM E2AP-CommonDataTypes

    maxnoofErrors,
    maxProtocolIEs
FROM E2AP-Constants;

-- A

```



```

-- *****
-- [New for E2AP v02.00] copied from 3GPP 38.413 (NGAP) IEs
-- *****
AMFName ::= PrintableString (SIZE(1..150, ...))

-- B
-- C
Cause ::= CHOICE {
    ricRequest          CauseRICrequest,
    ricService          CauseRICservice,
    e2Node              CauseE2node,
    transport           CauseTransport,
    protocol            CauseProtocol,
    misc                CauseMisc,
    ...,
    serviceLayer        CauseServiceLayer
}

CauseE2node ::= ENUMERATED {
    e2node-component-unknown,
    ...
}

CauseMisc ::= ENUMERATED {
    control-processing-overload,
    hardware-failure,
    om-intervention,
    unspecified,
    ...
}

CauseProtocol ::= ENUMERATED {
    transfer-syntax-error,
    abstract-syntax-error-reject,
    abstract-syntax-error-ignore-and-notify,
    message-not-compatible-with-receiver-state,
    semantic-error,
    abstract-syntax-error-falsely-constructed-message,
    unspecified,
    ...
}

CauseRICrequest ::= ENUMERATED {
    ran-function-id-invalid,
    action-not-supported,
    excessive-actions,
    duplicate-action,
    duplicate-event-trigger,
    function-resource-limit,
    request-id-unknown,
    inconsistent-action-subsequent-action-sequence,
    control-message-invalid,
    ric-call-process-id-invalid,
    control-timer-expired,
    control-failed-to-execute,
    system-not-ready,
    unspecified,
    ...,
    ric-subscription-end-time-expired,
    ric-subscription-end-time-invalid,
    duplicate-ric-request-id,
    eventTriggerNotSupported,
    requested-information-unavailable,
    invalid-information-request
}

CauseRICservice ::= ENUMERATED{
    ran-function-not-supported,
    excessive-functions,
    ric-resource-limit,
    ...
}

CauseServiceLayer ::= SEQUENCE {
    serviceLayerCause    ServiceLayerCause,
    ...
}

```

```

CauseTransport ::= ENUMERATED {
    unspecified,
    transport-resource-unavailable,
    ...
}

-- *****
-- copied from 3GPP 38.413 (NGAP) IEs
-- note: ie-Extensions removed
-- *****
CriticalityDiagnostics ::= SEQUENCE {
    procedureCode          ProcedureCode          OPTIONAL,
    triggeringMessage      TriggeringMessage      OPTIONAL,
    procedureCriticality    Criticality            OPTIONAL,
    ricRequestorID         RICrequestID           OPTIONAL,
    iEsCriticalityDiagnostics CriticalityDiagnostics-IE-List OPTIONAL,
    ...
}

CriticalityDiagnostics-IE-List ::= SEQUENCE (SIZE(1..maxnoofErrors)) OF CriticalityDiagnostics-IE-Item

CriticalityDiagnostics-IE-Item ::= SEQUENCE {
    iECriticality          Criticality,
    iE-ID                 ProtocolIE-ID,
    typeOfError            TypeOfError,
    ...
}

-- D
-- E

-- Following IE used to carry 3GPP defined SETUP and RAN Configuration messages defined in FlAP, ElAP,
XnAP, etc.
E2nodeComponentConfiguration ::= SEQUENCE{
    e2nodeComponentRequestPart  OCTET STRING,
    e2nodeComponentResponsePart OCTET STRING,
    ...
}

E2nodeComponentConfigurationAck ::= SEQUENCE{
    updateOutcome      ENUMERATED {success, failure, ...},
    failureCause       Cause          OPTIONAL,
    ...
}

E2nodeComponentInterfaceType ::= ENUMERATED {ng, xn, e1, f1, w1, s1, x2,...}

E2nodeComponentID ::= CHOICE{
    e2nodeComponentInterfaceTypeNG  E2nodeComponentInterfaceNG,
    e2nodeComponentInterfaceTypeXn  E2nodeComponentInterfaceXn,
    e2nodeComponentInterfaceTypeE1  E2nodeComponentInterfaceE1,
    e2nodeComponentInterfaceTypeF1  E2nodeComponentInterfaceF1,
    e2nodeComponentInterfaceTypeW1  E2nodeComponentInterfaceW1,
    e2nodeComponentInterfaceTypeS1  E2nodeComponentInterfaceS1,
    e2nodeComponentInterfaceTypeX2  E2nodeComponentInterfaceX2,
    ...
}

E2nodeComponentInterfaceE1 ::= SEQUENCE{
    gNB-CU-UP-ID      GNB-CU-UP-ID,
    ...
}

E2nodeComponentInterfaceF1 ::= SEQUENCE{
    gNB-DU-ID          GNB-DU-ID,
    ...
}

E2nodeComponentInterfaceNG ::= SEQUENCE{
    amf-name           AMFName,
    ...
}

E2nodeComponentInterfaceS1 ::= SEQUENCE{
    mme-name           MMename,
    ...
}

E2nodeComponentInterfaceX2 ::= SEQUENCE{

```

```

    global-eNB-ID          GlobalENB-ID    OPTIONAL,
    global-en-gNB-ID       GlobalenGNB-ID  OPTIONAL,
    ...
}

E2nodeComponentInterfaceXn ::= SEQUENCE{
    global-NG-RAN-Node-ID    GlobalNG-RANNode-ID,
    ...
}

E2nodeComponentInterfaceW1 ::= SEQUENCE{
    ng-eNB-DU-ID            NGENB-DU-ID,
    ...
}

-- *****
-- copied from 3GPP 36.423 (X2AP) IEs
-- note: ie-Extensions removed
-- *****
ENB-ID ::= CHOICE {
    macro-eNB-ID            BIT STRING (SIZE (20)),
    home-eNB-ID             BIT STRING (SIZE (28)),
    ... ,
    short-Macro-eNB-ID      BIT STRING (SIZE (18)),
    long-Macro-eNB-ID       BIT STRING (SIZE (21))
}

-- *****
-- copied from 3GPP 38.423 (XnAP) IEs
-- note: choice-extension removed
-- *****
ENB-ID-Choice ::= CHOICE {
    enb-ID-macro            BIT STRING (SIZE (20)),
    enb-ID-shortmacro       BIT STRING (SIZE (18)),
    enb-ID-longmacro        BIT STRING (SIZE (21)),
    ...
}

-- *****
-- copied from 3GPP 36.423 (X2AP) IEs
-- note: ie-Extensions removed
-- Note: to avoid duplicate names with XnAP, GNB-ID renamed ENGNB-ID, GlobalGNB-ID renamed GlobalenGNB-ID
-- *****
ENGNB-ID ::= CHOICE {
    gNB-ID                  BIT STRING (SIZE (22..32)),
    ...
}

-- F
-- G
GlobaleE2node-ID ::= CHOICE{
    gNB                     GlobaleE2node-gNB-ID,
    en-gNB                  GlobaleE2node-en-gNB-ID,
    ng-eNB                  GlobaleE2node-ng-eNB-ID,
    eNB                     GlobaleE2node-eNB-ID,
    ...
}

GlobaleE2node-en-gNB-ID ::= SEQUENCE{
    global-en-gNB-ID        GlobalenGNB-ID,
    en-gNB-CU-UP-ID         GNB-CU-UP-ID    OPTIONAL,
    en-gNB-DU-ID            GNB-DU-ID       OPTIONAL,
    ...
}

GlobaleE2node-eNB-ID ::= SEQUENCE{
    global-eNB-ID           GlobalENB-ID,
    ...
}

GlobaleE2node-gNB-ID ::= SEQUENCE{
    global-gNB-ID           GlobalgNB-ID,
    global-en-gNB-ID        GlobalenGNB-ID  OPTIONAL,
    gNB-CU-UP-ID            GNB-CU-UP-ID    OPTIONAL,
    gNB-DU-ID               GNB-DU-ID       OPTIONAL,
    ...
}

GlobaleE2node-ng-eNB-ID ::= SEQUENCE{
    global-ng-eNB-ID        GlobalngeNB-ID,
    global-eNB-ID           GlobalENB-ID    OPTIONAL,
    ngENB-DU-ID             NGENB-DU-ID     OPTIONAL,
    ...
}

```

```

}
-- *****
-- copied from 3GPP 36.423 (X2AP) IEs
-- note: ie-Extensions removed
-- *****

GlobalENB-ID ::= SEQUENCE {
    pLMN-Identity      PLMN-Identity,
    eNB-ID             ENB-ID,
    ...
}
-- *****
-- copied from 3GPP 36.423 (X2AP) IEs
-- Note: to avoid duplicate names with XnAP, GNB-ID renamed ENGNB-ID, GlobalGNB-ID renamed GlobalenGNB-ID
-- *****
GlobalenGNB-ID ::= SEQUENCE {
    pLMN-Identity      PLMN-Identity,
    gNB-ID             ENGNB-ID,
    ...
}
-- *****
-- copied from 3GPP 38.423 (XnAP) IEs
-- note: choice-extension removed
-- *****
GlobalgNB-ID      ::= SEQUENCE {
    plmn-id            PLMN-Identity,
    gnb-id             GNB-ID-Choice,
    ...
}

-- *****
-- copied from 3GPP 38.423 (XnAP) IEs
-- note: choice-extension removed
-- *****
GlobalngeNB-ID    ::= SEQUENCE {
    plmn-id            PLMN-Identity,
    enb-id             ENB-ID-Choice,
    ...
}

-- *****
-- [NEW for E2AP v02.00] copied from 3GPP 38.423 (XnAP) IEs
-- Note: extension field removed
-- *****

GlobalNG-RANNode-ID ::= CHOICE {
    gNB                GlobalgNB-ID,
    ng-eNB             GlobalngeNB-ID,
    ...
}

GlobalRIC-ID ::= SEQUENCE{
    pLMN-Identity      PLMN-Identity,
    ric-ID             BIT STRING (SIZE (20)),
    ...
}

-- *****
-- copied from 3GPP 37.483 (E1AP) IEs
-- *****
GNB-CU-UP-ID ::= INTEGER (0..68719476735)

-- *****
-- copied from 3GPP 38.473 (F1AP) IEs
-- *****
GNB-DU-ID ::= INTEGER (0..68719476735)

-- *****
-- copied from 3GPP 38.423 (XnAP) IEs
-- note: choice-extension removed
-- *****
GNB-ID-Choice ::= CHOICE {
    gnb-ID             BIT STRING (SIZE(22..32)),
    ...
}
-- H
-- I
-- J

```

```
-- K
-- L

ListedRecordsOnly ::=          ENUMERATED {true, ...}

LoadMeasurementID ::= INTEGER(1..4095,...)

-- M

-- *****
-- [New for E2AP v02.00] copied from 3GPP 36.413 (S1AP) IEs
-- *****
MMENAME ::= PrintableString (SIZE (1..150,...))

-- N

-- *****
-- copied from 3GPP 37.473 (W1AP) IEs
-- *****
NGENB-DU-ID ::= INTEGER (0..68719476735)

-- O
-- P
-- *****
-- copied from 3GPP 36.423 (X2AP) IEs
-- *****
PLMN-Identity ::= OCTET STRING (SIZE(3))

-- Q
-- R
-- *****
-- Following IE defined in E2SM
-- *****
RANfunctionDefinition ::= OCTET STRING

RANfunctionID ::= INTEGER (0..4095)

RANfunctionOID ::= PrintableString(SIZE(1..1000,...))

RANfunctionRevision ::= INTEGER (0..4095)

RegistrationRequest ::= ENUMERATED{start, stop, add, ...}

RegistrationRequest-E2 ::= ENUMERATED{start, stop, add, ...}

ReportCharacteristics ::= BIT STRING (SIZE (8))

ReportingPeriodicity ::= ENUMERATED{ms500, ms1000, ms2000, ms5000, ms10000, ...}

ReportingPeriodicity-E2 ::= ENUMERATED{ms500, ms1000, ms2000, ms5000, ms10000, ...}

-- *****
-- Following IE defined in E2SM
-- *****
RICactionDefinition ::= OCTET STRING

-- new in E2AP-v03.00
RICactionExecutionOrder ::= INTEGER (0..255, ...)

RICactionID ::= INTEGER (0..255)

RICactionType ::= ENUMERATED{
    report,
    insert,
    policy,
    ...
}
-- *****
-- Following IE defined in E2SM
-- *****
RICassistanceHeader ::= OCTET STRING

-- *****
-- Following IE defined in E2SM
-- *****
RICassistanceMessage ::= OCTET STRING
```

```
-- *****
-- Following IE defined in E2SM
-- *****
RICassistanceOutcome ::= OCTET STRING

RICassistanceSN ::= INTEGER (0..65535)

RICassistanceUpdate ::= ENUMERATED{
    true,
    ...
}

RICassistanceUpdateNumber ::= INTEGER (0..65535)

-- *****
-- Following IE defined in E2SM
-- *****
RICcallProcessID ::= OCTET STRING

RICcontrolAckRequest ::= ENUMERATED{
    noAck,
    ack,
    ...
}

-- *****
-- Following IE defined in E2SM
-- *****
RICcontrolHeader ::= OCTET STRING

-- *****
-- Following IE defined in E2SM
-- *****
RICcontrolMessage ::= OCTET STRING

-- *****
-- Following IE defined in E2SM
-- *****
RICcontrolOutcome ::= OCTET STRING

-- *****
-- Following IE defined in E2SM
-- *****
RICeventTriggerDefinition ::= OCTET STRING

-- *****
-- Following IE defined in E2SM
-- *****
RICindicationHeader ::= OCTET STRING

-- *****
-- Following IE defined in E2SM
-- *****
RICindicationMessage ::= OCTET STRING

RICindicationSN ::= INTEGER (0..65535)

RICindicationType ::= ENUMERATED{
    report,
    insert,
    ...
}

RICloadConfirm ::= ENUMERATED{
    true,
    ...
}

RICloadInformation ::= SEQUENCE{
    loadStatus          ENUMERATED{overload, notoverload, ...}    OPTIONAL,
    loadEstimate        INTEGER (0..100)                          OPTIONAL,
    ...
}

RICloadRequest ::= ENUMERATED{
    true,
    ...
}
```

```

RICrequestID ::= SEQUENCE {
    ricRequestorID      INTEGER (0..65535),
    ricInstanceID       INTEGER (0..65535),
    ...
}

RICserviceLoadConfirm ::= SEQUENCE{
    ricServiceReportLoadConfirm      RICloadConfirm      OPTIONAL,
    ricServiceInsertLoadConfirm      RICloadConfirm      OPTIONAL,
    ricServiceControlLoadConfirm     RICloadConfirm      OPTIONAL,
    ricServicePolicyLoadConfirm      RICloadConfirm      OPTIONAL,
    ricServiceQueryLoadConfirm       RICloadConfirm      OPTIONAL,
    ...
}

RICserviceLoadInformation ::= SEQUENCE{
    ricServiceReportLoadInformation   RICloadInformation OPTIONAL,
    ricServiceInsertLoadInformation   RICloadInformation OPTIONAL,
    ricServiceControlLoadInformation  RICloadInformation OPTIONAL,
    ricServicePolicyLoadInformation   RICloadInformation OPTIONAL,
    ricServiceQueryLoadInformation    RICloadInformation OPTIONAL,
    ...
}

RICserviceLoadRequest ::= SEQUENCE{
    ricServiceReportLoadRequest      RICloadRequest      OPTIONAL,
    ricServiceInsertLoadRequest      RICloadRequest      OPTIONAL,
    ricServiceControlLoadRequest     RICloadRequest      OPTIONAL,
    ricServicePolicyLoadRequest      RICloadRequest      OPTIONAL,
    ricServiceQueryLoadRequest       RICloadRequest      OPTIONAL,
    ...
}

RICsubscriptionTime ::= OCTET STRING (SIZE(8))

RICsubsequentAction ::=SEQUENCE{
    ricSubsequentActionType          RICsubsequentActionType,
    ricTimeToWait                    RICtimeToWait,
    ...
}

RICsubscriptionAuditFlag ::= SEQUENCE{
    listedRecordsOnly                ListedRecordsOnly      OPTIONAL,
    ...
}

RICsubsequentActionType ::= ENUMERATED{
    continue,
    wait,
    ...
}

-- *****
-- Following IE defined in E2SM
-- *****
RICQueryHeader ::= OCTET STRING

-- *****
-- Following IE defined in E2SM
-- *****
RICQueryDefinition ::= OCTET STRING

-- *****
-- Following IE defined in E2SM
-- *****
RICQueryOutcome ::= OCTET STRING

RICtimeToWait ::= ENUMERATED{
    w1ms,
    w2ms,
    w5ms,
    w10ms,
    w20ms,
    w30ms,
    w40ms,
    w50ms,
    w100ms,

```

```

w200ms,
w500ms,
w1s,
w2s,
w5s,
w10s,
w20s,
w60s,
...
}
-- S

-- *****
-- Following IE defined in E2SM
-- *****

ServiceLayerCause ::= OCTET STRING

-- T
-- *****
-- copied from 3GPP 38.413 (NGAP) IEs
-- *****
TimeToWait ::= ENUMERATED {v1s, v2s, v5s, v10s, v20s, v60s, ...}

TNLInformation ::= SEQUENCE{
    tnlAddress      BIT STRING (SIZE(1..160,...)),
    tnlPort         BIT STRING (SIZE(16))  OPTIONAL,
    ...
}

TNLUsage ::= ENUMERATED{ric-service, support-function, both, ...}

TransactionID ::= INTEGER (0..255,...)

-- *****
-- copied from 3GPP 38.413 (NGAP) IEs
-- *****
TypeOfError ::= ENUMERATED {
    not-understood,
    missing,
    ...
}

-- U
-- V
-- W
-- X
-- Y
-- Z

END
-- ASN1STOP

```

9.3.6 Common definitions

```

-- ASN1START
-- *****
--
-- Common definitions
-- Derived from 3GPP 38.413 (NGAP)
--
-- *****

E2AP-CommonDataTypes {
iso(1) identified-organization(3) dod(6) internet(1) private(4) enterprise(1) 53148 e2(1) version2 (2)
e2ap(1) e2ap-CommonDataTypes (3) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

Criticality      ::= ENUMERATED { reject, ignore, notify }

Presence        ::= ENUMERATED { optional, conditional, mandatory }

ProcedureCode    ::= INTEGER (0..255)

```



```

ProtocolIE-ID      ::= INTEGER (0..65535)

TriggeringMessage  ::= ENUMERATED { initiating-message, successful-outcome, unsuccessful-outcome }

END
-- ASN1STOP

```

9.3.7 Constant definitions

```

-- ASN1START
-- *****
--
-- Constant definitions
--
-- *****

E2AP-Constants {
iso(1) identified-organization(3) dod(6) internet(1) private(4) enterprise(1) 53148 e2(1) version2 (2)
e2ap(1) e2ap-Constants (4) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

IMPORTS
    ProcedureCode,
    ProtocolIE-ID
FROM E2AP-CommonDataTypes;

-- *****
--
-- Elementary Procedures
--
-- *****
id-E2setup                ProcedureCode ::= 1
id-ErrorIndication        ProcedureCode ::= 2
id-Reset                  ProcedureCode ::= 3
id-RICcontrol             ProcedureCode ::= 4
id-RICindication          ProcedureCode ::= 5
id-RICserviceQuery        ProcedureCode ::= 6
id-RICserviceUpdate       ProcedureCode ::= 7
id-RICsubscription        ProcedureCode ::= 8
id-RICsubscriptionDelete  ProcedureCode ::= 9
id-E2nodeConfigurationUpdate ProcedureCode ::= 10
id-E2connectionUpdate     ProcedureCode ::= 11
id-RICsubscriptionDeleteRequired ProcedureCode ::= 12
id-E2removal              ProcedureCode ::= 13
id-RICsubscriptionModification ProcedureCode ::= 14
id-RICsubscriptionModificationRequired ProcedureCode ::= 15
id-RICquery               ProcedureCode ::= 16
id-RICsubscriptionAudit   ProcedureCode ::= 17
id-RICserviceLoadStatus   ProcedureCode ::= 18
id-RICserviceLoadUpdate   ProcedureCode ::= 19
id-RICsubscriptionStateControl ProcedureCode ::= 20
id-RICassistance          ProcedureCode ::= 21
id-RICassistanceIndication ProcedureCode ::= 22
id-RICassistanceHalt      ProcedureCode ::= 23
id-E2resourceStatusReportingInitiation ProcedureCode ::= 24
id-E2resourceStatusUpdate ProcedureCode ::= 25

-- *****
--
-- Extension constants
--
-- *****

maxProtocolIEs                INTEGER ::= 65535

-- *****
--
-- Lists
--
-- *****
maxnoofErrors                INTEGER ::= 256
maxofE2nodeComponents        INTEGER ::= 1024
maxofRANfunctionID           INTEGER ::= 256

```

```

maxofRICactionID          INTEGER ::= 16
maxofTNLA                 INTEGER ::= 32
maxofRICrequestID         INTEGER ::= 1024
maxofRICsubscriptions     INTEGER ::= 2147483648

```

```

-- *****
--
-- IEs
--
-- *****
id-Cause                      ProtocolIE-ID ::= 1
id-CriticalityDiagnostics    ProtocolIE-ID ::= 2
id-GlobalE2node-ID          ProtocolIE-ID ::= 3
id-GlobalRIC-ID              ProtocolIE-ID ::= 4
id-RANfunctionID            ProtocolIE-ID ::= 5
id-RANfunctionID-Item        ProtocolIE-ID ::= 6
id-RANfunctionIEcause-Item   ProtocolIE-ID ::= 7
id-RANfunction-Item          ProtocolIE-ID ::= 8
id-RANfunctionsAccepted      ProtocolIE-ID ::= 9
id-RANfunctionsAdded         ProtocolIE-ID ::= 10
id-RANfunctionsDeleted       ProtocolIE-ID ::= 11
id-RANfunctionsModified      ProtocolIE-ID ::= 12
id-RANfunctionsRejected      ProtocolIE-ID ::= 13
id-RIAction-Admitted-Item    ProtocolIE-ID ::= 14
id-RIActionID                ProtocolIE-ID ::= 15
id-RIAction-NotAdmitted-Item ProtocolIE-ID ::= 16
id-RIActions-Admitted        ProtocolIE-ID ::= 17
id-RIActions-NotAdmitted     ProtocolIE-ID ::= 18
id-RIAction-ToBeSetup-Item   ProtocolIE-ID ::= 19
id-RIccallProcessID          ProtocolIE-ID ::= 20
id-RIccontrolAckRequest      ProtocolIE-ID ::= 21
id-RIccontrolHeader          ProtocolIE-ID ::= 22
id-RIccontrolMessage         ProtocolIE-ID ::= 23
id-RIccontrolStatus          ProtocolIE-ID ::= 24
id-RIccindicationHeader      ProtocolIE-ID ::= 25
id-RIccindicationMessage     ProtocolIE-ID ::= 26
id-RIccindicationSN          ProtocolIE-ID ::= 27
id-RIccindicationType        ProtocolIE-ID ::= 28
id-RIccrequestID             ProtocolIE-ID ::= 29
id-RIccsubscriptionDetails   ProtocolIE-ID ::= 30
id-TimeToWait                ProtocolIE-ID ::= 31
id-RIcccontrolOutcome        ProtocolIE-ID ::= 32
id-E2nodeComponentConfigUpdate ProtocolIE-ID ::= 33
id-E2nodeComponentConfigUpdate-Item ProtocolIE-ID ::= 34
id-E2nodeComponentConfigUpdateAck ProtocolIE-ID ::= 35
id-E2nodeComponentConfigUpdateAck-Item ProtocolIE-ID ::= 36
id-E2connectionSetup         ProtocolIE-ID ::= 39
id-E2connectionSetupFailed    ProtocolIE-ID ::= 40
id-E2connectionSetupFailed-Item ProtocolIE-ID ::= 41
id-E2connectionFailed-Item    ProtocolIE-ID ::= 42
id-E2connectionUpdate-Item    ProtocolIE-ID ::= 43
id-E2connectionUpdateAdd      ProtocolIE-ID ::= 44
id-E2connectionUpdateModify   ProtocolIE-ID ::= 45
id-E2connectionUpdateRemove   ProtocolIE-ID ::= 46
id-E2connectionUpdateRemove-Item ProtocolIE-ID ::= 47
id-TNLIinformation           ProtocolIE-ID ::= 48
id-TransactionID              ProtocolIE-ID ::= 49
id-E2nodeComponentConfigAddition ProtocolIE-ID ::= 50
id-E2nodeComponentConfigAddition-Item ProtocolIE-ID ::= 51
id-E2nodeComponentConfigAdditionAck ProtocolIE-ID ::= 52
id-E2nodeComponentConfigAdditionAck-Item ProtocolIE-ID ::= 53
id-E2nodeComponentConfigRemoval ProtocolIE-ID ::= 54
id-E2nodeComponentConfigRemoval-Item ProtocolIE-ID ::= 55
id-E2nodeComponentConfigRemovalAck ProtocolIE-ID ::= 56
id-E2nodeComponentConfigRemovalAck-Item ProtocolIE-ID ::= 57
id-E2nodeTNLassociationRemoval ProtocolIE-ID ::= 58
id-E2nodeTNLassociationRemoval-Item ProtocolIE-ID ::= 59
id-RIccsubscriptionToBeRemoved ProtocolIE-ID ::= 60
id-RIccsubscription-withCause-Item ProtocolIE-ID ::= 61
id-RIccsubscriptionStartTime ProtocolIE-ID ::= 62
id-RIccsubscriptionEndTime   ProtocolIE-ID ::= 63
id-RIccEventTriggerDefinitionToBeModified ProtocolIE-ID ::= 64
id-RIccactionsToBeRemovedForModification-List ProtocolIE-ID ::= 65
id-RIccaction-ToBeRemovedForModification-Item ProtocolIE-ID ::= 66
id-RIccactionsToBeModifiedForModification-List ProtocolIE-ID ::= 67
id-RIccaction-ToBeModifiedForModification-Item ProtocolIE-ID ::= 68
id-RIccactionsToBeAddedForModification-List ProtocolIE-ID ::= 69
id-RIccaction-ToBeAddedForModification-Item ProtocolIE-ID ::= 70

```

```

id-RIActionsRemovedForModification-List      ProtocolIE-ID ::= 71
id-RIAction-RemovedForModification-Item      ProtocolIE-ID ::= 72
id-RIActionsFailedToBeRemovedForModification-List ProtocolIE-ID ::= 73
id-RIAction-FailedToBeRemovedForModification-Item ProtocolIE-ID ::= 74
id-RIActionsModifiedForModification-List      ProtocolIE-ID ::= 75
id-RIAction-ModifiedForModification-Item      ProtocolIE-ID ::= 76
id-RIActionsFailedToBeModifiedForModification-List ProtocolIE-ID ::= 77
id-RIAction-FailedToBeModifiedForModification-Item ProtocolIE-ID ::= 78
id-RIActionsAddedForModification-List         ProtocolIE-ID ::= 79
id-RIAction-AddedForModification-Item         ProtocolIE-ID ::= 80
id-RIActionsFailedToBeAddedForModification-List ProtocolIE-ID ::= 81
id-RIAction-FailedToBeAddedForModification-Item ProtocolIE-ID ::= 82
id-RIActionsRequiredToBeModified-List         ProtocolIE-ID ::= 83
id-RIAction-RequiredToBeModified-Item         ProtocolIE-ID ::= 84
id-RIActionsRequiredToBeRemoved-List          ProtocolIE-ID ::= 85
id-RIAction-RequiredToBeRemoved-Item          ProtocolIE-ID ::= 86
id-RIActionsConfirmedForModification-List      ProtocolIE-ID ::= 87
id-RIAction-ConfirmedForModification-Item      ProtocolIE-ID ::= 88
id-RIActionsRefusedToBeModified-List          ProtocolIE-ID ::= 89
id-RIAction-RefusedToBeModified-Item          ProtocolIE-ID ::= 90
id-RIActionsConfirmedForRemoval-List           ProtocolIE-ID ::= 91
id-RIAction-ConfirmedForRemoval-Item           ProtocolIE-ID ::= 92
id-RIActionsRefusedToBeRemoved-List           ProtocolIE-ID ::= 93
id-RIAction-RefusedToBeRemoved-Item           ProtocolIE-ID ::= 94
id-RIQueryHeader                             ProtocolIE-ID ::= 95
id-RIQueryDefinition                         ProtocolIE-ID ::= 96
id-RIQueryOutcome                           ProtocolIE-ID ::= 97
id-RISubscriptionAuditFlag                   ProtocolIE-ID ::= 98
id-RISubscriptionAuditList                   ProtocolIE-ID ::= 99
id-RISubscriptionAudit-Item                  ProtocolIE-ID ::= 100
id-RISubscriptionAuditAction-Item            ProtocolIE-ID ::= 101
id-RISubscriptionAuditConfirmedList           ProtocolIE-ID ::= 102
id-RISubscriptionAuditMissingList            ProtocolIE-ID ::= 103
id-RISubscriptionAuditUnkownList             ProtocolIE-ID ::= 104
id-E2nodeLoadMeasurementID                   ProtocolIE-ID ::= 105
id-RANfunctionLoad-Item                      ProtocolIE-ID ::= 106
id-RANfunctionLoad-List                      ProtocolIE-ID ::= 107
id-RANfunctionLoadConfirm-Item               ProtocolIE-ID ::= 108
id-RANfunctionLoadConfirm-List               ProtocolIE-ID ::= 109
id-RANfunctionLoadRequest-Item               ProtocolIE-ID ::= 110
id-RANfunctionLoadRequest-List               ProtocolIE-ID ::= 111
id-RegistrationRequest                       ProtocolIE-ID ::= 112
id-ReportingPeriodicity                      ProtocolIE-ID ::= 113
id-RIActionLoad-Item                         ProtocolIE-ID ::= 114
id-RIActionLoadConfirm-Item                  ProtocolIE-ID ::= 115
id-RIActionLoadRequest-Item                  ProtocolIE-ID ::= 116
id-RIcloadMeasurementID                      ProtocolIE-ID ::= 117
id-RISubscriptionLoad-Item                   ProtocolIE-ID ::= 118
id-RISubscriptionLoadConfirm-Item             ProtocolIE-ID ::= 119
id-RISubscriptionLoadRequest-Item             ProtocolIE-ID ::= 120
id-RANfunctionStateConfirm-Item              ProtocolIE-ID ::= 121
id-RANfunctionStateConfirm-List              ProtocolIE-ID ::= 122
id-RANfunctionStateControl-Item              ProtocolIE-ID ::= 123
id-RANfunctionStateControl-List              ProtocolIE-ID ::= 124
id-RIActionList-Item                         ProtocolIE-ID ::= 125
id-RISubscriptionList-Item                   ProtocolIE-ID ::= 126
id-RIAssistanceHeader                       ProtocolIE-ID ::= 127
id-RIAssistanceMessage                       ProtocolIE-ID ::= 128
id-RIAssistanceOutcome                       ProtocolIE-ID ::= 129
id-RIAssistanceSN                           ProtocolIE-ID ::= 130
id-RIAssistanceUpdate                       ProtocolIE-ID ::= 131
id-RIAssistanceUpdateNumber                  ProtocolIE-ID ::= 132
id-RegistratinoRequest-E2                     ProtocolIE-ID ::= 133
id-ReportingPeriodicity-E2                   ProtocolIE-ID ::= 134
id-ReportCharacteristics                     ProtocolIE-ID ::= 135
id-TNLCapacityIndicator                     ProtocolIE-ID ::= 136
id-HardwareLoadIndicator                     ProtocolIE-ID ::= 137
id-RANfunctionsAcceptModified-List           ProtocolIE-ID ::= 138
id-RANfunction-AcceptModified-Item           ProtocolIE-ID ::= 139

```

```

END
-- ASN1STOP

```

9.3.8 Container definitions

```

-- ASN1START
-- *****
--

```

```
-- Container definitions
--
-- derived from 3GPP 38.413 (NGAP)
-- *****

E2AP-Containers {
iso(1) identified-organization(3) dod(6) internet(1) private(4) enterprise(1) 53148 e2(1) version2 (2)
e2ap(1) e2ap-Containers (5) }

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- *****
--
-- IE parameter types from other modules.
--
-- *****

IMPORTS

    Criticality,
    Presence,
    ProtocolIE-ID
FROM E2AP-CommonDataTypes

    maxProtocolIEs
FROM E2AP-Constants;

-- *****
--
-- Class Definition for Protocol IEs
--
-- *****

E2AP-PROTOCOL-IES ::= CLASS {
    &id                ProtocolIE-ID                UNIQUE,
    &criticality        Criticality,
    &Value,
    &presence           Presence
}
WITH SYNTAX {
    ID                &id
    CRITICALITY        &criticality
    TYPE                &Value
    PRESENCE           &presence
}

-- *****
--
-- Class Definition for Protocol IEs
--
-- *****

E2AP-PROTOCOL-IES-PAIR ::= CLASS {
    &id                ProtocolIE-ID                UNIQUE,
    &firstCriticality    Criticality,
    &FirstValue,
    &secondCriticality    Criticality,
    &SecondValue,
    &presence           Presence
}
WITH SYNTAX {
    ID                &id
    FIRST CRITICALITY    &firstCriticality
    FIRST TYPE            &FirstValue
    SECOND CRITICALITY    &secondCriticality
    SECOND TYPE            &SecondValue
    PRESENCE             &presence
}

-- *****
--
-- Container for Protocol IEs
--
-- *****
```

```

ProtocolIE-Container {E2AP-PROTOCOL-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (0..maxProtocolIEs)) OF
        ProtocolIE-Field {{IEsSetParam}}

ProtocolIE-SingleContainer {E2AP-PROTOCOL-IES : IEsSetParam} ::=
    ProtocolIE-Field {{IEsSetParam}}

ProtocolIE-Field {E2AP-PROTOCOL-IES : IEsSetParam} ::= SEQUENCE {
    id                E2AP-PROTOCOL-IES.&id                {{IEsSetParam}},
    criticality        E2AP-PROTOCOL-IES.&criticality        {{IEsSetParam}}{@id}},
    value              E2AP-PROTOCOL-IES.&Value              {{IEsSetParam}}{@id}}
}

-- *****
--
-- Container for Protocol IE Pairs
--
-- *****

ProtocolIE-ContainerPair {E2AP-PROTOCOL-IES-PAIR : IEsSetParam} ::=
    SEQUENCE (SIZE (0..maxProtocolIEs)) OF
        ProtocolIE-FieldPair {{IEsSetParam}}

ProtocolIE-FieldPair {E2AP-PROTOCOL-IES-PAIR : IEsSetParam} ::= SEQUENCE {
    id                E2AP-PROTOCOL-IES-PAIR.&id                {{IEsSetParam}},
    firstCriticality   E2AP-PROTOCOL-IES-PAIR.&firstCriticality   {{IEsSetParam}}{@id}},
    firstValue         E2AP-PROTOCOL-IES-PAIR.&FirstValue         {{IEsSetParam}}{@id}},
    secondCriticality  E2AP-PROTOCOL-IES-PAIR.&secondCriticality  {{IEsSetParam}}{@id}},
    secondValue        E2AP-PROTOCOL-IES-PAIR.&SecondValue        {{IEsSetParam}}{@id}}
}

-- *****
--
-- Container Lists for Protocol IE Containers
--
-- *****

ProtocolIE-ContainerList {INTEGER : lowerBound, INTEGER : upperBound, E2AP-PROTOCOL-IES : IEsSetParam} ::=
    SEQUENCE (SIZE (lowerBound..upperBound)) OF
        ProtocolIE-SingleContainer {{IEsSetParam}}

ProtocolIE-ContainerPairList {INTEGER : lowerBound, INTEGER : upperBound, E2AP-PROTOCOL-IES-PAIR :
IEsSetParam} ::=
    SEQUENCE (SIZE (lowerBound..upperBound)) OF
        ProtocolIE-ContainerPair {{IEsSetParam}}

END
-- ASN1STOP

```

9.4 Message transfer syntax

E2AP shall use the ASN.1 Basic Packed Encoding Rules (BASIC-PER) Aligned Variant as transfer syntax, as specified in Recommendation ITU-T X.691 [15].

9.5 Timers

The following Timers are defined for use over the E2 interface in Near-RT RIC and E2 Node.

T_{RICEVENTcreate}

- Specifies the maximum time for the RIC Subscription procedure in the Near-RT RIC.

T_{RICEVENTdelete}

- Specifies the maximum time for the RIC Subscription Deletion procedure in the Near-RT RIC.

T_{RICEVENTmodify}

- Specifies the maximum time for the RIC Subscription Modification procedure in the Near-RT RIC.

$T_{\text{RICcontrol}}$

- Specifies the maximum time for the RIC Control procedure in the Near-RT RIC.

 T_{RICquery}

- Specifies the maximum time for the RIC Query procedure in the Near-RT RIC.

10 Handling of unknown, unforeseen and erroneous protocol data

Clause 10 of 3GPP TS 36.413 [24] is applicable for the purposes of the present document.

Annex (informative): Change history/Change request (history)

Date	Revision	Description
2020.01.22	01.00.00	Specification renamed v01.00.00 for approval
2020.01.28	01.00.00	Initial version
2020.07.08	01.00.01	Addition of CR adopted during meeting #60
2020.07.13	01.00.02	ASN.1 corrections, Table correction to align with ASN.1
2020.07.15	01.01	Editorial and functional corrections
2021.01.13	02.00.01	Addition of CR <RSY-2021.01.13-WG3-CR-0001-E2AP Spec v1.01 Corrections-v14> agreed at WG3#80 meeting, plus editorial corrections
2021.04.21	02.00.02	Addition of CRs <NOK-2021.03.02-WG3-E2AP-CR-0002-TNLA removal-v01.docx > agreed at WG3#88 <NOK.AO-2021.01.26-WG3-CR-0001-E2AP-RANconfig-v04.docx> agreed at WG3#94
2021.05.27	02.00.03	Addition of CR <INT-2021.05.26-WG3-CR-0005-E2AP-RICsubs_delete.docx> agreed at WG3#99. Endorsed WG3#100
2021.06.09	02.00.04	Addition of CR: < NOK.AO-2021.05.26-WG3-CR-0003-E2AP-RIC control-v01> agreed at WG3#100 Re-implementation of part of <NOK.AO-2021.01.26-WG3-CR-0001-E2AP-RANconfig-v04.docx> agreed at WG3#94 correcting error introduced in v02.00.02
2021.07.11	02.00.05	NOK-2021-06-09.WG3.CR-0004-E2AP-v02.00.04editorials-v01
2021.08.10	02.00	New features: RIC Subscription Delete, TNLA Removal. Corrections to all RIC service procedures. Change to ASN.1 version
2021.09.20	02.01.00	Addition of CR < NOK-2021.09.01-WG3-CR-0006-E2APv2.0-errata9.3.7-v01 >
2021.11.02	02.01.01	Addition of CR <SAM-2021.10.19-WG3-CR-0001-E2AP_E2Removal-v03 > approved WG3#117 Aligned format for July21 publication changes
2021.11.22	02.01.02	Corrections based on E2APv02.01 WG3 approval review process
2022.02.07	02.01	New feature: E2 removal. Editorial and functional corrections
2022.02.08	02.02.01	Addition of CR < NOK.AO-2022.01.05-WG3-CR-0008-E2AP-Global-gNB-ID-v01 > approved at WG3#127
2022.03.25	02.02.02	Addition of CRs approved at WG3#134 < NOK.AO-2022.02.21-WG3-CR-0010-E2AP-RIC Service Query clarification-v01> < NEC-2022.02.28-WG3-CR-0002-E2AP- RIC CONTROL FAILURE -v03> < NOK.AO-2022.03.03-WG3-CR-0012-E2AP-RIC Service Update Ack clarification-v02> < NOK-2022.01.03-WG3-CR-0011-E2AP-Reducing MAX limits-v02> approved at WG3#137 Note: This version contains non-backward compatible change with respect to v02.01 impacting RIC Subscription Delete Required message
2022.04.04	02.02.03	Editorial changes based on remarks during WG3 approval
2022.06.29	02.02	Clarification to RIC Service Update, RIC Control. Editorial and functional corrections
2022.06.20	02.02.04	Addition of CR < NOK.AO-2022.05.24-WG3-CR-0013-E2AP-EditorialCorrections-v02 >
2022.07.15	02.02.05	Addition of <NOK.AO-2022.06.24-WG3-CR-0015-E2AP-RIC Service Update Ack (ASN.1)-v03>
2022.07.20	02.03	Editorial and functional corrections
2022.11.02	02.03.01	Addition of CR: < NOK.AO-2022.09.02-WG3-CR-0017-E2AP-RIC Subscription duration limit-v06 > < NOK-2022.09.01-WG3-CR-0016-E2AP-E1AP reference correction-v03 >, error in CR corrected (ref [21] should be 37.483) Addition of "skin rose" to all PlantUML code to restore O-RAN look
2022.11.10	02.03.02	Editorial changes: - Correction to ASN.1 (correcting error in E2APv02.03.01) - Rearrangement of Table 8.1-1: Class 1 Elementary Procedures - Added missing Timer definitions Addition of CRs: < NOK-2022.06.21-WG3-CR-0014-E2AP-RIC subscription handling improvements-v06>, error in CR corrected (ASN.1 for Cause value aligned to name in 9.2.1) < MAV.AO-2022.03.03.WG3-CR-0014.E2AP 02.03 RIC_Subscription_Modification procedure-v15> <QCM.AO-2022.09.08-WG3-CR-0002-E2AP-RIC_Query_Procedure_v11>, error in CR corrected (ASN.1 for Cause value aligned to name in 9.2.1)

Date	Revision	Description
2022.11.16	02.03.03	Changes reflecting remarks received during WG3 approval process - Alignment to latest O-RAN template - Added R003 to file name - Updated copyright year - Removed "RIC Action Type" from RIC Subscription Modification to align with late submitted revised CR (impacts 8.2.5, 9.1.1.11, 9.3.4) - Corrected errors in CRs implemented in previous drafts - Corrected ASN.1 label "gNB-CU-CP-ID" to read "gNB-CU-UP-ID" in section 9.3.5
2022.12.07	03.00	New features: RIC Subscription Modification, RIC Query. Clarifications to RIC Subscription. Editorial and functional corrections
2023.01.27	03.00.01	CR <NOK-2023.01.09-WG3-CR-0019-E2AP-PAS step1-v01> approved WG3#171
2023.02.16	03.00.02	CR <NOK-2023.02.15-WG3-CR-0020-E2AP-PAS step2-v02> approved Prague F2F 16/2/2023
2023.03.17	03.00.03	CR <NOK-2023.03.13-WG3-CR-0021-E2AP-PAS step3-v4> approved by correspondence after WG3#176
2023.03.24	03.00.04	Inclusion of corrections agreed during WG3 approval process as per < O-RAN.WG3.E2AP-R003-v03.00.03-approvalChanges-v3 >
2023.03.24	03.01	Alignment of O-RAN Drafting Rules (ODR) in preparation for ETSI PAS submission
2023.05.15	03.01.01	CR <NOK-2023.04.28-WG3-CR-0022-E2AP-Correction to clause 2.1-v01> approved WG3#183
2023.06.27	03.01.02	CR <NOK-2023.05.26-WG3-CR-0024-E2AP-Critical ASN.1 correction-v01> approved WG3#189
2023.07.11	03.01.03	CR <SAM.AO-2023.05.10-WG3-CR-0001-E2AP-E2 Nod Component Configuration-v04> approved WG3#189
2023.07.26	03.01.04	Editorial corrections based on comments received during WG3 poll.
2023.07.28	03.01.05	Editorial changes to align to O-RAN TS template v01.01
2023.07.28	04.00	Clarification on E2 Node components configuration. Editorial and functional corrections
2023.10.04	04.00.01	CR <NOK.AO-2023.05.22-WG3-CR-0023-E2AP-SubscriptionAudit-v10.docx> approved WG3#201 and editorial corrections to figure title text.
2023.11.15	04.00.02	Editorial corrections implementing WG3 voting period feedback
2023.11.21	04.00.03	Restored full revision history into Annex: Change History and deletion of Annex: History
2023.11.21	05.00	New feature: RIC Subscription Audit. Editorial and functional corrections
2024.02.29	05.00.01	CR <NOK-2023.11.24-WG3-CR-0025-E2AP-Cleanup-v01> approved WG3#207 CR <NOK-2024.02.13-WG3-CR-0026-E2AP-Change history-v01> approved F2F Athens
2024.05.22	05.00.02	CR <NOK-2024.04.09-WG3-CR-0027-E2AP-Editorial corrections for PAS-step4-v04> approved WG3#225 Also includes editorial changes to align with ETSI "Edit Help" changes during PAS processing of v04.00
2024.06.21	05.00.03	CR <NOK-2025.06.04-E2AP-CR-0028-Clause 8.2.1.3 inconsistency term-v01> approved WG3 Incheon F2F CR <NOK-2025.06.04-E2AP-CR-0029-Clause 8.2.3.2 changes for PAS-v01> approved WG3 Incheon F2F CR <NOK-2025.06.04-E2AP-CR-0030-Clause 8.2.5.3+8.2.6.3 changes for PAS-v02 > approved WG3 Incheon F2F CR <NOK-2025.06.04-E2AP-CR-0031-Clause 8.2.7.3 changes for PAS-v01> approved WG3 Incheon F2F CR <NOK-2025.06.19-E2AP-CR-0032-Splitting Clause 8.3.4 RIC Service Update-v01> approved WG3#230
2024.07.26	05.00.04	Editorial corrections implementing review comments collected during July24 train approval
2024.07.26	06.00	Alignment to ETSI Drafting Rules and implementation of all agreed ETSI PAS comments
2024.10.09	06.00.01	CR <NOK-2024.09.27-E2AP-CR-0034-EditHelp alignment-v1> approved WG3#241

Date	Revision	Description
2024.11.22	06.00.02	Editorial changes to align with O-RAN WORKPROC v04.00 CR <NOK-2024.10.23-WG3-CR-0036-E2AP-Handling LS from MSG-v04> approved F2F Montreal CR < NOK-2024.10.1-E2AP-CR-0035-ErrorHandling added in Cause - v06> approved WG3 ArchTG#113 CR <NOK.AO-2024.09.16-WG3-CR-0033-E2AP-E2 Overload-v05> approved WG3 ArchTG#113 CR < NOK.AO-2024.09.16-WG3-CR-0037-E2AP-RIC Subscription State Control-v4> approved WG3 ArchTG#113 CR < NOK-2024.11.13-WG3-CR-0039-E2AP-Clause 9.3.1 correction-v1> approved WG3#245 CR < NOK-2024.11.08-E2AP-CR-0038-RIC Assistance-v06> approved WG3#245 Editorial corrections to clause 9.2 and ASN.1
2024.12.03	06.00.03	CR <LGE-2024.10.25-WG3-CR-0002-E2AP-Correction_RICaudit_v01> approved F2F Montreal Editorial corrections implementing comments received during Nov24 train approval process
2024.12.04	07.00	New features: RIC Subscription State Control, RIC Assistance, Service Level error handling. Implementation of changes related to ETSI MSG Liaison
2025.07.09	07.00.01	Implemented the following CRs with the rapporteur's clean-ups: - CR <RSYS.AO-2024.12.12-WG3-CR-0001-E2AP- Support of Resource Status Report-v07> approved at WG3#251 - CR <LGE-2025.03.12-WG3-CR-0006-E2AP-RICassist_onE2Setup-v01(withASN1)> approved at WG3#255 - CR <LGE-2025.05.26-WG3-CR-0007-E2AP-RICassist_onRICServiceUpdate-v01(withASN1)> approved at June 2025 F2F Fukuoka
2025.07.28	08.00	New features: Support for E2 resource status reporting and ASSISTANCE service capability exposure